8 Ecological Impact

8.1 Legislation, Standards and Guidelines

8.1.1 Local Legislation, Standards and Guidelines

8.1.1.1 The relevant legislation and associated guidelines applicable to the present study for the assessment of ecological impacts include:

- Forests and Countryside Ordinance (Cap. 96) and its subsidiary legislation, the Forestry Regulations;
- Wild Animals Protection Ordinance (Cap. 170);
- Environmental Impact Assessment Ordinance (Cap. 499) and relevant annexes 8, 11, 16, 20 and 21 of the associated Technical Memorandum;
- EIA Study Brief No. ESB-246/2012
- Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586) and its subsidiary legislation;
- Hong Kong Planning Standards and Guidelines (HKPSG) Chapter 10, "Conservation";
- Planning, Environmental & Lands Branch Technical Circular No. 1/97, Works Branch Technical Circular No. 4/97, "Guidelines for Implementing the Policy of Off-site Ecological Mitigation Measures";
- EIAO Guidance Note No. 6/2010 – Some Observations on Ecological Assessment from the Environmental Impact Assessment Ordinance Perspectives;
- EIAO Guidance Note No. 7/2010 – Ecological Baseline Survey for Ecological Assessment;
- EIAO Guidance Note No. 10/2010 – Methodologies for Terrestrial and Freshwater Ecological Baseline Survey

8.1.2 International Conventions and Guidelines

8.1.2.1 International conventions and guidelines potentially relevant to the assessment include:

- Convention on International Trade in Endangered Species of Wild Fauna and Flora ("CITES"). This Convention was first signed in 1973 and entered into force in 1975 to regulate the import and export of endangered species through a permit system. Depending on the degree of threat posed by international trade, CITES classifies over 30,000 endangered species of animals and plants into three Appendices. Hong Kong's obligation under this Convention is enforced via the Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586).
- International Union for Conservation of Nature (IUCN) and its associated Species Survival Commission (SCC) has set global standards and

definitive international standard for species extinction risk – the IUCN Red List of Threatened Species.

- United Nations Convention on Biological Diversity – this convention requires parties to regulate or manage biological resources important for the conservation of biological diversity whether within or outside protected areas, with a view to ensuring their conservation and sustainable use. It also requires parties to promote the protection of ecosystems, natural habitats and the maintenance of viable populations of species in natural surroundings. The People's Republic of China (PRC) ratified the Convention on Biological Diversity on 5th January 1993. The HKSAR Government has stated that it is "committed to meeting the environmental objectives" of the Convention. The Convention and Protocol were extended to Hong Kong on 9th May 2011.

8.2 Ecological Context

8.2.1 Existing Environment

8.2.1.1 The Assessment Area and the PDA of the Project are dominated by developed areas with extensive areas of low-rise village buildings and rural industrial developments. The PDA covers approximately 223.5ha of Yuen Long South Area. The Assessment Area (i.e. area within 500m from the boundary of all proposed development within PDA and the associated infrastructural works in the RODP) covers approximately 1,542.05 ha of the Yuen Long South area.

8.2.1.2 Within the Assessment Area there are several ecologically sensitive receivers including:

- The active egretry at Tai Tong and the breeding egrets and herons, including their flight lines to the foraging grounds;
- Wetlands and seasonal wetlands, including marshes, fishponds, wet agricultural lands in particular those southwest of Shan Ha Tsuen. Such habitats are potentially favoured by a range of herpetofauna species along with odonate and wetland-dependent avifauna species;
- Natural and semi-natural stream courses, in particular the Ecologically Important Stream (EIS) at Yeung Ka Tsuen, riparian vegetation and associated odonates;
- Secondary woodlands, in particular those near Tong Yan San Tsuen and south of Tong Yan San Tsuen Road, along the hillsides adjoining the western boundary of the PDA, those lowland woodlands behind Shan Ha Tsuen, near Lam Hau Tsuen and Lam Hau Wai Tsuen, and near Yeung Ka Tsuen respectively;
- Birds, in particular farmland and wetland-dependent birds including the Greater Painted-Snipe *Rostratula benghalensis*;
- Other fauna groups, including but not limited to mammals, herpetofauna, butterflies, odonates (e.g. Tawny Hooktail *Paragomphus carpricornis* and Club-tailed Cruiser *Macromia urania*), freshwater fishes and macro-invertebrates (e.g. *Caridina apodosis*) (Wilson 1997, Cai & Ng 1999)

- Several recognized sites of conservation importance in the vicinity of the Project, including Tai Lam Country Park (~30ha in Assessment Area), "Conservation Area" (~107ha in Assessment Area) and "Green Belt" zones (a total of ~188ha in Assessment Area), and the EIS at Yeung Ka Tsuen (~550m in Assessment Area).

8.2.1.3 **Figure 8.1** illustrates the extent of RODP and the recognized sites of conservation importance within the 500m Assessment Area for ecological impact assessment.

8.3 Field Survey and Assessment Methodology

8.3.1 Survey Methodology and Assessment Area

8.3.1.1 As stipulated in the EIA Study Brief No. ESB-246/2012 issued in June 2012, necessary field surveys for duration of at least 9 months to adequately cover the wet and dry seasons and the ardeid breeding season shall be conducted to fulfil the objectives of the EIA study. The Assessment Area for ecological impact assessment shall include all areas within 500m from site boundary of the Project Area (i.e. the PDA and the extent of the associated infrastructural works/ road facilities outside PDA) as shown in **Figure 8.1**. Detailed ecological field surveys were conducted from January to September 2013. During the formation of the RODP, there are new proposals for the infrastructures/road works outside PDA. The corresponding 500m assessment area of these new proposals was not covered in the survey conducted in 2013. As such, additional surveys were conducted from January through September 2014 to supplement the surveys in 2013. More additional surveys were conducted between September and December 2016 to update and verify the ecological baseline condition of the new Assessment Area as a result of revisions to RODP to reflect the latest development in the RODP.

8.3.1.2 All methodologies of ecological surveys followed those recommendations in EIAO Guidance notes No. 7/2010, 10/2010 and 11/2010 and as set out in the Inception Report under this Study. **Table 8.1** summarises the survey programme of flora and faunal surveys.

8.3.2 Literature Review

8.3.2.1 A detailed desktop study of any relevant information including past studies, surveys and literature was undertaken to identify key ecological issues and information gaps within the Assessment Area.

8.3.2.2 For the desktop review of status of the known egretry at Tai Tong, reference was made to the summary reports (2003 to 2016) prepared by the Hong Kong Bird Watching Society submitted to Agriculture, Fisheries and Conservation Department, Hong Kong Special Administrative Region Government and other updated information from the government.

8.3.2.3 The review also includes the EIA and TIA Studies for the Stage 2 of PWP Item No. 215DS-Yuen Long and Kam Tin Sewerage and Sewage Disposal (YLKTSSD) (AEIAR-078/2004) and the approved EIA of Hung Shui Kiu New Development Area Planning and Engineering Study – Investigation (AEIAR no. 203/2016). These are the only available EIA studies containing ecological survey data collected within or near the Assessment Area of the current Study. The study area(s) of previous EIA studies are shown in **Figure 8.1**.

8.3.2.4 Based on the available information reviewed, systematic surveys were carried out to cover flora and all key groups of fauna and their associated habitats in the Assessment Area to fill the information gaps. Efforts were made to search for any species or habitats of conservation importance and update the conditions of any known sites of conservation significance identified based on the desktop study.

8.3.3 Habitat Mapping and Vegetation Survey

8.3.3.1 A reconnaissance survey was performed at the commencement of the survey period to characterize habitats and to select representative areas for detailed baseline surveys. Habitat maps of suitable scale (1:5000) were prepared with reference to recent aerial photographs and confirmed by local ground-truthing. A habitat and vegetation survey was conducted during the period from March to September 2013, from March to September 2014, and from September to December 2016 covering both the dry and the wet seasons to record the dominant and notable plant species, their status in Hong Kong and relative abundance. General characteristics of the flora community present in each habitat were noted for use in habitat description and evaluation. All recognized sites of conservation importance, locations of habitats and species of conservation importance recorded in the assessment area were mapped. Representative colour photographic records of each habitat type and the identified ecological features were taken and presented with the map.

8.3.4 Terrestrial Mammal Surveys

8.3.4.1 Surveys for terrestrial mammals were conducted simultaneously with daytime herpetofauna surveys. Signs of mammal activities such as droppings, diggings or burrows, scats, paw prints and tracks were photographed and identified as accurately as possible from the field signs encountered. Bat surveys were conducted by visual surveys at dusk and at night in suitable foraging locations and through the checking for suitable roost sites in suitable plants (*Livistona chinensis*) potentially used by Short-nosed Fruit Bats *Cynopterus sphinx* during the wet season. Additional bat surveys were conducted in the Additional Assessment Area(s) in January, April, June and September 2014, and from September through December 2016 to supplement the original survey findings. Observed or echolocating bats were identified to the lowest taxonomic level possible. During the bat surveys a transect was followed through suitable habitats, recording the output of the bat detector for later analysis. Any other mammals observed during nocturnal surveys through direct observation with torchlight were also recorded. Nomenclature for terrestrial mammals followed Shek (2006).

8.3.5 Bird Surveys

8.3.5.1 Bird surveys were conducted to investigate the suitability and/or utilization of the Assessment Area and its environs by bird species, especially for species restricted to certain habitats or defined home ranges that are known to occur on or near site. The bird surveys were conducted monthly between January and September 2013 covering both the dry and wet season. Additional bird surveys were conducted in the additional assessment areas in January, April, June and September 2014, and from September to December 2016 to supplement the original survey findings. To coincide with the peak hour of bird activities bird surveys were conducted within one hour from sunrise. A fixed transect was walked by an experienced bird surveyor with observation aided by 8x or 10x binoculars. All observed or heard

bird species were identified and recorded according to the habitat in which the bird was observed/ heard, and species of conservation importance and/or dependence of farmland and wetland-dependent birds including Greater Painted-snipe was enumerated and mapped according to the habitat from which they were recorded. Four nocturnal bird surveys were also conducted, in particular to focus on calling night birds e.g. Greater Painted Snipe, Savanna Nightjar and Owls during the breeding season. Notable behaviour, in particular breeding was noted and nests of any bird species of conservation importance will be mapped. Nomenclature followed the latest official Hong Kong Bird list available, whereas conservation status will follow Carey *et al.* (2001) and Fellowes *et al.* (2002).

8.3.6 Egretry and Bird Flight-line Surveys

8.3.6.1 The egretry at Tai Tong is at the southern limits of the Assessment Area. During the breeding season, foraging birds will travel through and use the Assessment Area as part of their feeding range. As such, the foraging areas and flight lines of any active roosting/breeding egrets at Tai Tong egretry were determined by way of four flight-line surveys conducted between April and July 2013. Updated surveys were conducted between May and June 2014 at both Tai Tong Egretry and the newly discovered (July 2013) egretry to update the status of both. Further verification surveys were conducted in June and July 2016 to verify the findings of the newly discovered egretry. Counts commenced within one hour of dawn, and covered the period of maximum activity of foraging egrets. During this period, an observer positioned at a suitable location close or at the Project site (as shown in **Figure 8.9**) and between the egretry and potential foraging areas, recorded birds seen leaving the egretry. The species, number of individuals, time of observation were recorded. Individual flight-lines were recorded onto a map for later analysis; if/when birds are seen to land, the location and, wherever possible, the habitat were recorded. The known egretry location at Tai Tong and the location of the newly discovered egretry (i.e. the potential egretry identified in 2013 surveys) are shown in **Figure 8.1**.

8.3.7 Herpetofauna Surveys

8.3.7.1 Day-time reptile and amphibian surveys were conducted monthly from January to September 2013, in 2014 (in January, April, June and September) and 2016 (September to December). During the surveys careful searches of appropriate microhabitats and refuges for reptiles were undertaken and all reptiles observed were identified and counted. In addition to active searching, exposed, basking or foraging reptiles observed at a greater distance from the fixed transect were also recorded. Wetland or riparian habitats were searched for herpetofauna, and all species seen and vocalising individuals heard were recorded. Night-time surveys were also undertaken monthly at target sites to check for nocturnal reptiles and vocalizing amphibians from April to July 2013; at the additional Assessment Area in April, June and September 2014; and to update the entire Assessment Area from September to December 2016. Nomenclature and status for reptiles followed Karsen *et al.* (1998) and AFCD (2006) while that of amphibians followed AFCD (2005).

8.3.8 Dragonfly and Butterfly Surveys

8.3.8.1 Nine dragonfly and butterfly surveys were conducted from January to September 2013 on a monthly basis covering the main period of activity of these groups. Supplementary surveys were conducted in January, April, June and September 2014 to cover the additional Assessment Area. Additional surveys from September through December 2016 were conducted to verify the baseline conditions in the updated Assessment Area. Attention was paid to those habitats which provide suitable habitat for these species groups. All dragonflies and butterflies observed were identified and enumerated with the aid of binoculars. Nomenclature for butterflies followed Lo and Hui (2010), and odonate nomenclature followed Tam et. al. (2011).

8.3.9 Stream Surveys

8.3.9.1 In-stream fauna were surveyed in March and August 2013 (with some additional surveys in November 2013) along watercourses in the Assessment Area of PDA. Supplementary stream surveys were conducted in March, June, July and September 2014, and the period from September to November 2016 to update the Assessment Area. Various qualitative sampling techniques, including direct observation, active searching, sweep sampling and kick sampling, were utilised depending on the local site conditions. The streams were also surveyed for other wildlife during other surveys over the 9-month period.

8.3.10 Survey Programme

8.3.10.1 The 9-month survey programme for ecological field surveys conducted in 2013 for this EIA Study is outlined in **Table 8.1a** below. Updated egretry flightline surveys were also conducted between May and June 2014 at both Tai Tong Egretry and the egretry (discovered in July 2013) to update the status of both. **Table 8.1b** shows the additional 9-month survey programme for surveys to cover additional Assessment Area due to new proposals outside PDA. Additional 4-month surveys from September to December 2016 were conducted to verify and update the baseline conditions within the updated 500m Assessment Area (**Table 8.1c**). Coverage and transect routes for faunal surveys are presented in **Figure 8.2**. Locations and coverage of stream surveys area shown in **Figure 8.3**.

Table 8.1a Programme for Ecological Field Surveys in 2013

Season	**Dry Season (2013)**			**Wet Season (2013)**					
Survey Group	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**
Habitats and Vegetation									
Mammal (day)									
Mammal (night)									
Bird (day)									
Bird (night)									
Bird (Egretry and Flight-line)									
Herpetofauna (Day)									
Herpetofauna (Night)									
Dragonfly and Butterfly									
Stream Surveys									

Table 8.1b Programme for Additional Ecological Field Surveys in 2014

Season	Dry Season (2014)			Wet Season (2014)					
Survey Group	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**
Habitats and Vegetation									
Mammal (day)									
Mammal (night)									
Bird (day)									
Bird (night)									
Bird (Egretry and Flight-line)									
Herpetofauna (Day)									
Herpetofauna (Night)									
Dragonfly and Butterfly									
Stream Surveys									

Table 8.1c Programme for Verification Ecological Field Surveys in 2016

Season	Wet Season (2016)				Transition (2016)	Dry Season (2016)	
Survey Group	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Habitats and Vegetation							
Mammal (day)							
Mammal (night)							
Bird (day)							
Bird (night)							
Bird (Egretry and Flight-line)							
Herpetofauna (Day)							
Herpetofauna (Night)							
Dragonfly and Butterfly							
Stream Surveys							

8.3.11 Nomenclature

8.3.11.1 It should be noted that for faunal groups, scientific names are not included in the main text if the species occurs within the Appendices and the common English name is used. Exceptions to this are those species with no common English name.

8.4 Ecological Baseline Conditions

8.4.1 Habitats and Vegetation

Literature Review

8.4.1.1 Habitats within the PDA are dominated by developed areas with extensive areas of low-rise village buildings and rural industrial developments. The proposed infrastructural road works/ improvement works beyond the PDA are all located along existing roads or new roads extending from or along Yuen Long Highway,

except for the small section running through a wooded area (orchard) connecting Shan Ha Tsuen and the southern section of Lam Tai West Road (**Figure 8.1**).

8.4.1.2 The Study Area boundary of the EIA and TIA Studies YLKTSSD (AEIAR-078/2004) covered the majority of Areas 2 and 3 and the eastern part of Area 1 of PDA of this current Study (**Figure 8.1**).

8.4.1.3 A small portion of the Assessment Area of the EIA Study of Hung Shui Kiu New Development Area (HSKNDA) (AEIAR-203/2016) overlaps with the western part of the Assessment Area of this Study (Figure 8.1). Relevant results of these previous EIA report were reviewed. No rare or protected plant species were recorded during the EIA Study for YLKTSSD and the EIA Study for HSKNDA within the current Assessment Area of this Study. Watercourses from the HSKNDA that fall within the Assessment Area for this Project are all largely modified and of low ecological value.

8.4.1.4 A review of the botanical survey conducted by Xing et al. (2000) identified that a rare orchid, *Spiranthes hongkongensis* was previously recorded in Yuen Long area, and presumably this species may occur within or close to the Assessment Area of the current Study. This species is protected by Forests and Countryside Ordinance (Cap. 96) and Protection of Endangered Species of Animals and Plants Ordinance (Cap. 586).

8.4.1.5 The study of fung shui woods by Webb (1996) based on examination of historical aerial photographs compiled a list of 337 fung shui woods, including fung shui woodland in Shan Ha Tsuen. The fung shui woods in Shan Ha Tsuen described in Webb (1996) covered approximately 3.5ha area on a knoll and close to village buildings (exact location is not certain). This fung shui woodland in Shan Ha Tsuen has not been included in other updated studies of fung shui woods (e.g. Chu, 1998; Yip *et. al.*, 2004).

8.4.1.6 Approximately 30ha of Tai Lam Country Park is located just outside the southern limit of Area 3 of the PDA and within the 500m Assessment Area. Tai Lam Country Park occupies a total of approximately 5,412ha of the western New Territories and is the second largest Country Park in Hong Kong. Tai Lam Country Park covers the area extending from Tsuen Wan to Tuen Mun, bounded by Route Twisk in the east; the countryside north of Tuen Mun Highway in the south; Shek Kong, the Ho Pui Catchments and Tai Tong in the north; and Tuen Mun and Lam Tei Reservoir in the west. Similar to the majority hillside area of Tai Lam Country Park, the Country Park area located to the south of PDA are dominated by shrubby grassland communities in the upland areas, and plantation woodland dominated by *Acacia confusa*, *Lophostemona confertus*, *Eucalyptus robusta* and *Pinus elliottii* on the lower slope.

8.4.1.7 Approximately 107ha of Conservation Area (CA) is located adjacent to the western boundary of PDA in the 500m Assessment Area. This zoning is intended to protect and retain the existing natural landscape, ecological or topographical features of the area for conservation, educational and research purposes and to separate sensitive environment such as Country Park from the adverse effects of development. The majority area of CA in the 500m Assessment Area is dominated by shrubby grassland, with a minor northern portion dominated by exotic plantation comprising species such as *Acacia confusa*, *Casuarina equisetifolia*, *Eucalyptus robusta*, *Lophostemon confertus* and *Pinus elliottii*.

8.4.1.8 Approximately 188ha of Green Belt (GB) is located in the 500m Assessment Area, mainly located along the western margin of PDA, to the north of Yuen Long Highway and scattered in Ha Tsuen. The planning intention of this zone is primarily for defining the limits of urban and sub-urban development areas by natural features and to contain urban sprawl as well as to provide passive recreational outlets. This zone in the Study Area is dominated by upland shrubby grassland, and lowland habitats including scattered woodlands interspersed with village plantation/ orchards and developed area.

Field Survey

8.4.1.9 A total of 21 habitat types were identified in the Assessment Area as presented in the Habitat Map (**Figure 8.4**). Representative photographs taken for each of the identified habitat types are presented in **Appendix 8.1**. A full list of flora species recorded and the relative abundance within each habitat is provided in **Appendix 8.2**. **Table 8.2** below displays the types of habitats in the Assessment Area and the Project Area. As most of the watercourses in the Assessment Area are narrow and the streambeds are of varying widths due to the topography and sloping profile, and could not be precisely mapped across the dominant habitats due to the existing overgrown vegetation, the areas for watercourses (~17,110m in length within 500m Assessment Area) have been grouped into the dominant habitat type(s)). The areas of the watercourses presented separately in **Table 8.2** represent a relatively broad-brush estimate for the total areas within the Project Area and Assessment Area respectively. Details of each of the habitat types are described in the following sub-sections.

Table 8.2 Habitats within the Assessment Area

Habitats[1]	**500m Assessment Area[5] (ha)**	**Project Area (ha)**	
		PDA (ha)	**Infrastructural Works Outside PDA[5] (ha)**
Watercourse[3,4]	3.75 (~17,110m)	1.02 (~3,600m)	0.00
Key catchments running through PDAs:			
Catchment 1	314m	-	-
Catchment 2	1114m	-	-
Catchment 3	420m	72m	-
Catchment 4	1426m	740m	-
Catchment 5	2006m	204m	-
Catchment 6	1650m	639m	-
Catchment 7	1023m	956m	-
Catchment 8	242m	242m	-
Catchment 9	1640m	755m	
Catchment 10	499m	-	-
Catchment 11	1775m	-	-
Other Watercourses	~5,000m	-	-
Channelised Watercourse	21.22	5.03	0.58
Pond	4.00	0.41	0.00
Inactive Fish Pond	0.25	0.00	0.00
Dry Agricultural Land	32.54	9.57	0.03
Wet Agricultural Land	3.72	0.76	0.00
Mixed Dry & Wet Agricultural Land	2.73	2.33	0.01
Inactive Agricultural Land	3.39	1.54	0.01

Habitats[1]	500m Assessment Area[5] (ha)	Project Area (ha)	
		PDA (ha)	Infrastructural Works Outside PDA[5] (ha)
Seasonally Wet Grassland	8.07	1.26	0.00
Reedbed	0.60	0.28	0.00
Marsh	0.09	0.00	0.00
Grassland	22.17	0.30	0.00
Grassland/Shrubland	174.42	3.36	0.00
Secondary Woodland	58.67	8.62	0.53
Plantation	46.56	0.59	0.30
Lychee Woodland	0.27	0.06	0.00
Orchard	17.56	3.73	0.00
Nursery	0.78	0.19	0.00
Village Area	242.23	16.52	0.19
Wasteland	76.50	5.25	0.99
Developed Area	826.13	163.75	13.22
Total[2]	**1541.89**	**223.54**	**15.86**

Notes:
1. Areas for each of the habitats (excluding watercourses) include the watercourses running through these habitats.
2. To avoid double-counting of the habitat areas, areas of watercourses have been excluded from the total areas.
3. Watercourse lengths are approximate measurements only and include a combination of main and channel and tributaries.
4. Due to their very nature, lengths of seasonal watercourse have not been measured.
5. The areas are calculated based on latest RODP.

Watercourses

8.4.1.10 Truly natural watercourses are limited to the hillslopes outside of the PDA boundaries, where there has been little or no human impact to the channels or banks of these streams. Given the topography, particularly in the west, many of these streams are seasonal, with water restricted to pools or slow flow in the dry season. Semi-natural shrubby grasslands are the dominant riparian vegetation of these seasonal streams in the upland (**Figure 8.4**). Lengths of seasonal watercourse are indicative only, due to their ephemeral nature.

8.4.1.11 Within the PDA, approximately 3,600m of permanently wet watercourses, (including ~750m of highly modified watercourses with no direct upstream linkages) which run through the Project Area before discharge have largely been heavily modified or influenced by human development and agricultural activities. In the true sense of the word, they are no longer natural but semi-natural. However, some do retain natural characteristics and a natural source of flow but it should be noted that they are heavily impacted/covered/piped/modified in some part of their lengths. They are of varying quality, depending on the land use that they flow through. Whilst some streams are subject to extensive pollution and dumping (e.g. in areas of industry or residential area) and are fragmented by developments and have been piped, others do retain open channels, natural substrates, and good water quality and in-channel vegetation. These watercourses have been heavily impacted historically by land use within the PDA footprint, where sections have been covered or piped under industrial units and residential developments, or have been modified and lost much of their ecological function.

8.4.1.12 The higher quality streams in turn provide habitat for the endemic crab *Somanniathelphusa zanklon* (locations of this species recorded in the Assessment Area are indicated in **Figure 8.5**), listed as Endangered by IUCN (Esser & Cumberlidge 2008a) and considered to be of Global Concern (Fellowes et al. 2002), though appears to be reasonably widespread in this Assessment Area.

8.4.1.13 For ease of assessment, these streams in the main Project Area have been split into 'catchments' (with assigned reference nos. as shown in **Figure 8.3**) as they need to be treated as a complete system.

8.4.1.14 Lowland watercourses in the eastern part of the Assessment Area generally run through village areas and lowland agriculture. Whilst subject to some point pollution, these are generally of reasonable water quality and also support populations of *S. zanklon*. These watercourses are shown in **Figure 8.3**.

8.4.1.15 Most of these lowland stream sections are very exposed and with riparian vegetation dominated by common herbs including *Alocasia macrorrhizos*, *Bidens alba*, *Commelina diffusa*, *Kyllinga polyphylla*, *Microstegium ciliatum*, *Mikania micrantha*, *Musa x paradisiaca*, *Neyraudia reynaudiana*, *Oxalis debilis* subsp. *corymbosa*, *Polygonum chinense*, and common native tree species such as *Celtis sinensis*, *Ficus hispida*, *Macaranga tanarius* var. *tomentosa* and *Microcos nervosa*.

Channelized Watercourse

8.4.1.16 Several large, trapezoidal and boxed concrete channels run through the Assessment Area. These are largely concrete lined with a dry weather flow channel at the base. These channels are designed specifically for flood capacity and have few features with real ecological potential. Part of the lower section of the main drainage channel (to the north of the Assessment Area) has been lined with grasscrete and there are areas of riparian vegetation which have some limited ecological value. Common herbaceous vegetation were recorded along these channelized watercourses, including *Alocasia macrorrhizos*, *Alternanthera philoxeroides*, *Alternanthera sessils*, *Bidens alba*, *Brachiaria mutica*, *Colocasia esculenta*, *Cyperus imbricatusm*, *Kyllinga polyphylla*, and *Neyraudia reynaudiana*. *Ficus hispida*, *Ficus microcarpa* and *Ficus subpisocarpa* are the dominant tree species that had established along the narrow embankment of some channelized watercourses. Water quality appears to be of poor quality, a result of pollution by domestic and industrial effluent directly discharged to these channels.

8.4.1.17 Part of the Yuen Long Bypass Floodway occurs at the eastern end of the Assessment Area. Grasscrete cells in the channel bed allow the growth of wetland macrophytes and provide a series of microhabitats for wetland fauna. A dry weather flow channel maintains flow throughout the year.

8.4.1.18 Smaller concrete channels and ditches occur through the Assessment Area and ultimately link up with these larger channels. Generally these channels have little or no ecological value due to a lack of microhabitats and low water levels.

Pond and Inactive Fish Pond

8.4.1.19 A total of 34 ponds were identified within or close to the boundary of the Assessment Area as listed in **Table 8.3** below and their locations are shown in **Figure 8.6**, of these a total of five shallow farm ponds (Ponds #5 - #9) and a small, and a very shallow pond (Pond #17) were identified in Area 2 within the PDA. All the ponds (Ponds #5 – 9) in Area 1 of the PDA are shallow farm ponds for

irrigation, with the pond bund vegetation dominated by common herbaceous vegetation of cultivated areas and marshes, including *Alternanthera philoxeroides*, *Bidens alba*, *Commelina diffusa*, *Musa x paradisiaca*, *Panicum repens*, *Pouzolzia zeylanica*, *Solanum americanum* and *Spermacoce stricta*, and a few planted fruit trees such as *Eriobotrya japonica*, *Mangifera indica*, and the weedy shrub species *Lantana camara*. Pond #17 was found containing very shallow water (not suitable for fisheries/ irrigation purposes) and partially fenced. The pond appeared to be actively managed but the use of this pond was uncertain. There are some other farm ponds (Ponds #1, #10, #12 and #13) located in the Assessment Area but outside of the Project Area. These farm ponds are either shallow ponds for retention of rainwater for irrigation or inactive wet-fields for wet agriculture.

8.4.1.20 There are a total of two small fish ponds/ managed ponds (Ponds #3 and #4) identified within the 500m Assessment area, located behind the dense woodland behind Shan Ha Tsuen and outside of the PDA. These two ponds were being operated by the local villages for cultivation of Nile Tilipia *Oreochromis niloticus*. The use of the fish stock is uncertain.

8.4.1.21 There are two Fung Shui Ponds/ Landscaped Ponds located in Fui Sha Wai (Pond #2) and Ying Lung Wai (Pond #21) respectively, both are outside of the proposed PDA. The ponds were surrounded with metal fences. Common herbaceous vegetation and a few fruit trees including *Morus alba* and *Mangifera indica* were recorded along the pond bunds of Pond #2. The pond bund area of Pond #21 was vegetated with ornamental species including the palm tree *Archontophoenix alexandrae*.

8.4.1.22 There is one inactive fish pond located near Fui Sha Wai (Pond #22). Common herbaceous species such as *Bidens alba*, *Commelina diffusa*, *Mikania micrantha*, *Panicum maximum* and *Phragmites australis*, fruit trees *Artocarpus heterophyllus*, and *Psidium guajava*, and a few common native tree species such as *Celtis sinensis*, *Listea glutinosa*, were the most dominant vegetation species along the pond bund.

8.4.1.23 There are six stormwater ponds including Pond #14 at Kiu Tau Wai Stormwater Pumping Station, Pond #16 at Sheung Cheung Wai Stormwater Pumping Station, Ponds #48 and #49 at Sik Kong Wai Stormwater Pumping Station, Pond #50 in Sik Kong Wai Tsuen, and Pond #51 next to Lo Uk Tsuen, all are located outside PDA or any associated works areas. These ponds are entirely man-made and largely concreted ponds.

Table 8.3 Ponds within and in the vicinity of the Assessment Area

Pond no.	Area (ha)	Habitat	Within PDA (Yes/ No)	Within 500m Assessment Area (Yes/ No)
1	0.18	Pond (appeared to be a former farm pond continuously flooded due to close proximity to watercourses)	No	Yes
2	0.16	Fung Shui Pond/ Landscaped Pond	No	Yes
3	0.10	Pond (small fish pond)	No	Yes
4	0.02	Pond (small fish pond)	No	Yes
5	0.03	Pond (active farm pond for irrigation)	Yes	Yes
6	0.03	Pond (active farm pond for irrigation)	Yes	Yes

Pond no.	Area (ha)	Habitat	Within PDA (Yes/ No)	Within 500m Assessment Area (Yes/ No)
7	0.18	Pond (active farm pond for irrigation)	Yes	Yes
8	0.10	Pond (active farm pond for irrigation)	Yes	Yes
9	0.03	Pond (active farm pond for irrigation)	Yes	Yes
10	0.12	Pond (appeared to be active farm pond for irrigation)	No	Yes
11	0.03	Pond (small abandoned pond within wooded area)	No	Yes
12	0.12	Pond (active farm pond for irrigation)	No	Yes
13	0.21	Pond (inactive farm pond)	No	Yes
14	0.43	Pond (Stormwater Pond in Kiu Tau Wai Stormwater Pumping Station)	No	Yes (partially)
15	0.60	Pond (Marshy Pond)	No	No
16	0.63	Pond (Stormwater Pond in Sheung Cheung Wai Stormwater Pumping Station)	No	No
17	0.04	Small, managed pond (shallow pond); nature and commercial use uncertain	Yes	Yes
18	0.05	Pond (overgrown)	No	Yes
19	0.17	Pond (overgrown)	No	Yes (partially)
20	0.31	Recently filled pond	No	No
21	0.09	Fung Shui Pond/ Landscaped Pond	No	Yes
22	0.25	Inactive Fish Pond	No	Yes
40	0.13	Landscaped Pond	No	Yes
41	0.31	Overgrown with vegetation	No	Yes
42	0.21	Active farm pond for irrigation	No	Yes
43	0.07	Mitigation pond	No	Yes
44	0.08	Mitigation pond	No	Yes
45	0.09	Overgrown with vegetation	No	Yes
46	0.18	Mitigation pond	No	Yes
47	0.13	Mitigation pond	No	Yes
48	0.34	Stormwater pond in Sik Kong Wai Stormwater Pumping Station	No	Yes
49	0.31	Stormwater pond in Sik Kong Wai Stormwater Pumping Station	No	Yes
50	0.18	Stormwater pond in Sik Kong Tsuen	No	Yes
51	0.32	Stormwater pond next to Lo Uk Tsuen	No	Yes

Dry Agricultural Land

8.4.1.24 Approximately 32.54ha of active and dry agricultural land are present in the Assessment Area, of which approximately 9.57ha is within the PDA and approximately 0.03ha within other proposed works area outside of PDA. These

active and dry agricultural lands are mainly located within the village areas on the east of Kiu Hing Road (including those near Ma Tin Pok, Fraser Village and Tin Liu Tsuen, west of Shui Tsiu San Tsuen Road, and in Yeung Ka Tsuen), north of Tong Yan San Tsuen Interchange, near Fui Shan Wai, southeast of Sha Tseng Tsuen, and southwest of Shan Ha Tsuen. Among these the agricultural land adjoining Shan Ha Tsuen is the largest and continuous dry agricultural land in the PDA and the Assessment Area. Dominant crop species of these dry agricultural lands include *Allium fistulosum*, *Aloe vera*, *Benincasa hispida*, *Brassica juncea*, *Ipomoea batatas*, *Lactuca sativa*, *Lycopersicon esculentum*, *Solanum melongena* and *Solanum tuberosum*. Fruit trees such as *Artocarpus heterophyllus*, *Carica papaya*, *Clausena lansium*, *Dimocarpus longan*, *Litichi chinensis* and *Mangifera indica* are commonly planted in this habitat type.

8.4.1.25 The field bunds are dominated by common and pioneer herbaceous vegetation with varying coverage and density depending on the frequency of site management. Dominant species include *Ageratum conyzoides*, *Alocasia macrorrhizos*, *Amaranthus viridis*, *Bidens alba*, *Chloris barbata*, *Cuscuta chinensis*, *Cyperus iria*, *Eleusine indica*, *Ipomoea cairica*, *Ipomoea triloba*, *Kyllinga brevifolia*, and *Solanum americanum*.

8.4.1.26 A patch of approximately 6m x 3m of the cultivated herb *Houttuynia cordata* was recorded in the active agricultural land in Fui Sha Wai in 2013 (as shown in **Figure 8.5**). The wild population of this herb species is considered as "very rare" in its local distribution by Fellowes *et al.* (2002). However, this species is commonly cultivated in village areas in Hong Kong. Two individuals of *Ceratopteris thalictroides* were recorded in the dry agricultural land located north of Shek Po Tsuen and outside of the PDA or any other associated works areas. This species is regarded as "Rare" in Xing et al. (2000). Status in China as "Vulnerable", under State protection (Category II) in China (AFCD 2003).

Wet Agricultural Land

8.4.1.27 Wet agriculture (~3.72ha in the Assessment Area including ~0.76ha in PDA) is scattered through the large agricultural land to the southwest of Shan Ha Tsuen, to the southeast of Sha Tseng Tsuen, to the southwest of Fui Shan Wai, and near Yau Tin West Road. *Ipomoea aquatica* is the main crop species recorded in these wet agricultural lands. The bunds separating the wet-fields are subject to high level of human disturbance (e.g. weeding and trampling) with very sparse vegetation dominated by common herbaceous species such as *Hedyotis diffusa*, *Oxalis corniculata* and *Oxalis debilis* subsp. *corymbosa*.

Mixed Dry & Wet Agricultural Land

8.4.1.28 Predominantly mixed dry and wet agricultural lands (~2.73ha in the Assessment Area) are mainly located to the southeast of Sha Tseng Tsuen (~2.33ha in Area 1 of the PDA) and on Long Ho Road opposite to Yeung Uk Tsuen. This habitat comprises a mosaic of wet and dry fields and the use of fields can be changed rapidly. Dry agriculture is often temporary and these fields may be flooded in other seasons according to the preferences of the farmers. The main crop species include *Aloe vera*, *Ipomoea aquatica* and *Zea mays*. Due to frequent weed control and site management vegetation along the field bunds is rather sparse and dominated by common pioneer herbaceous species such as *Alternanthera sessils*, *Amaranthus viridis*, *Ipomoea cairica*, *Plantago major*, *Polygonum chinense*, *Pueraria lobata* and *Solanum americanum*.

Inactive Agricultural Land

8.4.1.29 Approximately 3.39ha of inactive agricultural lands were identified in the Assessment Area, among which approximately 1.54ha is present in the PDA, and approximately 0.01ha is present in the proposed works areas outside of PDA. These fallow fields were commonly found near existing villages outside PDA. Within the PDA, inactive agricultural land parcels were scattered within the large agricultural land adjoining the southwest of Shan Ha Tsuen (within the PDA) respectively. These inactive agricultural fields are dominated by common pioneer herbs such as *Ageratum conyzoides*, *Amaranthus viridis*, *Bidens alba*, *Conyza canadensis*, *Cuscuta chinensis*, *Ipomoea cairica*, *Ipomoea triloba*, *Kyllinga polyphylla*, *Mikania micrantha*, *Polygonum chinense*, *Rumex trisetifer* and *Solanum americanum*, remnants of abandoned crops of *Lycium chinense*, and weedy trees and shrubs such as *Leucaena leucocephala* and *Lantana camara*. When these follow fields are left abandoned over time they will be increasingly overgrown with weedy vegetation.

Seasonally Wet Grassland

8.4.1.30 Seasonally wet grasslands (~8.07ha) in the Assessment Area are concentrated in Yeung Ka Tsuen adjoining existing active agricultural lands and/or in close proximity to existing watercourses, and within the mixed wet and dry agricultural land in Sha Tseng Tsuen. A few relatively larger seasonally wet grasslands are present along Shui Tsui San Tsuen Road and Lam Tai West Road respectively, where they are located close to existing channels. There are a few small seasonally wet grasslands recorded in the village area in Shan Ha Tsuen and some within low-rise developments. This habitat has been derived from poorly drained fallow fields with rainwaters retained or flooded by overflows from nearby watercourses during the wet season periods. It can support growth of wetland plant species including dominant species such as *Cyclosorus interruptus*, *Commelina diffusa*, *Hedychium coronarium*, *Leersia hexandra*, *Ludwigia octovalvis*, *Ludwigia perennis*, *Microstegium ciliatum*, *Polygonum barbatum*, *Polygonum chinense*, *Polygonum perfoliatum*, *Phragmites australis*, and *Pycreus polystachyos*.

Reedbeds

8.4.1.31 A total of approximately 0.60ha of reedbeds are scattered within the Assessment Area. This habitat is derived from colonization of *Phragmites australis* into the poorly drained, low-lying areas or fallow fields where rainwater has been retained or being flooded by overflows from nearby watercourses during wet seasons.

Marsh

8.4.1.32 A small area of freshwater marsh (~0.09ha) was recorded in a low-lying area on Siu Sheung Road near Pok Oi Hospital and outside of the Project Area. This marsh habitat was overgrown with reed *Phragmites australis* and some common wetland grass such as *Brachiaria mutica*, common native grass species such as *Neyraudia reynaudiana*, weedy tree species *Leucaena leucocephala* and common native tree species *Macaranga tanarius* var. *tomentosa*. This habitat is derived from a poorly-drained, low-lying fallow field where rainwater had been retained and the wet-field was later colonized by common wetland plants and weedy species.

Grassland

8.4.1.33 Approximately 22.17ha grasslands are present in the Assessment Area, including 0.3ha in the PDA. All of them are either lowland grasslands scattered through

village areas or the grasslands on lower slopes adjoining the hillside uplands. These lowland grasslands are dominated by common grassy vegetation including *Neyraudia reynaudiana*, *Panicum maximum*, *Digitaria ciliaris*, *Eleusine indica*, *Imperata cylindrica* var. *major*, and *Kyllinga polyphylla*. This habitat is chronically disturbed lowland habitat and is considered of low floristic diversity and low ecological value.

Grassland/Shrubland

8.4.1.34 Approximately 174.42ha grassland/ shrubland habitat was mainly found in the continuous upland adjoining the western boundary of the PDA (including ~3.36ha within the PDA). There is a small patch of grassland/ shrubland on the small knoll within the developed area close to the southern end of Lam Tai West Road. The woody climber species *Gnetum luofuense* was recorded in the hillside shrubby grassland to the west of PDA. This species is regarded as Very Common by Xing *et al*. (2000), classified as "Near Threatened" in IUCN Red List but locally very common in Hong Kong.

8.4.1.35 This habitat comprises native shrub species including *Baeckea frutescens*, *Glochidion eriocarpum*, *Helicteres angustifolia*, *Melastoma dodecandrum*, *Melastoma sanguineum*, *Rhaphiolepis indica*, and *Rhodomyrtus tomentosa*, and native grass species such as *Arundinella nepalensis*, *Neyraudia reynaudiana*, *Paspalum conjugatum*, *Dicranopteris pedata*, *Imperata cylindrica* var. *major*, *Ischaemum barbatum*, *Miscanthus floridulus* and *Miscanthus sinensis*. This habitat type is considered as semi-natural habitats intermittently disturbed and maintained by anthropogenic hill fire. Although this habitat is considered of low to moderate floristic diversity, it is considered of high naturalness compared to the heavily disturbed lowland grasslands, and will progressively turn into secondary woodland with the control of hill fire.

Secondary Woodland

8.4.1.36 Approximately 58.67ha of secondary woodlands are present in the Assessment Area, including ~8.62ha in the PDA and ~0.53ha within the footprint of the proposed infrastructural works area outside of the PDA. These secondary woodlands include ~19.09ha established on the continuous upland adjoining the western boundary of PDA, ~13.23ha adjoining south of Shan Ha Tsuen, ~4.28ha in or near Tong Yan San Tsuen, ~3.30ha near Lam Hau Tsuen and Lam Hau Wai Tsuen, ~1.29ha under the pylon near Yeung Ka Tsuen, and ~17.48ha of small woodlands scattered through the village or developed areas. Distribution of these woodlands within the Assessment Area is shown in **Figure 8.7.** Details of these woodlands are described separately as follows:

8.4.1.37 Woodland behind Shan Ha Tsuen (~13.23ha): This dense secondary woodland comprises a moderate diversity of mature trees of typical native woodland species including *Acronychia pedunculata*, *Aporusa dioica*, *Bischofia javanica*, *Breynia fruticosa*, *Bridelia tomentosa*, *Celtis sinensis*, *Cinnamomum burmannii*, *Cinnamomum camphora*, *Ficus variegata*, *Litsea glutinosa*, *Litsea monopetela*, *Microcos nervosa*, *Rhus chinensis*, *Rhus succedanea*, *Sapium sebiferum* and *Schefflera heptaphylla*, with dense understory dominated by native shrubs such as *Mussaenda pubescens*, *Psychotria asiatica*, and *Rubus reflexus*, and woody climber *Desmos chinensis*. The floristic composition suggests that the woodland is relatively mature and diverse, with moderate to high ecological value due to its maturity and large size.

8.4.1.38 A small number of seedlings of the protected tree species *Aquilaria sinensis* were recorded in this woodland area (**Figure 8.5**). This species is used medicinally as well as for ornamental purposes. Potential threats include destruction of habitats and over-exploitation, but the species is common in lowland areas in Hong Kong (AFCD 2003). This species is listed as Near Threatened (NT) and is under State protection (Category II) in China (AFCD 2003) and is classified as Vulnerable (VU) on the IUCN Red List of Threatened Species (IUCN, 2017), and currently protected by Cap. 586. The woody climber species *Gnetum luofuense* was recorded in this woodland. This species is regarded as Very Common by Xing et al. (2000), classified as "Near Threatened" in IUCN Red List but locally very common in Hong Kong.

8.4.1.39 Hillside woodland (~19.09ha): This habitat type includes woodland patches located on lower slopes abutting the developed area on the western boundary of the PDA and adjoining the upper plantation and shrubby grassland habitats. These woodlands comprise some generally younger native secondary woodland naturally regenerated in the absence of hill fire and/or intentional clearance, and some more mature lowlands that have remained undisturbed from previous developments. These native woodlands are generally of moderate floristic diversity and dominated by native tree species such as *Cratoxylum cochinchinense*, *Ficus variegata*, *Litsea cubeba*, *Litsea glutinosa*, *Macaranga tanarius* var. *tomentosa*, *Mallotus paniculatus*, *Rhus chinensis*, *Rhus succedanea*, *Sapium discolor*, *Zanthoxylum nitidum*, and moderately dense understory dominated by *Psychotria asiatica* and *Sarcandra glabra*. Exotic plantation species such as *Acacia confusa* are also scattered through some of these woodland areas. This habitat type is considered of moderate ecological value due to its naturalness (semi-natural upland habitats) and remoteness (away from intentional human disturbance/ clearance).

8.4.1.40 Woodlands near Tong Yan San Tsuen (~4.28ha): These woodlands comprise abandoned orchards, remnants of native woodlands and planted ornamental/ amenity trees. Dominant tree species include the native trees *Celtis sinensis*, *Ficus hispida*, *Ficus microcarpa*, *Ficus virens*, *Ligustrum sinense*, *Litsea glutinosa*, *Litsea monopetala*, *Macaranga tanarius* var. *tomentosa*, *Microcos nervosa*, planted exotic trees such as *Bauhinia variegata*, and a number of common fruit trees including *Clausena lansium*, *Dimocarpus longan* and *Litchi chinensis*. For some of these woodlands, the understory has been occupied with small and scattered village houses preventing natural regeneration of the understory of these woodlands. These woodlands support generally low diversity of common tree species; however, they are considered of some potential for ecological rehabilitation through removal of the village houses in the understory and enhancement planting.

8.4.1.41 Woodlands near Lam Hau Tsuen and Lam Hau Wai Tsuen (~3.30ha): These are woodland remnants in the villages of Lam Hau Tsuen and Lam Hau Wai Tsuen. The southern woodland at Lam Hau Tsuen comprises a low to moderate diversity of common native tree species such as *Bischofia javanica*, *Bridelia tomentosa*, *Cinnamomum camphora*, *Ficus hirta*, *Ficus variegata*, *Listea glutinosa*, *Macaranga tanarius* var. *tomentosa*, and *Sterculia lanceolata*, and some common fruit trees including *Carica papaya*, *Dimocarpus longan* and *Litchi chinensis*. The understory of the woodland is relative sparse and lack of native shrub species as it has been used as burial grounds by the local villagers. The northern woodland at Lam Hau Wai Tsuen comprises predominantly abandoned orchards of common fruit trees with much fewer native tree species. *Celtis sinensis* and *Macaranga tanarius* var. *tomentosa* are the only dominant native trees recorded in this

woodland. There is a small abandoned and fenced orchard of *Clausena lansium* located in the centre of this woodland. Outside of the fenced orchard much of the woodland understory has been used as burial grounds. The overall ecological value of these woodlands is considered low to moderate due to the presence of burial grounds which has prevented natural regeneration of the understory vegetation and reduced potential for ecological enhancement.

8.4.1.42 Woodland under pylon near Yeung Ka Tsuen (~1.29ha): There is a small wooded area under an existing pylon near Yeung Ka Tsuen. This secondary woodland is comprised of relatively low diversity of common native trees (e.g. *Aporusa dioica*, *Bridelia tomentosa*, *Ficus hispida*, *Litsea glutinosa*, *Macaranga tanarius* var. *tomentosa*, *Mallotus paniculatus*) interspersed with exotic plantation species such as *Acacia confusa* and *Casuarina equisetifolia* and invasive tree *Leucaena leucocephala*. The woodland is considered of low to moderate ecological value.

8.4.1.43 Other woodlands scattered in developed/ village areas (~17.48ha): These woodlands mainly comprise abandoned orchards and/or village plantations with few common native tree species dominated by *Bridelia tomentosa*, *Ficus hispida, Macaranga tanarius* var. *tomentosa* and *Mallotus paniculatus*. These isolated woodlands are considered of low to moderate ecological value due to high degree of habitat fragmentation by existing developments and high level of human disturbance.

Plantation

8.4.1.44 Approximately 46.56ha of plantation woodlands are present in the Assessment Area, including ~0.59ha in the PDA, and approximately 0.30ha outside of PDA (for the proposed Reclaimed Water Service Reservoir. These plantation woodlands are mainly found in the continuous hillside uplands to the west of the PDA, and comprise few exotic plantation species including *Acacia confusa*, *Casuarina equisetifolia*, *Eucalyptus robusta*, *Lophostemon confertus* and *Pinus elliottii*. The understory of these plantations is relatively sparse and lack of natural regeneration. These plantation woodlands have been established mainly for reforestation purposes by the government. Due to lack of native vegetation and natural regeneration the overall floristic diversity and the ecological value of plantation woodlands are low.

Lychee Woodland

8.4.1.45 There are two considerable sizes of Lychee orchards (~0.27ha in total) located at the junction between the mature secondary woodland and large agricultural lands in Shan Han Tsuen, including approximately 0.06ha falling in the PDA. These Lychee woodlands appeared to be abandoned orchards and comprise mature/ over-mature trees of *Litchi chinensis* and with very sparse understory and lack of native plant species. The ecological value of these mature abandoned Lychee woodlands is considered low.

Orchard

8.4.1.46 Orchards (~17.56ha of the Assessment Area including ~3.73ha in PDA) of varying sizes comprising common fruit trees dominated by *Clausena lansium*, *Dimocarpus longan*, *Litchi chinensis* and *Mangifera indica*. Rapidly growing, pioneer tree species such as the native tree of *Macaranga tanaria* var. *tomentosa* and invasive exotic tree of *Leucaena leucocephala* were found interspersed within some areas of over-mature/ abandoned orchards which have been subject to recent disturbance/

site clearance work. Two mature tree specimens (~12m height) of *Artocarpus hypargyreus* were recorded near the orchard area close to the western boundary of Area 1 (Figure 8.5). *Artocarpus hypargyreus* is regarded as Vulnerable in China Red Data Book (Pang et al. 2011) and IUCN (IUCN 2017), and "Near Threatened" by AFCD (2003). However, it is commonly found in lowland forest and not locally protected in Hong Kong. Orchards in the Assessment Area are generally intermittently disturbed, man-made habitats which offer minimal opportunities to wildlife. The overall ecological value of this habitat is considered low.

Nursery

8.4.1.47 Approximately ~0.78ha of nursery habitats are located in the Assessment Area. Approximately 0.19ha of the nursery located on Long Hon Road falls in the PDA. This nursery comprises a wide diversity of common ornamental species and is considered of having low ecological value due to its man-made nature and the existing high level of disturbance from active management.

Village Area

8.4.1.48 Approximately 242.23ha of the Assessment Area is covered by village areas, including ~16.52ha in the PDA, and ~0.19ha within the proposed footprint for associated infrastructural works outside of the PDA. This habitat type is mainly located east of Kiu Hing Road adjoining the eastern boundary of the PDA, and between Area 1 and Area 2 of PDA. This habitat comprises diverse plant diversity of planted fruit trees including *Citrus reticulata*, *Clausena lansium*, *Dimocarpus longan*, *Eriobotrya japonica*, *Litchi chinensis* and a wide range of ornamental species, with open fields dominated by common pioneer herbs such as *Bidens alba*, *Polygonum chinense*, *Solanum americanum*, *Youngia japonica*. The floristic diversity of this habitat is moderate however the floristic value is considered low due to lack of native vegetation communities and high degree of fragmentation and disturbance.

Wasteland

8.4.1.49 Wasteland (~76.50ha of the Assessment Area, including approximately 5.25ha in the PDA and ~0.99ha within footprint of proposed infrastructural works outside PDA) is mainly scattered within the village areas. This habitat is mainly derived from fallow cultivation and woodland clearance and has become overgrown with weedy vegetation after a long period of abandonment. This habitat comprising mainly pioneer herbaceous species such as *Bidens alba*, *Cynodon dactylon*, *Ipomoea cairica*, *Mikania micrantha*, *Kyllinga polyphylla*, and *Pueraria lobata* and is lack of tree stands. This overgrown habitat is considered of low ecological value due to high level of human disturbance and low floristic diversity and value. A mature tree specimen (~11m height) of *Artocarpus hypargyreus* of conservation importance was recorded in the wasteland area near the western boundary of Area 1 (Figure 8.5).

Developed Area

8.4.1.50 The remaining ~826.13ha of the Assessment Area (including ~163.75ha of the PDA and ~13.22ha within the proposed footprint of infrastructural works outside PDA) belongs to developed area dominated by low-rise rural residential developments, and rural industrial developments, and urban areas in Yuen Long. This habitat comprises low diversity of planted exotic trees such as *Bombax ceiba*,

Bauhinia variegata and wasteland vegetation type of low floristic diversity and ecological value.

8.4.1.51 A mature and planted specimen of the rare tree species *Ailanthus fordii* was recorded on Tong Yan San Tsuen Road and within existing rural industrial development and within PDA. The wild population of this tree species is considered "rare" in Fellowes *et al*. (2002) and protected by Cap. 96A.

8.4.1.52 Approximately 10 young saplings of the protected tree species *Aquilaria sinensis* (protected under Cap. 586) were found planted in ground in a small private nursery near the southern end of Lam Tei West Road (Figure 8.5f) in the survey conducted in December 2016. The location of this recently established nursery is within the proposed road connecting Area 1 and Areas 2/3 of PDA.

8.4.1.53 In addition to the six rare or protected plant species recorded in the Assessment Area as mentioned above, there were another seven plant species listed in the IUCN Red List (2017), including the exotic shrub species *Aglaia odorata* ("Lower Risk/ Near Threatened", the exotic tree species *Araucaria heterophylla* ("Vulnerable"), the fruit tree *Dimocarpus longan* ("Lower Risk/ Near Threatened"), the exotic tree *Platycladus orientalis* ("Near Threatened"), the exotic tree *Podocarpus macrophyllus* var. *maki* ("Near Threatened"), the exotic tree *Khaya senegalensis* ("Vulnerable"), and the native herb species *Sonchus arvensis* ("Near Threatened"). All the six shrub/ tree species are widely cultivated locally. The native herb *S. arvensis* is a very common species commonly found in wasteland or abandoned field. None of these species are considered to be of conservation concern in Hong Kong.

8.4.2 Mammals

Literature Review

8.4.2.1 There is little information for mammals in the Assessment Area. Japanese Pipistrelle was recorded in fishponds and cultivated areas in the study area of EIA and TIA for YLKTSSD (AEIAR-078/2004). This species is protected under Wild Animals Protection Ordinance (Cap. 170). Burrows of rodents (*Rattus* spp.) were recorded on grassy hillsides in the study area of EIA and TIA of YLKTSSD. These rodent species are commonly found near residential areas (Shek 2006). Within the Study area for the Hung Shui Kiu New Development Area (EIA 248/2016), seven mammal species were recorded with five of conservation importance: Short-nosed Fruit Bat *Cynopterus sphinx*, Japanese Pipistrelle *Pipistrellus abramus*, Chinese Noctule *Nyctalus plancyi*, *Myotis* spp., and Small Indian Civet *Viverricula indica.*

Field Surveys

8.4.2.2 Nine species of mammals (including six species of bats) have been recorded from within the 500m Assessment Area, five of which are listed as being of conservation concern (**Appendix 8.3**).

8.4.2.3 Many of the habitats within the PDA have limited potential for mammal species. Numbers of mammals were greatest in the area of mixed agriculture, secondary woodland and watercourses in the west of the PDA.

8.4.2.4 Of the bats, Leschenault's Rousette, Short-nosed Fruit Bat and Himalayan Leaf-nosed Bat were all found dead in illegal mist nets located within dry agriculture in the west of the PDA. Leschenault's Rousette and Himalayan Leaf-nosed Bat are

both listed as being of Local Concern by Fellowes *et al.* (2002) on account of their breeding/roosting populations.

8.4.2.5 By far the most abundant bat was apparently Japanese Pipistrelle[1], recorded across all habitats but in greatest numbers over agriculture and channelised watercourses in the PDA. This species is common and widespread in Hong Kong (Shek 2006) and is not considered to be of conservation concern (though it should be noted all bats are protected under Cap.170). Other unidentified bat species were recorded foraging along channelized watercourses in the east of the Assessment Area.

8.4.2.6 Chinese Noctule was regularly recorded foraging above agricultural fields within the PDA. A single Least Horseshoe was recorded from riparian woodland mosaic outside of the PDA. Both species are listed as being of Potential Regional Concern (Fellowes *et al.* 2002) and a wide distribution in Hong Kong (Shek 2006).

8.4.2.7 Whilst no bat roosts were recorded during surveys, Japanese Pipistrelle and Chinese Noctule will readily utilise buildings and holes in trees to form roost sites. Other insectivorous bats may also roost in Bamboos. Short-nosed Fruit Bat roost in Chinese Fan-palms which occur throughout the Project Area.

8.4.2.8 Red Muntjac was recorded in two locations in close proximity; footprints were recorded from a stream to the west of the PDA and a direct observation made from the nearby secondary woodland. This small deer species is listed as being of Potential Regional Concern (Fellowes *et al.* 2002) and has a wide distribution in Hong Kong (Shek 2006).

8.4.3 Avifauna

Literature Review

Tai Tong Egretry

8.4.3.1 The egretry at Tai Tong was first documented in an egretry survey report by Wong & Woo (2003) and counts have been conducted there during each breeding season since 2003. This egretry was considered to be abandoned in 2014 (Anon 2014) and was not in use in 2015 (Anon. 2015). A summary of the numbers and species composition of egrets and herons at the Tai Tong egretry during past breeding seasons is shown in **Table 8.4a**.

Table 8.4a Number of Nests Recorded at Tai Tong Egretry between 2003 and 2015

Year/Species	Little Egret *Egretta garzetta*	Chinese Pond Heron *Ardeola bacchus*	Eastern Cattle Egret *Bubulcus coromandus*	Total No. of Nests Recorded
2003	-	8	2	10
2004	-	16	15	31

[1] The known call structure and peak frequency of Hong Kong bat species is currently limited to the Horseshoe Bats (*Rhinolophus* spp.) and Leaf-nosed Bats (*Hipposideros* sp.) as per Shek and Lau (2006); there is not yet a reference for the call structure of other Hong Kong bats so identification of these species is not always possible. Most species in this study could not be identified to species level. Abundance and assigning species to particular habitats is also difficult given the mobility and sometimes numerous individuals present.

Year/Species	Little Egret *Egretta garzetta*	Chinese Pond Heron *Ardeola bacchus*	Eastern Cattle Egret *Bubulcus coromandus*	Total No. of Nests Recorded
2005	1	17	10	28
2006	1	10	15	26
2007	-	24	15	39
2008	-	10	9	19
2009	-	5	11	16
2010	-	7	12	19
2011	-	4	5	9
2012	-	4	-	4
2013	-	2	-	2
2014	-	-	-	0
2015	-	-	-	0

Source: (Wong & Woo 2003, Anon 2004, Anon 2005, Anon 2006, Anon 2007, Anon 2008, Anon 2009, Anon 2010, Anon 2011, Anon 2012, Anon 2013, Anon 2014, Anon 2015)

8.4.3.2 For the newly discovered Tai Tong (Pak Sha Tsuen) egretry, the numbers and species composition of egrets and herons are summarised in **Table 8.4b.**

Table 8.4b Number of Nests Recorded at Tai Tong (Pak Sha Tsuen) Egretry between 2014 and 2016

Year/Species	Little Egret *Egretta garzetta*	Chinese Pond Heron *Ardeola bacchus*	Eastern Cattle Egret *Bubulcus coromandus*	Total No. of Nests Recorded
2014	3	12	2	17
2015	7	16	11	34
2016	12	13	5	30

Source: (Anon 2014, Anon 2015, Anon 2016)

Other Bird Species

8.4.3.3 Thirty-three bird species were recorded in the EIA and TIA studies for YLKTSSD (AEIAR-078/2004) most being common and widespread in Hong Kong. Species of conservation importance included common ardeid species and White-throated Kingfisher.

8.4.3.4 Within the Study area for the Hung Shui Kiu New Development Area (EIA 248/2016), a total of 92 avifauna species were recorded with 23 species being of conservation importance, of which 16 are wetland-dependent. Bird species of conservation importance recorded within 500m of the current Assessment Area included Chinese Pond Heron, Little Egret, Greater Coucal and Red-billed Starlings. These species are generally considered to be common in a Hong Kong context.

Field Surveys

8.4.3.5 A total 94 bird species were recorded from this Study, with 24 species listed as being of conservation concern (**Appendix 8.4**). Most birds species are common and widespread in Hong Kong and associated with habitats with high levels of disturbance

8.4.3.6 Within the PDA, 14 species of conservation concern were recorded in low numbers during this Study.

8.4.3.7 A single Yellow Bittern was recorded from mixed dry and wet agricultural land within PDA (Near Tong Yan San Tsuen) and is listed as being of Local Concern on account of its breeding sites (Fellowes et al. 2002). This uncommon summer visitor and passage migrant which primarily occur in the Deep Bay area where it shows a preference for reedmarsh and mangrove habitats (Carey *et al.* 2001); this habitat is largely absent from the Assessment Area.

8.4.3.8 Black-crowned Night Heron is listed as being of Local Concern (Fellowes *et al.* 2002) and were recorded in low numbers near watercourses within the PDA.

8.4.3.9 Eastern Cattle Egret is listed as being of Local Concern on account of its breeding population (Fellowes *et al.* 2002). Recorded from various types of agriculture and from Secondary Woodland (where they were associating with domestic cattle), this species is an uncommon to common in Hong Kong (Carey *et al.* 2001). Historically this species bred at Tai Tong egretry, but has not been recorded there since 2011. It has been recorded at Tai Tong (Pak Sha Tsuen) Egretry since 2014. This species was only recorded breeding from five egretries in Hong Kong in 2016 (Anon. 2016).

8.4.3.10 Chinese Pond Heron is listed as Potential Regional Concern (Fellowes *et al.* 2002) but is abundant in Hong Kong (Carey *et al.* 2001); recorded in low numbers in various habitats but generally in the watercourses of the eastern extent of the Assessment Area. It has been recorded breeding at Tai Tong egretry between 2004 and 2013 and at Tai Tong (Pak Sha Tsuen) Egretry since 2014.

8.4.3.11 A single Grey Heron, listed as Potential Regional Concern (Fellowes *et al.* 2002), was recorded in the channelized watercourse in the eastern extent of the Assessment Area. This habitat is typical for this abundant winter visitor (Carey *et al.* 2002).

8.4.3.12 A single Great Egret, listed as Potential Regional Concern (Fellowes *et al.* 2002), was recorded in low numbers in the channelized watercourses in the eastern extent of the Assessment Area. This habitat is typical for this common to abundant waterbird (Carey *et al.* 2002).

8.4.3.13 Little Egret is listed as Potential Regional Concern (Fellowes *et al.* 2002) but is abundant in Hong Kong (Carey *et al.* 2001). Recorded from a wide range of habitats in the PDA and Assessment Area, it was formerly a breeding species at Tai Tong egretry in low numbers,- has not been recorded there since 2007. It has been recorded breeding at Tai Tong (Pak Sha Tsuen) Egretry since 2014.

8.4.3.14 Little Ringed Plover is listed as being of listed as being of Local Concern on account of its breeding population (Fellowes *et al.* 2002). Recorded from channelised watercourses, a typical habitat for this species, it is a locally common winter visitor and passage migrant in Hong Kong (Carey *et al.* 2001), though breeding records are more common.

8.4.3.15 A single Greater Painted-snipe was recorded from a channelised watercourse in the PDA during this Study. It is listed as being of listed as being of Local Concern (Fellowes et al. 2002), and usually inhabits wet agricultural fields and freshwater marshes in Hong Kong, where it is locally uncommon. During supplementary survey visits, up to 20 Greater Painted-snipe were recorded from ponds in the agricultural mosaic in the eastern extent of the Assessment Area.

8.4.3.16 A single Greater Coucal was recorded from Inactive Agriculture; this is a common Resident (Carey *et al.* 2001).

8.4.3.17 Singles of White-throated Kingfisher were recorded from mixed dry and wet agricultural land within PDA. (Near Tong Yan San Tsuen). This species is listed as being of Local Concern on account of its breeding sites (Fellowes *et al.* 2002). This locally common resident is widespread in Hong Kong (Carey *et al*. 2001).

8.4.3.18 A single Zitting Cisticola was recorded from dry agricultural land within PDA (Shan Ha Tsuen). A common passage migrant, recorded from open areas with long grass, such as fish ponds and agriculture (Carey *et al*. 2002) this specie sis listed as being of Local Concern by Fellowes *et al*. (2002).

8.4.3.19 Two White-cheeked Starlings were recorded from dry agricultural land within PDA (Shan Ha Tsuen). Listed as being of Potential Regional Concern (Fellowes *et al*. 2002) it is a common but localised winter visitor (Carey *et al.* 2001) with recent breeding records (Welch 2016).

8.4.3.20 Red-throated Pipit was recorded from dry agricultural land, a typical habitat for this passage migrant and winter visitor. It is of listed as being of Local Concern (Fellowes *et al*. 2002).

Egretry

8.4.3.21 From surveys conducted between 2013-16 under this Study, no breeding activity was recorded from the Tai Tong egretry. This egretry has been in steady decline and was revealed that Tai Tong egretry was abandoned during 2014 and 2015 breeding seasons (Anon. 2014, Anon. 2015). Verification surveys in 2016 confirmed no breeding activity at this site.

8.4.3.22 In July 2013, four ardeid nests were recorded in a stand of bamboo just outside the PDA in the centre of the Assessment Area (see **Figures 8.1**). This has subsequently been referred to as Tai Tong (Pak Sha Tsuen) (Anon. 2014)with a maximum of 30 nests belonging to Eastern Cattle Egret, Little Egret and Chinese Pond Heron recorded in 2016 (Table 8.5; Anon. 2016).

8.4.3.23 This egretry is located on the slopes of a small knoll, used for burial grounds, and is isolated from suitable foraging habitat by developed area, made up of wasteground, residential and industrial yards and storage with varies degrees of disturbance. The egretry is also located close to industrial units and wasteground which are subject to regular human activity. The closest suitable foraging habitat (channelised nullah) is at least 350m away to the east. Findings of average numbers of ardeids recorded from suitable foraging habitats during the breeding season 2013 (Apr- Jul) are shown in Table 8.4c.

8.4.3.24 Flightline surveys at Tai Tong (Pak Sha Tsuen) were conducted on four dates in May and June 2014, designed to coincide with the peak activity of adult birds foraging to support dependant young, observed a total of three Ardeid species; Chinese Pond Heron (119 records of flights to and from the egretry), Cattle Egret

(54) and Little Egret (44). Six broad flightlines (each consisting of more than 10 records across four surveys) were identified (**Figure 8.9**). Verification surveys in 2016 were conducted across two dates to ensure that birds were broadly following similar flightlines as those identified in 2014.

8.4.3.25 Flightline 1 heads north before crossing Yuen Long (West) Nullah and heading northwest towards agricultural land and beyond. A large cow was also present in the Woodland behind Shan Ha Tsuen; Eastern Cattle Egrets were observed in association with this cow on occasions. Flightline 2 heads directly north following the Yuen Long (West) Nullah. Flightline 3 heads north before birds veered off northeast towards the Yuen Long Highway and continued flying towards the direction of the Kam Tin River. Flightline 4 follows a similar route with birds disappearing from view, well beyond the 500m Assessment Area to the east. Flightline 5 follows an easterly direction, with birds flying above village houses and across the Yuen Long Nullah; occasionally birds appeared to drop into the nullah. Flightline 6 follows a southerly route to the mix of agricultural land in this area. Very few birds headed south perhaps due to the paucity of suitable habitats in this area for foraging.

8.4.3.26 From the observation point, foraging grounds could not be confirmed due to a combination of factors including the topography, built up areas and distance travelled by departing birds. It should be noted that the current land use, i.e. industrial storage units may already cause periodic disturbances to flightlines due to the operational activities of these units through the use of tower cranes and long-boomed vehicles.

8.4.3.27 Table 8.4c summarises the findings of the three breeding species from potential foraging habitat during the 2013 core breeding season from within the Assessment Area. Overall numbers of individuals recorded per survey are very low, for example, only two Little Egrets were recorded from Channelised Watercourses within the PDA across a four month period i.e. April to July 2013.

Table 8.4c Average number of Ardeids per survey recorded from suitable foraging habitats during the breeding season 2013 (Apr- Jul)

	Within PDA				**Outside PDA**							
Species	**CW**	**Gr**	**MA**	**Wc**	**Cw**	**DA**	**MA**	**Wc**	**Or**	**Po**	**V**	**Wa**
Chinese Pond Heron		0.25			0.25	0	0.25	0.75	0.5	0.5	0.25	0.25
Eastern Cattle Egret						0.5				0.5		
Little Egret	0.5		0.25	0.5				0.25			0.25	

CW – Channelised Watercourse; DA – Developed Area; G – Grassland; MA – Mixed Dry & Wet Agriculture; Or – Orchard; Po – Pond; V – Village; Wa – Wasteground; Wc – Watercourse.

8.4.3.28 Verification surveys of flightlines from Tai Tong (Pak Sha Tsuen) egretry in June and July 2016 revealed broadly similar departures (**Figure 8.9**) as those seen in 2014 Surveys.

8.4.4 Herpetofauna

Literature Review

8.4.4.1 Six species of reptiles including Red-eared Slider, Chinese Gecko, Bowring's Gecko, Changeable Lizard, Chinese Skink and Long-tailed Skink, were recorded in the study area of the EIA and TIA studies for YLKTSSD (AEIAR-078/2004). All of them are common and widespread in Hong Kong (Karsen *et al*. 1998). Within the Study area for the Hung Shui Kiu New Development Area (EIA 248/2016), 10 reptile species were recorded, none were of conservation importance. Five species of amphibians including Asian Common Toad, Gunther's Frog, Paddy Frog, Brown Tree Frog and Asiatic Painted Frog, were recorded in the study area of the EIA and TIA studies for YLKTSSD (AEIAR-078/2004). All of them are common and widespread in Hong Kong (Karsen *et al*. 1998). Within the Study area for the Hung Shui Kiu New Development Area (EIA 248/2016), seven amphibian species were recorded, none being of conservation importance.

Field Surveys

Reptiles

8.4.4.2 Fourteen reptile species were recorded from the Study, with four species, Reeve's Terrapin, Indo-chinese Rat Snake, Common Rat Snake and Buff-striped Keelback, listed as being of conservation concern.

8.4.4.3 Reeve's Terrapin is listed as being Endangered (IUCN 2017) and of Global Concern (Fellowes *et al.* 2002). Recorded from wet agriculture in the west of the PDA, this species is declining in Hong Kong as it is replaced by the exotic Red-eared Slider *Trachemys scripta elegans* (Karsen *et al*. 1998).

8.4.4.4 A single Indo-chinese Rat Snake, listed as being of Potential Regional Concern (Fellowes *et al.* 2002) was recorded from village area near Yeung Ka Tsuen EIS in Assessment Area. This species is considered to be common and widespread in Hong Kong (Karsen *et al.* 1998).

8.4.4.5 Common Rat Snake is listed as being of Potential Regional Concern (Fellowes *et al.* 2002). Recorded from inactive agriculture within the PDA, this species is typical of this habitat type and is considered to be common and widespread in Hong Kong (Karsen *et al.* 1998).

8.4.4.6 Buff-striped Keelback is listed as being of Local Concern (Fellowes *et al.* 2002) and was recorded in Dry Agricultural Land at Shan Ha Tsuen within PDA and also in waste ground in the eastern Assessment Area. It is typical of lowland habitats inhabiting the environs of stream and irrigation channels in agricultural fields (Karsen *et al*. 1998)

8.4.4.7 All other reptiles recorded, except Grass Lizard which is uncommon, are considered to be common and widespread in Hong Kong (**Appendix 8.5**).

Amphibians

8.4.4.8 The only amphibian of conservation concern recorded from within the PDA was Chinese Bullfrog *Hoplobatrachus chinensis*, considered to be of Potential Regional Concern (Fellowes *et al*. 2002), from the agricultural mosaic in the west. This area supports a good assemblage of nine (excluding Greenhouse Frog *Eleutherodactylus planirostris*) other lowland amphibian species, all of which are considered to be

common and widespread in Hong Kong (Chan *et al*. 2005). Records were also made of the exotic Greenhouse Frog, a recent addition to the Hong Kong list and potential pest species.

8.4.4.9 Outside of the PDA, tadpoles of Lesser Spiny Frog *Quasipaa exilispinosa* were recorded from an upland stream. This frog is listed as Vulnerable (IUCN 2017) and considered to be of Potential Global Concern (Fellowes *et al.* 2002), though is common and widespread in upland habitats in Hong Kong (Chan *et al.* 2005). This species requires clean and fast-flowing water for breeding sites.

8.4.5 Dragonflies

Literature Review

8.4.5.1 The lowland stream at Yeung Ka Tsuen is listed as an EIS by AFCD (AFCD 2011a). More than twenty species of dragonfly have been recorded there at Tai Tong including the following uncommon species Tawny Hooktail *Paragomphus capricornis*, Hong Kong Clubtail *Leptogomphus hongkongensis* and a number of stream specialists such as Club-tailed Cruiser *Macromia urania*, Giant Hooktail *Megalogomphus sommeri* and Dog-legged Clubtail *Burmagomphus vermicularis* (Wilson 1997; Tam *et al.* 2011).

8.4.5.2 The Club- tailed Cruiser *Macromia urania* and Tawny Hooktail *Paragomphus capricornis* are listed as Global and Regional Concern by Fellowes *et al*. (2002). Club-tailed Cruiser was once listed as Endangered in 1996 but, is now Least Concern in the IUCN Red list since 2006 and listed as common in Hong Kong (Wilson 2011; Tam *et al.* 2011). Giant Hooktail *Megalogomphus sommeri* and Dog-legged Clubtail *Burmagomphus vermicularis* are listed as Local Concern by Fellowes *et al.* (2002).

8.4.5.3 Up-to-date information regarding other odonates in the Assessment Area is lacking. In 2002, AFCD established the Dragonfly Working Group to monitor the status of dragonflies and carry out intensive and territory-wide surveys on dragonflies in Hong Kong (Tam *et al.* 2008). The Working Group recorded a total of 116 dragonfly species in Hong Kong in 2002 – 2010 (Tam *et al.* 2011). Odonate species documented as occurring in the general area of the Assessment Area by Wilson (1997) in An Annotated Checklist of Hong Kong Dragonflies with Recommendations for their Conservation in 1997 are shown in **Table 8.5**.

Table 8.5 Dragonfly Species Recorded in the Tai Tong and Tai Lam Country Park

No.	Species	Location[(1)]	Commonness[(2)]
1.	Indochinese Copperwing *Mnais mneme*	Tin Fu Tsai, Tai Lam Country Park	C
2.	Chinese Greenwing *Neurobasis chinensis*	Tai Tong	C
3.	Orange- faced Sprite *Pseudagrion rubriceps*	Tai Tong	C
4,	Blue- tailed Shadowdamsel *Drepanosticta hongkongensis*	Tai Lam Country Park	C
5.	Fiery Emperor *Anax immaculifrons*	Tai Tong	C
6.	Small Clubtail *Stylogomphus chunliuae*	Yeung Ka Tsuen	C

No.	Species	Location[1]	Commonness[2]
7.	*Asiagomphus septimus*	Tai Tong	Not known
8.	Dog-legged Clubtail *Burmagomphus vermicularis*	Tai Tong	C
9.	Hong Kong Clubtail *Leptogomphus hongkongensis*	Tai Lam Country Park	C
10.	*Lamelligomphus hongkongensis*	Tai Tong	Not known
11.	Giant Hooktail *Megalogomphus sommeri*	Tai Tong	C
12.	Tawny Hooktail *Paragomphus capricornis*	Tai Tong	UC
13.	Common Flangetail *Ictinogomphus pertinax*	Tai Tong	C
14.	Golden Flangetail *Sinictinogomphus clavatus*	Tai Tong	C
15.	Club-tailed Cruiser *Macromia urania*	Yeung Ka Tsuen	C
16.	Scarlet Dwarf *Nannophya pygmaea*	Tai Tong	C
17.	Greater Blue Skimmer *Orthetrum melania*	Tsing Fai Tong, Tai Lam Country Park	UC
18.	Lesser Blue Skimmer *Orthetrum triangulare*	Tai Tong	UC
19.	Pied Skimmer *Pseudothemis zonata*	Tai Lam Country Park	C
20.	Amber-winged Glider *Hydrobasileus croceus*	Tai Tong	C
21.	Sapphire Flutterer *Rhyothemis triangularis*	Tai Tong	C
22.	Evening Skimmer *Tholymis tillarga*	Tai Tong	C
23.	Dingy Dusk-darter *Zyxomma petiolatum*	Tai Tong	C
24.	Emerald Cascader *Zygonyx iris*	Tai Tong	A

Source: Wilson (1997)
Notes:
1. The location follows Wilson (1997).
2. The status of commonness and distribution follows Tam et al. (2011):.R- Rare, UC- Uncommon, C- Common and A- Abundant.

8.4.5.4 Seven dragonfly species recorded in the study area of the EIA and TIA studies for YLKTSSD (AEIAR-078/2004) all being considered to be common in Hong Kong.

8.4.5.5 Within the Study area for the Hung Shui Kiu New Development Area (EIA 248/2016), 29 species of odonate were recorded with Scarlet Basker *Urothemis signata signata* the only species of conservation importance recorded.

Field Surveys

8.4.5.6 Thirty-six species of odonate were recorded from the Study (**Appendix 8.6**) with three species listed as being of conservation concern. Blue-spotted Dusk-hawker

Gynacantha japonica, Blue Chaser *Potamarcha congener* and Emerald Cascader *Zygonyx iris* are all listed by Fellowes *et al.* (2002) as being of conservation concern.

8.4.5.7 The only dragonfly species of conservation concern recorded within PDA was Blue Chaser, which is listed as being of Local Concern by Fellowes *et al*. (2002). This species is considered to be common and widespread in Hong Kong (Tam *et al*., 2011). Blue Chaser was recorded from the wet agriculture in the north of the PDA, and was also recorded from a watercourse in the Assessment Area which has subsequently been decked over.

8.4.5.8 Blue-spotted Dusk-hawker and Emerald Cascader, which are listed as Local Concern and Potential Global Concern by Fellowes *et al*. (2002), were recorded from streams outside of the PDA.

8.4.5.9 Within the PDA, a good assemblage of odonate species were recorded from wetter habitats in the agricultural mosaic and from watercourses, all being common and widespread in Hong Kong (Tam *et al.* 2011).

8.4.6 Butterflies

Literature Review

8.4.6.1 Eight butterfly species recorded in the study area of the EIA & TIA studies for YLKTSSD (AEIAR-078/2004) all considered to be common and widespread in Hong Kong.

8.4.6.2 Within the Study area for the Hung Shui Kiu New Development Area (EIA 248/2016), 66 butterfly species were recorded. Six species were of conservation importance (single observations of each species): Dark Brown Ace *Halpe porus*, Grass Demon *Udaspes folus*, Common Rose *Pachliopta aristolochiae goniopeltis*, Cornelian *Deudorix epijarbas menesicles*, Common Cerulean *Jamides celeno celeno* and Danaid Egg-fly *Hypolimnas misippus*. All are considered either "uncommon", "rare" or "very rare" in Hong Kong (AFCD, 2014). Among them, Dark Brown Ace and Danaid Egg-fly are of Local Concern (Fellowes *et al*. 2002).

Field Surveys

8.4.6.3 A total of 89 species were recorded from the Study (**Appendix 8.7**), with eleven species considered to be Rare or Very Rare (Chan *et al.* 2011). These are Common Dart *Potanthus pseudomaesa*, Pale Palm Dart *Telicota colon*, Grass Demon *Udaspes folus*, Purple and Gold Flitter *Zographetus satwa*, Spotted Angle *Caprona dlida*, Forget-me-not *Catochrysops strabo*, Metallic Cerulean *Jamides alecto*, Tiny Grass Blue *Zizula hylax*, Courtesan *Euripus nyctelius*, Swallowtail *Papilio xuthus* and Small Cabbage White *Pieris rapae*.

8.4.6.4 Four were recorded within the PDA, including Forget-me-not, Metallic Cerulean, Tiny Grass Blue and Swallowtail.

8.4.6.5 Others were recorded from watercourse, grassland/shrubland, secondary woodland, village area and/or seasonally wet grassland habitats outside of the PDA.

8.4.6.6 Common Dart, Pale Palm Dart and Spotted Angle are also listed as being of Local Concern, while Purple and Gold Flitter is listed as being of Regional Concern (Fellowes *et al.* 2002).

8.4.6.7 Within the PDA, the mosaic of agricultural habitats, orchard and secondary woodland at Sha Ha Tsuen supported the highest diversity of butterfly species.

8.4.7 Stream Fauna

Literature Review

8.4.7.1 There is little information available for in-stream fauna in the Assessment Area. The EIA & TIA studies for YLKTSSD (AEIAR-078/2004) recorded Tilapia *Oreochromis* spp. in the stream in Tai Tong near the EIS in Yeung Ka Tsuen. This species is widespread and non-native (Lee *et al*. 2004). Within the Study area for the Hung Shui Kiu New Development Area (EIA 248/2016), the only aquatic fauna of conservation importance recorded was Predaceous Chub *Parazacco spilurus*.

8.4.7.2 The shrimp *Caridina apodosis* has been recorded from a mountain stream near Tai Tong Village (Cai & Ng 1999); it is only known from the type series, collected in 1994 and is considered to be Critically Endangered by IUCN (2017). As the exact location of the stream from which the specimens were obtained is not known, there is a clear danger that it may already have been incorporated into the urban landscape and be ecologically unsuitable for this species (De Grave 2013). Given that this species has not been seen again since its discovery, this proves to be a significant information gap. Urgent surveys are required to establish if the species is still extant and if so to identify the distribution of the species in the wider area (De Grave 2013).

Field Surveys

8.4.7.3 Of the entire stream fauna recorded (**Appendix 8.8**); the most notable records are of two endemic crab species; *S. zanklon* and *Cryptopotamon anacoluthon*.

8.4.7.4 Populations of *S. zanklon* were recorded from several streams/catchments in the PDA. These streams are fed by runoff from the slopes of the hills to the west. Numbers of this crab are much higher in those streams where there are natural substrates and banks with good riparian vegetation cover, with no sources of point pollution. This crab was also recorded from watercourses and irrigation channels flowing through agriculture and village areas in the eastern Assessment Area. One watercourse where this crab was recorded has been decked over during this Study. It is listed as being of Global Concern (Fellowes *et al.* 2002) and is considered to be Endangered by IUCN (Esser & Cumberlidge 2008a). *Somanniathelphusa zanklon* is widely distributed within Hong Kong, recorded from the northwestern and northeastern New Territories and Lantau Island (Stanton & Leven 2016).

8.4.7.5 A dead individual of *C. anacoluthon* was recorded from a drainage channel where it presumably was washed down from upstream sections following a period of heavy rain. This was the only individual recorded during the Study but is presumed to occur in clean hillside streams to the south and west of the PDA. This species shows a preference for clean, fast-flowing stream (Stanton *et al.* 2017) Listed as Potential Global Concern by Fellowes *et al.* (2002) it is considered to be Vulnerable by IUCN (Esser & Cumberlidge 2008b).

8.4.7.6 Predaceous Chub was recorded in the Yeung Ka Tsuen EIS. Predaceous Chub is considered as Vulnerable in Mainland China (Wang *et al.* 1998), although it is common and widespread in Hong Kong (Lee *et al*. 2004). It is listed as Least Concern in China by Jiang *et al.* (2016).

8.4.7.7 Small Snakehead was recorded from sections of good quality watercourses in in catchments 6 & 7 within PDA. Listed as being of Local Concern by Fellowes et al. (2002), it is uncommon in the wild (AFCD 2017). Records have come from a few streams in North district and on Lantau Island. The fish is also cultivated in some fish farms and are available from fish market (Lee et al. 2004, AFCD 2017).

8.4.7.8 Larvae of dragonflies of conservation importance were recorded from Catchment 6 (Watercourse near Shan Ha Tsuen Agriculture). These include Tawny Hooktail listed as Regional Concern, Club-tailed Cruiser, Dancing Shadow-emerald and Emerald Cascader *Zygonyx iris insignis*. Tawny Hooktail is listed as Regional Concern (Fellowes *et al.* 2002) and Uncommon in Hong Kong found in woodland streams with fine to course gravel beds (Tam *et al*. 2011, AFCD 2017). Club-tailed Cruiser and Dancing Shadow-emerald are both listed under (Fellowes *et al.* 2002) but are Common in Hong Kong (Tam *et al*. 2011, AFCD 2017).

8.4.7.9 Larvae of Scarlet Basker *Urothemis signata* was recorded from Catchment 9 in a heavily fragmented watercourse. Whilst listed under Fellowes *et al*. (2002) it is considered to be Common by Tam *et al.* (2011).

8.4.7.10 Fauna in the more disturbed and polluted channels running through the PDA were dominated by exotic and introduced species, tolerant of poor water conditions.

8.5 Evaluation of Habitats and Species

8.5.1 Habitats Evaluation

Watercourses

8.5.1.1 The watercourses surveyed in this Study have been separated into broad catchment areas for ease of assessment (**Figure 8.3**). As most lowland streams have been modified or influenced by humans in some way, they are not strictly 'natural'; however, they may retain some natural characteristics or hydrology. They have been grouped together in the following tables where different catchments are broadly similar in ecological value.

8.5.1.2 The shrimp *Caridina apodosis* was recorded from a mountain stream near Tai Tong Village (Cai & Ng 1999) and is considered to be Critically Endangered by IUCN (2017). As the exact location of the stream from which the specimens were obtained is not known, and as it has not been seen since 1994, there is a clear danger that the location may already have been incorporated into the urban landscape and be ecologically unsuitable for this species. While this proves to be a significant information gap in the presence or absence of this species in the Assessment Area, it is considered that there is no gap in the assessment methodology as these sections of upland ('mountain') streams with suitable water quality also harbours other species of conservation importance which are accounted for in the following sections.

8.5.1.3 To further confirm locations of watercourses, the aquatic fauna sampling points from Figure 8.3 have been included, where available, to help clarity.

Catchment 1, 8 and 10

8.5.1.4 These small watercourses are generally very narrow and shallow, and all pass through areas subject to heavy human influences. Streams have been heavily

modified along the banks but in some cases have a natural substrate. Water quality generally appears to be poor and subject to point pollution. Catchment 1 is a concrete u-channel, though does have natural substrate in the bottom.

8.5.1.5 Fish are largely restricted to mosquito fish, guppies and Tilapia and the dominant invertebrate was Apple Snail. No fish or invertebrate species of conservation significance were found in two of the watercourses during surveys conducted under the current study; however, there is potential for the crab *S. zanklon* to occur in Catchment 8, given its proximity to Catchments 6 and 7 where the crab has been recorded. This species is listed as of global conservation concern, and has been found in several watercourses in the Study.

Table 8.6 Ecological Evaluation of Watercourses in the Assessment Area (Catchments 1, 8 and 10)

	Catchment		
	1 **(Sampling points 10 & 11)**	**8** **(Sampling points 42)**	**10**
Naturalness	Man-made, concrete channel with some natural substrates, though this may have been washed in.	Narrow ditch with natural substrate but heavily modified channel course and banks. Evidence of direct discharge of domestic effluent.	Semi-natural, natural beds but heavily modified banks and influenced by direct discharge of domestic effluent.
Size	Narrow channel, less than 1m wide and approx. 300 in length. Outside of the PDA.	Small shallow ditch, less than 1m wide with very low water flow. Approx. length 240m in PDA though is not wet throughout the year	Two fragmented sections (approx. 230m & 270m in length).
Diversity	Low diversity of species.		
Rarity	N/a	Potential for *S. zanklon.*	N/a
Re-creatability	Readily re-created.		
Fragmentation	Generally heavily fragmented by culverts and pipes to other sections of the same watercourse.	Fragmented from downstream section.	Generally heavily fragmented by culverts and pipes to other sections of the same watercourse.
Ecological Linkage	Limited linkages as passes through developed area.	Linkages with hillside habitats.	Limited linkages.
Potential Value	Low. Highly fragmented.	Limited. Could be improved by addressing water quality issues, enhancement of channel and provision of linkages in piped/culverted sections.	
Nursery/ Breeding Ground	Not known though unlikely given the apparent poor condition of most of these streams.		
Age	Unknown.		

	Catchment		
	1 **(Sampling points 10 & 11)**	**8** **(Sampling points 42)**	**10**
Abundance/ Richness of Wildlife	Very low abundance and richness of fauna.		
Ecological Value	**Low**	**Low**	**Low**

Catchment 2 - Yeung Ka Tsuen EIS

8.5.1.6 The lowland stream at Yeung Ka Tsuen is an Ecologically Important Stream (EIS), on account of the diversity of stream specialist dragonflies recorded. In its current state, the stream channel is heavily choked with grasses and ruderal vegetation. Flow is slow and subject to local pollution from adjacent agricultural land uses. No flora or fauna of conservation concern were recorded from this stream during this Study.

Table 8.7 Ecological Evaluation of Watercourses in the Assessment Area (Catchment 2 - Yeung Ka Tsuen EIS)

Criteria	**Catchment 2 - Yeung Ka Tsuen EIS** **(Sampling point 25)**
Naturalness	Semi-natural - natural beds but modified banks and influenced by direct discharge of domestic effluent.
Size	Large (approx. 1110m in total for the permanently wet watercourses)
Diversity	Records of high diversity of odonata including stream specialists. Flora diversity very low.
Rarity	Species of Conservation Interest are based on the results of Literature Review. Club-tailed Cruiser and Tawny Hooktail listed as Global and Regional Concern (Fellowes *et al*. 2002). Club-tailed Cruiser now listed as Least Concern in the IUCN Red list since 2006 and listed as common in Hong Kong (Wilson 2011; Tam *et al.* 2011). Giant Hooktail and Dog-legged Clubtail are listed as Local Concern (Fellowes *et al*. 2002). None of these species were recorded in recent surveys. Predaceous Chub has also been recorded from this EIS.
Re-creatability	Readily re-created.
Fragmentation	Not fragmented hydrologically.
Ecological Linkage	Ecologically linked to upstream areas and adjacent farmland, and downstream to channelised watercourse.
Potential Value	Moderate to high following improvement in water quality, vegetation clearance and enhancement/sympathetic management of channel.
Nursery/ Breeding Ground	None recorded in this Study. Potential for a wide range of uncommon odonate species.
Age	Unknown.
Abundance/ Richness of Wildlife	Abundance and diversity of species is low-moderate.

Criteria	Catchment 2 - Yeung Ka Tsuen EIS (Sampling point 25)
Ecological Value	**Moderate.**

Catchment 3

8.5.1.7 This watercourse collects in the hills to the south of Tai Tong and passes through a chicken farm/industrial area before discharging into the drainage channel. The stream has been heavily modified as it passes through (and under) the developed area within the PDA. Sections of streambed have been concreted to prevent erosion around footings of buildings. Several pipes were observed discharging effluent directly into the stream and water quality appeared poor with a malodorous smell.

8.5.1.8 The only species recorded from the lower section of stream were chironomid larvae, indicative of poor water quality. It should be noted however that a dead individual of the crab *C. anacoluthon* was recorded from the location where this stream meets the drainage channel. It is assumed this crab was washed downstream and deposited in this location following a period of heavy rainfall when the stream was in spate and would indicate that the upper reaches of this catchment are of good ecological value.

Table 8.8 Ecological Evaluation of Watercourses in the Assessment Area (Catchment 3)

Criteria	Catchment 3 (Sampling points 27)
Naturalness	Upstream section largely natural and free from human influence. Downstream section is heavily modified banks, shaded and influenced by direct discharge of domestic effluent.
Size	Approx. 100m passes through, or close to, the PDA in its downstream section; this section has been heavily modified.
Diversity	Low diversity of species.
Rarity	Potential for *C. anacoluthon* in upper reaches.
Re-creatability	Readily re-created.
Fragmentation	Not fragmented hydrologically. Lower section fragmented ecologically by adjacent land uses.
Ecological Linkage	Linkages with hillside habitats in upper sections and channelised watercourse in lower section.
Potential Value	Could be improved by addressing water quality issues, enhancement of channel and removal of adjacent land use in lower section.
Nursery/ Breeding Ground	Not known though unlikely given the apparent poor condition of most of these streams. Upper reaches are likely to support a population of *C. anacoluthon*.
Age	Unknown.
Abundance/ Richness of Wildlife	Low abundance and diversity of fauna.
Ecological Value	**Low** in lower section which pass through the PDA **Moderate** in upper sections.

Catchment 4

8.5.1.9 This catchment is fed by the hills to the southwest of the PDA to the west; these watercourses in the uplands are likely to be highly seasonal. The topography here results in fast-flowing stream with plenty of pools, riffles and cascades. Several fauna species of conservation concern, including Lesser Spiny Frog and Emerald Cascader, were recorded from these sections and are typical of this stream type.

8.5.1.10 As the watercourse flows through the PDA, several branches meet the main channel. Much of the streambeds are composed of natural sandy substrates with a good coverage of riparian vegetation. Water quality is highly variable throughout this watercourse and is dependent on adjacent land use. Where industrial and residential properties occur, outfall pipes freely discharge effluent directly into the main channel. The endemic crab *S. zanklon* was found in the cleaner channels of this catchment in sections of stream with good macrophyte coverage. Both adult and immature crabs were recorded, indicating a breeding population. This crab is considered to be of Global Concern (Fellowes *et al*. 2002) and is listed as Endangered by IUCN (2017), though it appears to be fairly widespread in the northern New Territories.

8.5.1.11 Two lower tributaries (identified by sampling locations 30 and 40 on **Figure 8.3**) do not have upstream connections and appear to be fed by drains and/or surface run-off from the developed area through which they flow. Access to both sections is limited by 2m high hoarding. Channel width for both sections is less than 1m, and much of the southernmost tributary (labelled 30 in **Figure 8.3**) appears to be subject to seasonal flow.

Table 8.9 Ecological Evaluations of Watercourses in the Assessment Area (Catchment 4)

Criteria	**Upstream Section** **(Sampling point 32)**	**Main Channel in PDA** **(Sampling points 29 & 31)**	**Tributaries in lower section of PDA** **(Sampling points 30 & 40)**
Naturalness	Natural.	Semi-natural, natural beds but modified banks and influence by direct discharge of domestic effluent.	
Size	Upstream sections are fairly extensive.	Approx. 475m passes through the PDA.	Narrow channels, <1m wide. Approx. length of 185m and 80m, though extent of permanent water appears to be variable.
Diversity	Low diversity of species.	Low diversity of species.	
Rarity	Lesser Spiny Frog Tadpoles, Blue-spotted Dusk-hawker, Emerald Cascader.	*S. zanklon* in good numbers in those watercourse sections with good water quality.	Potential for *S. zanklon* in wet sections.
Re-creatability	Difficult to re-create due to topography of hillside.	Readily re-created.	
Fragmentation	Not fragmented.	Largely continuous with upstream section.	No direct connect within upstream sections.

Criteria	**Upstream Section** **(Sampling point 32)**	**Main Channel in PDA** **(Sampling points 29 & 31)**	**Tributaries in lower section of PDA** **(Sampling points 30 & 40)**
Ecological Linkage	Functionally linked to downstream sections and adjacent hillside habitats.	Functionally linked to upstream sections and downstream drainage channel. Adjacent habitat of negligible ecological value.	No significant ecological linkages other than with the main channel. Adjacent land use is ecologically poor. Source of water not clear, presumed surface run-off.
Potential Value	Moderate	Moderate to high following improvement in water quality and enhancement of channel and creation of buffer zone.	Moderate if upstream linkages could be reinstated.
Nursery/ Breeding Ground	Lesser Spiny Frog and Emerald Cascader larvae recorded from this section.	Juvenile and immature crabs recorded.	
Age	Unknown.	Unknown but modified by human activity.	
Abundance/ Richness of Wildlife	Moderate abundance of wildlife but low diversity.	In areas of good quality habitat, abundance of species is high. Poor water quality results in low densities and diversity.	
Ecological Value	**Moderate to high** on account of the naturalness of this watercourse.	**Moderate to high** for main section which contains species of conservation importance and is connected to upstream sections.	**Low to moderate** as tributaries isolated from upstream sections and fed by run-off.

Catchment 5

8.5.1.12 The streams of this catchment all sit outside of the PDA. Where they enter the PDA, they are piped under industrial units, other than a short section of approx. 120m that discharges directly into a channelised watercourse; this section is less than 1m with low flow of water, but has natural banks and substrate.

8.5.1.13 These streams may have been modified in the past but they have natural banks and streambeds. The upper sections in hillside grassland are likely to be seasonal, given the low diversity and numbers of species recorded. The lower sections pass through shrubland, stands of woodland and abandoned agriculture. Water quality appears to be good with no sources of point pollution.

8.5.1.14 A dead individual of the crab *S. zanklon* was recorded from the drainage channel into which this catchment is linked. It is likely this individual was washed into the channel following heavy rainfall.

Table 8.10 Ecological Evaluation of Watercourses in the Assessment Area (Catchment 5)

Criteria	**Catchment 5** **(Sampling point 23)**
Naturalness	Largely natural in upstream sections with human modifications where watercourses meet the PDA.
Size	Large (approx. 2,000m in total for the permanently wet watercourses) for whole catchment. Approximately 200m (two sections of 150m + 50m) within PDA.
Diversity	Low diversity and abundance of species.
Rarity	*S. zanklon* likely to occur in these streams.
Re-creatability	Difficult to re-create.
Fragmentation	Fragmented from other catchments by drainage channel and developed area in lower section.
Ecological Linkage	Linkages with natural habitats outside of the PDA. No above ground linkages with other watercourses within PDA.
Potential Value	Moderate.
Nursery/ Breeding Ground	Not known, likely *S. zanklon* breeds here.
Age	Unknown.
Abundance/ Richness of Wildlife	Low abundance and diversity.
Ecological Value	**Moderate**

Catchments 6 and 7

8.5.1.15 Streams 6 and 7 would likely have formed part of the same catchment prior to the construction of the drainage channel in the lower section. Both streams have natural substrates for much of their lengths and natural banks with riparian vegetation. Stream 6 passes through a series of agricultural fields before it becomes channelised in the lower section; it is also fed by several smaller tributaries that arise in the hill slopes to the west of the PDA. Stream 7 passes through predominantly light industrial and container yards areas, though a short section flows through light agricultural land.

8.5.1.16 The endemic crab *S. zanklon* was recorded from both streams, notably in areas where aquatic vegetation was present.

Table 8.11 Ecological Evaluation of Watercourses in the Assessment Area (Catchment 6 and 7)

Criteria	Catchment 6 (Sampling points 37, 38 & 39)	Catchment 7 (Sampling point 35)
Naturalness	Largely natural. Heavily modified and influenced by humans but banks and substrate natural; apparent good water quality. Passes through various agricultural habitats and secondary woodland.	Heavily modified and influenced by humans. Banks and substrate natural in places. Apparent good water quality away from areas of direct discharge. Passes through various agricultural habitats and light industry.
Size	Wide riparian corridor (approx. 1, 700m in total for the permanently wet watercourses) with approx. 640m of watercourses within the PDA.	Narrow riparian corridor, hemmed in by industrial land uses (approx. 1,000m in total for the permanently wet watercourses) with approx. 960m of watercourses within the PDA.
Diversity	Low-moderate diversity of species.	
Rarity	*S. zanklon* recorded from both catchments. Small Snakehead (from Catchment 6) and good assemblage of dragonfly larvae recorded from these sites.	
Re-creatability	Readily re-created if source of water is maintained.	
Fragmentation	The two streams were formerly part of the same system but have been fragmented by the construction of the Drainage Channel.	
Ecological Linkage	Significant linkages with hillside habitats, secondary woodland and agricultural land.	Limited linkages due to adjacent land use.
Potential Value	Lots of potential for enhancement by way of creating riparian corridors and improving water quality.	
Nursery/ Breeding Ground	Immature *S. zanklon* recorded indicating stream is a breeding site.	
Age	Unknown but modified by human activity.	
Abundance/ Richness of Wildlife	Moderate abundance of riparian and stream fauna including species of conservation importance.	Low-moderate abundance of stream fauna including species of conservation importance.
Ecological Value	**Moderate to high** given its naturalness, linkages and presence of species of conservation importance.	**Moderate to high** in unpolluted sections which contain species of conservation importance.

Catchment 9

8.5.1.17 These two hydrologically isolated watercourses flow in a south to north direction through this catchment. The source of their water may be from the hillstreams in the western hills or from surface run-off from adjacent developed land; this is unconfirmed due to the local land use issues. Years of human modification has resulted these watercourses being culverted and/or piped underneath developments

and infrastructure, leaving open water sections heavily modified with limited natural features through much of their lengths.

8.5.1.18 These streams are subject to point pollution (e.g. direct discharge and dumping) particularly the western watercourse that passes by the Tai Tong (Pak Sha Tsuen) Egretry where water quality appeared extremely poor with a malodorous smell.

8.5.1.19 Species recorded were largely limited to exotic and introduced stream fauna common throughout the New Territories.

8.5.1.20 However, there is potential for the crab *S. zanklon* to occur particularly in the eastern watercourse as water quality appears to be moderate and banks and substrates are largely natural as it passes through the PDA.

Table 8.12 Ecological Evaluation of Watercourses in the Assessment Area (Stream Catchment 9)

Criteria	**Catchment 9**	
	Western **(Sampling points 15, 17 & 20)**	**Eastern** **(Sampling points 16, 18 & 19)**
Naturalness	Heavily modified following extensive human development in the area. Highly fragmented by development and poor water quality observed.	Heavily modified following extensive human development in the area. Some sections have natural substrate and apparently free from pollution.
Size	Narrow channels approx. 1m wide. Fragmented within and outside of the PDA with sections of open channel measuring approx. 420m (65m+39m+162m+94m+59m) in PDA and a further 330m outside of the PDA.	A continuous section of approx. 890m flowing through both PDA (approx. 340m) and outside PDA (approx. 550m). Channel itself ranges from 0.5m to 2m along its length.
Diversity	Low diversity of species.	
Rarity	None recorded.	None recorded. Potential for endemic crab species.
Re-creatability	Readily re-created.	
Fragmentation	Highly fragmented both hydrologically and by adjacent land uses.	
Ecological Linkage	Limited to linkages with drainage channels. No upstream linkages and adjacent land use is poor ecologically.	
Potential Value	Some potential for enhancement by way of creating and opening up riparian corridors, recreate streams by removing culverts/piped sections and improving water quality and channel conditions. Limited however by adjacent land uses and lack of ecological linkages.	
Nursery/ Breeding Ground	None recorded.	
Age	Unknown but modified by human activity.	
Abundance/	Large numbers of exotic stream fauna.	

Criteria	Catchment 9	
	Western **(Sampling points 15, 17 & 20)**	**Eastern** **(Sampling points 16, 18 & 19)**
Richness of Wildlife		
Ecological Value	**Low** – lack of continuous open channel combined with poor water quality and ecological linkages.	**Low to moderate** as some natural substrate and potential for endemic crab species.

Catchment 11

8.5.1.21 Several watercourses flow through the agricultural land and village areas of Catchment 11. Years of human modification have culverted and/or piped these streams underneath developments and infrastructure, leaving only a few short sections as open channels. These streams are subject to pollution (e.g. direct discharge and dumping), but appear to have moderate water quality in some sections. Species recorded were largely limited to exotic and introduced stream fauna common throughout the New Territories, though the endemic crab *S. zanklon* was recorded from both streams, notably in areas where aquatic vegetation was present.

Table 8.13 Ecological Evaluation of Watercourses in the Assessment Area (Stream Catchment 11)

Criteria	Catchment 11
Naturalness	Semi-natural, heavily modified flowing through agriculture and village areas
Size	Small
Diversity	Low diversity of species.
Rarity	Endemic crab *S. zanklon* recorded.
Re-creatability	Readily re-created.
Fragmentation	Some fragmentation.
Ecological Linkage	Some ecological linkages with channelized watercourses and agricultural land.
Potential Value	Lots of potential for enhancement by way of creating and opening up riparian corridors, recreate streams by removing culverts/piped sections and improving water quality and channel conditions.
Nursery/ Breeding Ground	None recorded.
Age	Unknown but modified by human activity.
Abundance/ Richness of Wildlife	Large numbers of exotic stream fauna.
Ecological Value	**Moderate** as scope for improvement, good linkages with agricultural land and present of endemic species.

Channelized Watercourse

8.5.1.22 Channelized watercourses are mainly located along existing roads of Kung Um Road/ Kiu Hing Road, Lam Tei West Road, Long Hong Road, Tong Yan San Tsuen Road, and across Fui Sha Wai. The Yuen Long Bypass Floodway channel extends to the east of the Assessment Area. These are entirely man-made habitats and are generally considered of low to moderate ecological value due to linkages with existing watercourses and high potential for ecological enhancement.

Table 8.14 Ecological Evaluation of Channelized Watercourses in the Assessment Area

Criteria	Channelized Watercourses
Naturalness	Man-made. Concrete banks and stream bed with no natural features
Size	Medium (~21.22ha Assessment Area)
Diversity	Low-moderate flora and fauna diversity.
Rarity	Low numbers of wetland dependant birds recorded including Little Ringed Plover (5), Greater Painted-snipe (1), Chinese Pond Heron (4), Grey Heron (1), Great Egret (1), Little Egret (12). Numbers in parentheses are total numbers recorded during the Study. Single records of a dead crab *S. zanklon* and *C. anacoluthon* in channels relate presumably to these individuals being washed down from more suitable stream courses following heavy rains and increased flow. Yuen Long Bypass Floodway provides habitat for moderate numbers of foraging bats.
Re-creatability	Readily re-created.
Fragmentation	Functionally connected to other channelized watercourses and adjacent habitats.
Ecological Linkage	Ecologically linked to natural streams.
Potential Value	Lots of scope for improving water quality and providing ecological enhancements in the channel.
Nursery/ Breeding Ground	None recorded
Age	Less than 30 years.
Abundance/ Richness of Wildlife	Moderate abundance and diversity.
Ecological Value	**Low to Moderate** as scope for ecological enhancements across as large area.

Ponds and Inactive Fish Pond

8.5.1.23 A total of 31 ponds and/or inactive fish pond(s) are present in the Assessment Area with sizes ranging from 0.02ha to 0.45ha among which only five farm ponds for irrigation within mixed dry and wet agricultural land near Sha Tseng Tsuen, and a small, managed pond (shallow pond) (Pond #17) in Area 1 are located within the PDA. Outside of PDA, there are two shallow farm ponds for irrigation/ wet agriculture near Fui Sha Wai and Yeung Uk Tsuen respectively, two small-scale

managed/ fish ponds for cultivation of Nile tilapia, an inactive fish pond in Fui Sha Wai, a stormwater pond in Kiu Tau Wai stormwater pumping station (outside of the PDA), and a few abandoned ponds. All the ponds are entirely man-made habitats with low floristic diversity, and they are generally considered of low to moderate ecological value in view of their linkages with adjacent wetland habitats/ agricultural land.

Table 8.15 Ecological Evaluation of Ponds in the Assessment Area

Criteria	**Ponds in Assessment Area**
Naturalness	Man-made.
Size	Small (ranging from 0.02ha to 0.45ha with a total area of ~4ha in the Assessment Area).
Diversity	Low floristic diversity; low-moderate faunal diversity.
Rarity	Common in northern New Territories. Up to ten Greater Painted-snipe recorded from ponds in the eastern extent of the Assessment Area.
Re-creatability	Readily re-created if suitable hydrological present.
Fragmentation	Fragmented by agricultural land and village development.
Ecological Linkage	Ecological and hydrological linkages with wetland habitats including natural/ channelized watercourses, seasonally wet grasslands and agricultural lands.
Potential Value	Potential for enhancement through dedicated management regime.
Nursery/ Breeding Ground	Not known.
Age	No known.
Abundance/ Richness of Wildlife	Low to moderate.
Ecological Value	**Low to Moderate.**

Table 8.16 Ecological Evaluation of Inactive Fish Pond in the Assessment Area

Criteria	**Inactive Fish Pond in Assessment Area**
Naturalness	Man-made.
Size	Small (0.25ha in Assessment Area).
Diversity	Low floristic and faunal diversity.
Rarity	Common in northern New Territories.
Re-creatability	Readily re-created if suitable hydrological present.
Fragmentation	Fragmented by agricultural land.
Ecological Linkage	Ecological and hydrological linkages with wetland habitats including channelized watercourses, seasonally wet grasslands and agricultural lands.
Potential Value	Potential for enhancement through dedicated management regime.
Nursery/ Breeding Ground	Not known.
Age	Not known.

Criteria	Inactive Fish Pond in Assessment Area
Abundance/ Richness of Wildlife	Low to moderate.
Ecological Value	**Low to Moderate.**

Wet Agricultural Land

8.5.1.24 The wet agricultural land in the Assessment Area is considered of moderate ecological value given its current linkages with other adjacent wetland habitats, but its potential value is limited by its small size.

Table 8.17 Ecological Evaluation of Wet Agricultural Land in the Assessment Area

Criteria	Wet Agricultural Land in Assessment Area
Naturalness	Man-made habitat with poorly established vegetation along field bunds.
Size	Small (~3.72ha in the Assessment Area)
Diversity	Low floristic diversity and low to moderate faunal diversity.
Rarity	Common habitat type in northern New Territories but declining due to development; can support uncommon wetland dependent species. Species of conservation interest include ardeids in low numbers, Chinese Noctule, Reeve's Terrapin and Chinese Bullfrog.
Re-creatability	Readily re-created with appropriate hydrological conditions.
Fragmentation	Fragmented in the Assessment Area.
Ecological Linkage	Linkages with the adjacent agricultural land.
Potential Value	Potential for enhancement through management but this is limited by its small size.
Nursery/ Breeding Ground	Not known.
Age	Not known.
Abundance/ Richness of Wildlife	Low to moderate.
Ecological Value	**Moderate** in those wet agricultural areas near Shan Ha Tsuen – due to species of conservation concern and proximity to other habitats of ecological value. **Low to Moderate** – others scattered through PDA.

Mixed Dry & Wet Agricultural Land

8.5.1.25 Mixed dry and wet agricultural land is mainly located near Sha Tseng Tsuen and on Long Ho Road opposite to Yeung Uk Tsuen. This is an entirely man-made habitat with some potential for habitat enhancement through dedicated management regime. However the potential value is limited by its small size and the isolation arising from existing low-rise rural development. The overall ecological value of this habitat is considered low to moderate.

Table 8.18 Ecological Evaluation of Mixed Dry & Wet Agricultural Land in the Assessment Area

Criteria	Mixed Dry & Wet Agricultural Land in Assessment Area
Naturalness	Man-made habitat with limited self-sown vegetation.
Size	Small (~2.73ha).
Diversity	Low floral and faunal diversity.
Rarity	Common habitat type in northern New Territories but declining due to development; can support uncommon wetland dependent species. Ardeids in low numbers have been recorded from this area.
Re-creatability	Readily re-created.
Fragmentation	Highly fragmented by existing development.
Ecological Linkage	Some linkages with the wetland/ agricultural lands in Shan Ha Tsuen but being fragmented by existing low-rise rural development
Potential Value	Potential for enhancement through management but this is limited by its small size and fragmentation by existing development.
Nursery/ Breeding Ground	Not known.
Age	Not known.
Abundance/ Richness of Wildlife	Low to moderate.
Ecological Value	**Low to Moderate.**

Dry Agricultural Land

8.5.1.26 Dry agricultural land is the predominant habitat in the low-lying area adjoining the southwest of Shan Ha Tsuen. This type of habitat is also scattered throughout the village areas to the east of the PDA and in Fui Shan Wai.

Table 8.19 Ecological Evaluation of Dry Agricultural Land in the Assessment Area

Criteria	Dry Agricultural Land in Assessment Area
Naturalness	Man-made habitat with self-sown vegetation established along the field bunds.
Size	Large (~32.54ha in the Assessment Area).
Diversity	Moderate floristic diversity and faunal diversity.
Rarity	Common habitat type in northern New Territories but declining due to development. A patch of approximately 6m x 3m of the cultivated herb *Houttuynia cordata* was recorded in the active agricultural land in Fui Sha Wai (outside PDA) in 2013. The wild population of this herb species is considered as "very rare" local distribution by Fellowes *et al*. (2002); this patch is assumed to be planted or cultivated. Two individuals of *Ceratopteris thalictroides* were recorded in the dry agricultural land located north of Shek Po Tsuen and outside of the PDA or any other associated works areas. This species is

Criteria	Dry Agricultural Land in Assessment Area
	regarded as "Rare" in Xing *et al.* (2000). Status in China as "Vulnerable", under State protection (Category II) in China (AFCD 2003).
Re-creatability	Readily re-created.
Fragmentation	Low in Shan Ha Tsuen; High in other village areas.
Ecological Linkage	Good linkages with existing secondary woodland and natural watercourses near Shan Ha Tsuen; low to moderate linkages with other wetland habitats.
Potential Value	Potential for enhancement through active management.
Nursery/ Breeding Ground	Not known.
Age	Not known.
Abundance/ Richness of Wildlife	Moderate.
Ecological Value	**Moderate** for agricultural land adjoining Shan Ha Tsuen; **Low to Moderate** for other scattered agricultural lands.

Inactive Agricultural Land

8.5.1.27 Inactive agricultural land is scattered through active agricultural land and villages in the Assessment Area. This habitat supports low floristic diversity but has good linkages with adjacent agricultural land and potential for ecological enhancement through appropriate active management. This habitat is considered of low to moderate ecological value.

Table 8.20 Ecological Evaluation of Inactive Agricultural Land in the Assessment Area

Criteria	Inactive Agricultural Land in Assessment Area
Naturalness	Man-made habitat with self-sown vegetation established along the field bunds.
Size	Small (~3.39ha in Assessment Area).
Diversity	Low floral and faunal diversity.
Rarity	Common habitat type in northern New Territories but declining due to development. Leschenault's Rousette, Short-nosed Fruit Bat and Himalayan Leaf-nosed Bat were all found dead in illegal mist nets located in this habitat.
Re-creatability	Readily re-created
Fragmentation	Low.
Ecological Linkage	Linkages with adjacent active agricultural land and/or wetland habitats.
Potential Value	Potential for enhancement through active management.
Nursery/ Breeding Ground	Not known.
Age	Not known.

Criteria	Inactive Agricultural Land in Assessment Area
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low to Moderate.**

Seasonally Wet Grassland

8.5.1.28 Most of the areas of seasonally wet grassland in the Assessment Area are subject to high levels of human disturbance as they are surrounded by frequently managed agricultural land or isolated by village house developments. The overall ecological value of this habitat is low to moderate due to its low diversity of flora and fauna species but with good linkages to other wetland habitats and potential for habitat enhancement.

Table 8.21 Ecological Evaluation of Seasonally Wet Grassland in the Assessment Area

Criteria	Seasonally Wet Grassland in Assessment Area
Naturalness	Semi-natural and developed through natural succession from man-made wetland habitats.
Size	Medium (~8.07ha in the Assessment Area).
Diversity	Low floristic and faunal diversity.
Rarity	Not uncommon in Hong Kong.
Re-creatability	Readily re-created with appropriate hydrological conditions.
Fragmentation	Fragmented by existing villages and roads.
Ecological Linkage	Linkages to nearby agricultural land and watercourses.
Potential Value	Potential for enhancement under appropriate management.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low to Moderate.**

Reedbeds

8.5.1.29 Two small reedbeds are located within the mixed dry and wet agricultural land in Sha Tseng Tsuen and on Long Ho Road near Pok Oi Hospital respectively. Due to their small size and low fauna usage they are considered to be of low to moderate ecological value.

Table 8.22 Ecological Evaluation of Reedbed in the Assessment Area

Criteria	Reedbed in Assessment Area
Naturalness	Semi-natural.
Size	Small (~0.60ha in the Assessment Area).
Diversity	Low floristic and faunal diversity.
Rarity	Rare and declining in Hong Kong.

Criteria	Reedbed in Assessment Area
Re-creatability	Can be recreated under suitable hydrological conditions and with appropriate management.
Fragmentation	Highly fragmented by existing village development.
Ecological Linkage	Linkages to adjacent wetland habitats (mixed dry and wet agricultural land and/or channels).
Potential Value	Potential for enhancement but limited by small size and existing fragmentation.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low to Moderate** due to its rarity.

Marsh

8.5.1.30 A small area of marsh is located in the low-lying area on Siu Sheung Road near Pok Oi Hospital (on the north-eastern margin of the Assessment Area). This fresh water marsh was derived from a low-lying wet-field (with retained rainwater) and overgrown with weedy and common wetland plant species.

Table 8.23 Ecological Evaluation of Marsh in the Assessment Area

Criteria	Marsh in Assessment Area
Naturalness	Semi-natural.
Size	Small (~0.09ha in the Assessment Area).
Diversity	Low floristic and faunal diversity.
Rarity	Rare and declining in Hong Kong.
Re-creatability	Can be recreated under suitable hydrological conditions and with appropriate management.
Fragmentation	Highly fragmented by existing buildings.
Ecological Linkage	No significant linkage to other wetland habitats.
Potential Value	Potential for enhancement is limited by its small size and existing fragmentation.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low** due to its small size and high degree of fragmentation by existing buildings.

Grassland

8.5.1.31 Grassland in the Assessment Area is low-lying, disturbed and scattered within the village areas. This habitat is considered of low floristic diversity and low ecological value

Table 8.24 Ecological Evaluation of Grassland in the Assessment Area

Criteria	Grassland in Assessment Area
Naturalness	Semi-natural but frequently disturbed.
Size	Medium (~22.17ha in Assessment Area).
Diversity	Low floristic and limited structural complexity.
Rarity	Common.
Re-creatability	Readily re-created.
Fragmentation	Highly fragmented by villages.
Ecological Linkage	Low.
Potential Value	Potential for enhancement through replanting.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

Grassland/Shrubland

8.5.1.32 Grassland/shrubland is concentrated on the continuous hill slopes to the west of the PDA. This is a semi-natural habitat formed after deforestation and maintained/ intermittently disturbed by anthropogenic hill fire.

Table 8.25 Ecological Evaluation of Grassland/ Shrubland in the Assessment Area

Criteria	Grassland/ Shrubland in Assessment Area
Naturalness	Semi-natural intermittently disturbed and maintained by anthropogenic hill fire.
Size	Large (~174.42ha in Assessment Area).
Diversity	Low to moderate.
Rarity	Common. The woody climber species *Gnetum luofuense* was recorded in the hillside shrubby grassland to the west of PDA. This species is regarded as Very Common by Xing *et al.* (2000), classified as "Near Threatened" in IUCN Red List but locally very common in Hong Kong.
Re-creatability	Can be re-created but considerable time required.
Fragmentation	Not fragmented.
Ecological Linkage	Linkages with continuous upland habitats of similar types.
Potential Value	Potential for enhancement through reforestation and control of hill fire.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.

Criteria	Grassland/ Shrubland in Assessment Area
Ecological Value	**Low to Moderate.**

Secondary Woodland

8.5.1.33 Approximately 58.67ha of secondary woodlands are present in the Assessment Area, including ~8.62ha within the PDA. For ease of assessment these woodlands are divided into several groups (**Figure 8.7**) according to their locations and nature, and their ecological values are described separately as follows:

Table 8.26 Ecological Evaluation of Secondary Woodland behind Shan Ha Tsuen

Criteria	Secondary Woodland behind Shan Ha Tsuen in Assessment Area
Naturalness	Semi-natural to natural through natural succession.
Size	Large (~13.23ha in the Assessment Area).
Diversity	Moderate.
Rarity	Large areas of lowland native secondary woodland are uncommon in Hong Kong. Reeve's Muntjac and Least Horseshoe Bat were recorded in or close to this woodland. A small number of seedlings of the protected tree species *Aquilaria sinensis* were recorded in this woodland but outside PDA. The woody climber species *Gnetum luofuense* was recorded in the secondary woodland. This species is regarded as Very Common by Xing *et al.* (2000), classified as "Near Threatened" in IUCN Red List but locally very common in Hong Kong.
Re-creatability	Can be re-created but considerable time is required to reach current stage
Fragmentation	Generally large and not fragmented.
Ecological Linkage	Linkages with existing semi-natural upland habitats.
Potential Value	Potential for become more mature woodland in the absence of clearance and disturbance.
Nursery/ Breeding Ground	Not known.
Age	Not known but mature woodland
Abundance/ Richness of Wildlife	Moderate.
Ecological Value	**Moderate to High** due to its large size and high maturity/ structural complexity.

Table 8.27 Ecological Evaluation of Hillside Secondary Woodland

Criteria	Hillside Secondary Woodland in Assessment Area
Naturalness	Semi-natural to natural through natural succession.
Size	Large (~19.09ha in the Assessment Area).
Diversity	Low to Moderate.
Rarity	Lowland native secondary woodland is uncommon in Hong Kong.

Criteria	Hillside Secondary Woodland in Assessment Area
Re-creatability	Can be re-created but considerable time is required to reach current state.
Fragmentation	Fragmented by other upland habitats.
Ecological Linkage	Linkages with existing semi-natural upland habitats.
Potential Value	Potential to become more mature woodland in the absence of clearance and disturbance.
Nursery/ Breeding Ground	Not known.
Age	Maintained by hill fire as young secondary woodland.
Abundance/ Richness of Wildlife	Moderate.
Ecological Value	**Moderate.**

Table 8.28 Ecological Evaluation of Secondary Woodland near Tong Yan San Tsuen

Criteria	Secondary Woodland near Tong Yan San Tsuen in Assessment Area
Naturalness	Semi-natural comprising planted vegetation and heavily disturbed (village developments in woodland understory).
Size	Small (~4.28ha in the Assessment Area).
Diversity	Low.
Rarity	Common.
Re-creatability	Can be re-created but considerable time is required to reach current stage.
Fragmentation	Fragmented by existing village development.
Ecological Linkage	Poor linkage to other natural habitats.
Potential Value	Potential for enhancement through removal of village houses in the understory and re-vegetation/ replanting.
Nursery/ Breeding Ground	Not known.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low to Moderate.**

Table 8.29 Ecological Evaluation of Woodland near Lam Hau Tsuen and Lam Hau Wai Tsuen

Criteria	Secondary Woodland near Lam Hau Tsuen and Lam Hau Wai Tsuen in Assessment Area
Naturalness	Semi-natural comprising planted fruit trees/ ornamentals and spontaneous vegetation; heavily disturbed with understory occupied by burial grounds.
Size	Small (~3.30ha in the Assessment Area).

Criteria	Secondary Woodland near Lam Hau Tsuen and Lam Hau Wai Tsuen in Assessment Area
Diversity	Low.
Rarity	Common.
Re-creatability	Can be re-created but considerable time is required to reach current stage.
Fragmentation	Fragmented by existing village development and burial grounds.
Ecological Linkage	Poor linkage to other natural habitats.
Potential Value	Limited potential for enhancement due to presence of burial grounds in the understory.
Nursery/ Breeding Ground	Not known.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low to Moderate.**

Table 8.30 Ecological Evaluation of Secondary Woodland under Pylon in Yeung Ka Tsuen

Criteria	Secondary Woodland under Pylon in Yeung Ka Tsuen in Assessment Area
Naturalness	Semi-natural comprising planted and naturally-established vegetation.
Size	Small (~1.29ha in the Assessment Area).
Diversity	Low.
Rarity	Common.
Re-creatability	Can be re-created but considerable time is required to reach current stage.
Fragmentation	Fragmented by existing village development.
Ecological Linkage	Poor linkage to other natural habitats.
Potential Value	Potential for enhancement through re-vegetation/ replanting.
Nursery/ Breeding Ground	Not known.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low to Moderate.**

Table 8.31 Ecological Evaluation of Other Scattered Woodlands

Criteria	Other Scattered Secondary Woodlands in Assessment Area
Naturalness	Semi-natural comprising planted and naturally-established vegetation.

Criteria	Other Scattered Secondary Woodlands in Assessment Area
Size	Small (total ~17.48ha in the Assessment Area).
Diversity	Low.
Rarity	Common.
Re-creatability	Can be re-created but considerable time is required to reach current stage.
Fragmentation	Fragmented by existing villages and roads.
Ecological Linkage	Poor linkage to other natural habitats.
Potential Value	Potential for enhancement through re-vegetation/ replanting.
Nursery/ Breeding Ground	Not known.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low to Moderate.**

Plantation

8.5.1.34 Plantation woodlands are spread through hillside uplands to the west of the PDA, and are composed of a few exotic plantation species including *Acacia confusa*, *Casuarina equisetifolia*, *Eucalyptus robusta*, *Lophostemon confertus* and *Pinus elliottii*. The understory of these plantations is relatively sparse and natural regeneration is very limited in some upland areas with poor soil.

Table 8.32 Ecological Evaluation of Plantation in the Assessment Area

Criteria	Plantation in Assessment Area
Naturalness	Man-made with some self-sown species.
Size	Large (~46.56ha in Assessment Area).
Diversity	Low.
Rarity	Common.
Re-creatability	Can be re-created but considerable time is required to reach its current state.
Fragmentation	Fragmented by other upland habitats.
Ecological Linkage	Good linkages with other upland habitats.
Potential Value	Potential for enhancement through afforestation.
Nursery/ Breeding Ground	Not found.
Age	~15-20 years.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

Lychee Woodland

8.5.1.35 There are two Lychee orchards of considerable size located at the junction between the mature secondary woodland and large agricultural lands in Shan Han Tsuen.

These appear to be abandoned orchards and comprise mature/ over-mature trees of *Litchi chinensis* with very sparse understory and lack of native plant species. The ecological value of these mature abandoned Lychee woodlands is considered low.

Table 8.33 Ecological Evaluation of Lychee Woodland in the Assessment Area

Criteria	Lychee Woodland in Assessment Area
Naturalness	Man-made.
Size	Small (~0.27ha in Assessment Area).
Diversity	Low.
Rarity	Common.
Re-creatability	Can be re-created but considerable time is required to reach its current state.
Fragmentation	Fragmented by woodlands and agricultural land.
Ecological Linkage	Linkages with existing secondary woodland.
Potential Value	Limited potential for enhancement due to lack of natural regeneration in the understory.
Nursery/ Breeding Ground	Not found.
Age	Mature/ over mature.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

Orchard

8.5.1.36 Orchards of varying sizes were commonly found in the village environs of the Assessment area and are entirely man-made habitats. Due to high degree of fragmentation by existing village settlements, high level of human disturbance/ active management, and generally very low floristic diversity, this habitat type is considered of low ecological value.

Table 8.34 Ecological Evaluation of Orchard in the Assessment Area

Criteria	Orchard in Assessment Area
Naturalness	Man-made.
Size	Medium (~17.56ha in Assessment Area).
Diversity	Low.
Rarity	Common. Two individual trees of *Artocarpus hypargyreus* were recorded at the orchard near the western boundary of Area 1. *Artocarpus hypargyreus* is regarded as "Vulnerable" in China Red Data Book (Pang et al. 2011) and IUCN (IUCN 2017), and "Near Threatened" by AFCD (2003). However, it is commonly found in lowland forest and not locally protected.
Re-creatability	Readily re-created.
Fragmentation	Fragmented by village development.
Ecological Linkage	Poor linkages with natural habitats.

Criteria	Orchard in Assessment Area
Potential Value	Limited potential for enhancement.
Nursery/ Breeding Ground	Not found.
Age	Young to semi-mature.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

Nursery

8.5.1.37 A number of plant nurseries were scattered in the Assessment Area. Similar to the existing orchards these nurseries are man-made habitats and are considered of having low ecological value due to their man-made nature and the existing high level of disturbance from active management.

Table 8.35 Ecological Evaluation of Nursery in the Assessment Area

Criteria	Nursery in Assessment Area
Naturalness	Man-made with self-sown species.
Size	Small (~0.78ha in Assessment Area).
Diversity	Low.
Rarity	Common.
Re-creatability	Readily re-created.
Fragmentation	Fragmented by suburban development.
Ecological Linkage	Poor ecological linkages.
Potential Value	Potential for enhancement through re-vegetation.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

Village Area

8.5.1.38 This habitat type is mainly located east of Kiu Hing Road adjoining the eastern boundary of the PDA, and between Area 1 and Area 2 of PDA. This habitat comprises diverse plant diversity of planted fruit trees and a wide range of ornamental species, with open fields dominated by common pioneer herbs. The floristic diversity of this habitat is moderate, however the floristic value is considered low due to lack of native vegetation communities and high degree of fragmentation and human disturbance.

Table 8.36 Ecological Evaluation of Village Area in the Assessment Area

Criteria	Village Area in Assessment Area
Naturalness	Man-made with self-sown species.
Size	Large (~242.23ha in Assessment Area).
Diversity	Moderate floristic diversity but low faunal diversity.
Rarity	Very common.
Re-creatability	Readily re-created.
Fragmentation	Fragmented by village houses.
Ecological Linkage	Some ecological linkages with wetland habitats and agricultural lands.
Potential Value	Potential for enhancement through re-vegetation and dedicated management regime. Generally low.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

Wasteland

8.5.1.39 This habitat is scattered through the Assessment Area and is mainly derived from fallow cultivation and woodland clearance and has become overgrown with weedy vegetation after a long period of abandonment. This overgrown habitat is considered of low ecological value due to high level of human disturbance and low floristic diversity and value.

Table 8.37 Ecological Evaluation of Wasteland in the Assessment Area

Criteria	Wasteland in Assessment Area
Naturalness	Man-made with self-sown species.
Size	Large (~76.50ha in Assessment Area).
Diversity	Low.
Rarity	Very common. One tree of *Artocarpus hypargyreus* was recorded at the wasteland near the western boundary of Area 1. *Artocarpus hypargyreus* is regarded as "Vulnerable" in China Red Data Book (Pang et al. 2011) and IUCN (IUCN 2017), and "Near Threatened" by AFCD (2003). However, it is commonly found in lowland forest and not locally protected.
Re-creatability	Readily re-created.
Fragmentation	Fragmented by village houses.
Ecological Linkage	Some ecological linkages with agricultural lands and village areas.
Potential Value	Potential for enhancement through re-vegetation and dedicated management regime.
Nursery/ Breeding Ground	Not found.

Criteria	Wasteland in Assessment Area
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

Developed Area

8.5.1.40 The majority of the Assessment Area belongs to developed area dominated by low-rise rural residential developments, and rural industrial developments, and urban areas in Yuen Long. Vegetation in this habitat is dominated by ornamental plants and common roadside plant communities. Due to its high level of human disturbance and man-made nature this habitat type is generally considered of low ecological value.

Table 8.38 Ecological Evaluation of Developed Area in the Assessment Area

Criteria	Developed Area in Assessment Area
Naturalness	Man-made.
Size	Large (~826.13ha in Assessment Area).
Diversity	Low.
Rarity	Very common; a mature and planted specimen of the rare tree species *Ailanthus fordii* was recorded on Tong Yan San Tsuen Road within existing rural industrial area and within PDA. The wild population of this tree species is considered "rare" in Fellowes *et al*. (2002) and protected by Cap. 96A. Approximately 10 young saplings of the protected tree species *Aquilaria sinensis* (protected under Cap. 586) were found planted in ground in a small private nursery near the southern end of Lam Tei West Road (Figure 8.5f); assumed that these have been planted/cultivated.
Re-creatability	Readily re-created.
Fragmentation	Fragmented.
Ecological Linkage	Some ecological linkages with agricultural lands and village areas.
Potential Value	Potential for enhancement through revegetation and dedicated management regime. Generally low.
Nursery/ Breeding Ground	Not found.
Age	Not known.
Abundance/ Richness of Wildlife	Low.
Ecological Value	**Low.**

8.5.2 Species Evaluation

8.5.2.1 A total of six plant, 24 bird, 11 butterfly, three dragonfly, two amphibian, four reptile species, nine aquatic fauna species, and five mammal species of conservation importance were identified in the Assessment Area and are listed in

Tables 8.39 to **8.46** below. Locations of plant species and non-avifauna species of conservation importance, and known egretry locations are shown in **Figure 8.5**.

Table 8.39 Flora Species of Conservation Importance

Species	Protection Status	Conservation Status	Distribution in Hong Kong[1-5]	Ecological Value
Ailanthus *Ailanthus fordii*	Cap. 96A	-	Rare in lowland evergreen forest	Low (as it is a single planted specimen within developed area)
Silver-back Artocarpus *Artocarpus hypargyreus*	Nil	Vulnerable[6]; "Near Threatened" [7]*	Commonly found in lowland forest and *fung shui* woods in Hong Kong (Xing *et al.* 2000; AFCD 2007)	Low to medium
Fishwort *Houttuynia cordata*	Nil	-	Very rare	Low (as it is cultivated and low in abundance)
Incense Tree *Aquilaria sinensis*	Cap. 586, State Protection (Category II) [7]	Vulnerable[6]; "Near Threatened" [7]*	Commonly found in lowland forest and *fung shui* wood	Medium (for wild population in woodland behind Shan Ha Tsuen) and Low (for planted population in developed area and in low abundance)
Luofushan Joint-fir *Gnetum luofuense*	Nil	Near Threatened[6]	Very Common[1]	Medium
Water Fern *Ceratopteris thalictroides*	Nil	"Vulnerable" [7]*	Rare[1]	Medium

[1]Xing et al. (2000) [2] AFCD (2007) [3] AFCD (2008)
[4] AFCD (2009) [5] AFCD (2011b) [6] IUCN (2017)
[7] AFCD (2003)
* Conservation/ Protection Status is stated in China Plant Red Data Book and Illustration of Rare & Endangered Plant in Guangdong Province as stipulated in AFCD (2003).

Fauna Species of Conservation Importance

Table 8.40 Bird Species of Conservation Importance Recorded in the PDA and Assessment Area (outside PDA)

Species	Conservation Status (Fellowes *et al.* 2002)[1]	IUCN Red List Status (IUCN 2017)	Max. count	Area/ Habitats Recorded
Eurasian Teal *Anas crecca*	Regional Concern	-	1	Assessment Area (Village Area)
Yellow Bittern *Ixobrychus sinensis*	(Local Concern)	-	1	PDA (Mixed Dry & Wet Agricultural Land)
Black-crowned	(Local	-	5 (2 at PDA	PDA

Species	Conservation Status (Fellowes *et al.* 2002)[1]	IUCN Red List Status (IUCN 2017)	Max. count	Area/ Habitats Recorded
Night Heron *Nycticorax nycticorax*	Concern)		and 3 at Assessment Area)	(Watercourse) Assessment Area (Pond)
Chinese Pond Heron *Ardeola bacchus*	Potential Regional Concern (Regional Concern)	-	11 (6 at PDA and 5 at Assessment Area)	PDA (Channelised Watercourse, Mixed Dry& Wet Agricultural Land, Dry Agricultural Land, Grassland/Shrubland, Inactive Agricultural Land, Watercourse) Assessment Area (Channelised Watercourse, Wet Agricultural Land, Dry Agricultural Land, Grassland, Grassland/Shrubland, Mixed Dry & Wet Agricultural Land, Watercourse, Orchard, Pond, Secondary Woodland, Village Area, Wasteland)
Eastern Cattle Egret *Bubulcus coromandus*	(Local Concern)	-	22 (20 at PDA and 2 at Assessment Area)	PDA (Dry Agricultural Land, Mixed Dry & Wet Agricultural Land) Assessment Area (Developed Area, Secondary Woodland)
Grey Heron *Ardea cinerea*	Potential Regional Concern	-	3 (1 at PDA and 2 at Assessment Area)	PDA (Channelised Watercourse) Assessment Area (Channelised Watercourse)
Great Egret *Ardea alba*	Potential Regional Concern (Regional Concern)	-	3 (1 at PDA and 2 at Assessment Area)	PDA (Channelised Watercourse) Assessment Area (Channelised Watercourse, Mixed Dry & Wet Agricultural Land,)
Little Egret *Egretta garzetta*	Potential Regional Concern (Regional Concern)	-	55 (5 at PDA and 50 at Assessment Area)	PDA (Channelised Watercourse, Pond, Developed Area, Dry Agricultural Land, Inactive Agricultural Land, Mixed Dry & Wet Agricultural Land, Watercourse) Assessment Area (Channelised Watercourse, Dry Agricultural Land, Grassland, Grassland/Shrubland,

Species	Conservation Status (Fellowes *et al.* 2002)[1]	IUCN Red List Status (IUCN 2017)	Max. count	Area/ Habitats Recorded
				Watercourse, Pond, Village Area)
Great Cormorant *Phalacrocorax carbo*	Potential Regional Concern	-	1 (overhead)	Assessment Area (Dry Agricultural Land)
Crested Serpent Eagle *Spilornis cheela*	(Local Concern)	Vu (China Red Data Book)	2	Assessment Area (Grassland/Shrubland, Secondary Woodland, Wasteland, In Flight)
Black Kite *Milvus migrans*	(Regional Concern)	-	1	Assessment Area (In Flight)
Little Ringed Plover *Charadrius dubius*	(Local Concern)	-	4 (3 at PDA and 1 at Assessment Area)	PDA (Channelised Watercourse) Assessment Area (Channelised Watercourse)
Greater Painted-snipe *Rostratula benghalensis*	Local Concern	-	21 (1 at PDA and 20 at Assessment Area)	PDA (Channelised Watercourse) Assesment Area (Pond)
Greater Coucal *Centropus sinensis*	-	Vu (China Red Data Book)	1 at PDA and 1 at Assessment Area	PDA (Inactive Agriculture) Assessment Area (Dry Agriculture, Mixed Agriculture, Orchard, Village, Wasteland)
Wood Sandpiper *Tringa glareola*	Local Concern	-	2	Assessment Area (Channelised Watercourse, Pond)
White-throated Kingfisher *Halcyon smyrnensis*	(Local Concern)	-	2 (1 at PDA and 1 at Assessment Area)	PDA (Mixed Dry & Wet Agricultural Land) Assessment Area (Dry Agricultural Land)
Collared Crow *Corvus torquatus*	Local Concern	Near Threatened	2	Assessment Area (Secondary Woodland)
Zitting Cisticola *Cisticola juncidis*	Local Concern	-	1	PDA (Dry Agricultural Land)
Golden-headed Cisticola *Cisticola exilis*	Local Concern	-	1	Assessment Area (Dry Agricultural Land)
Rufous-capped Babbler *Stachyridopsis ruficeps*	Local Concern	-	2	Assessment Area (Grassland/Shrubland, Secondary Woodland)
Red-billed Starling *Spodiopsar*	Global Concern	-	31	Assessment Area (Dry Agricultural Land, Orchard, Village Area)

Species	Conservation Status (Fellowes *et al.* 2002)[1]	IUCN Red List Status (IUCN 2017)	Max. count	Area/ Habitats Recorded
sericeus				
White-cheeked Starling *Spodiopsar cineraceus*	Potential Regional Concern	-	9 (8 at PDA and 1 at Assessment Area)	PDA (Dry Agricultural Land) Assessment Area (Dry Agricultural Land)
White-shouldered Starling *Sturnia sinensis*	(Local Concern)	-	4	Assessment Area (Wasteland)
Red-throated Pipit *Anthus cervinus*	Local Concern	-	3	PDA (Dry Agricultural Land)

Notes:

1. Conservation status in parentheses indicates that the assessment is on the basis of restrictedness in breeding and/or roosting sites rather than in general occurrence.

Table 8.41 Amphibian Species of Conservation Importance Recorded in the PDA and Assessment Area (outside PDA)

Species	Conservation Status (Fellowes *et al* 2002)	IUCN Red List Status (IUCN 2017)	Max. count	Area/ Habitats
Chinese Bullfrog *Hoplobatrachus chinensis*	Potential Regional Concern	-	3 (2 at PDA and 1 at Assessment Area)	PDA (Mixed Dry & Wet Agricultural Land, Dry Agricultural Land) Assessment Area (Seasonally Wet Grassland)
Lesser Spiny Frog *Paa exilispinosa*	Potential Global Concern	Vulnerable	2	Assessment Area (Watercourse)

Table 8.42 Reptile Species of Conservation Importance Recorded in the PDA and Assessment Area (outside PDA)

Species	Conservation Status (Fellowes *et al* 2002)	IUCN Red List Status (IUCN 2017)	Max. count	Area/ Habitats
Reeves' Terrapin *Chinemys reevesii*	Global Concern	Endangered	1	PDA (Wet Agricultural Land)
Indo-chinese Rat Snake	Potential Regional Concern	-	1	PDA (Village Area)
Common Rat Snake *Ptyas mucosus*	Potential Regional Concern	-	2 (1 at PDA and 1 at Assessment Area)	PDA (Inactive Agricultural Land) Assessment Area (Village Area)

Species	Conservation Status (Fellowes *et al* 2002)	IUCN Red List Status (IUCN 2017)	Max. count	Area/ Habitats
Buff-striped Keelback *Amphiesma stolatum*	Local Concern	-	2 (1 at PDA and 1 at Assessment Area)	PDA (Dry Agricultural Land) Assessment Area (Wasteland)

Table 8.43 Dragonfly Species of Conservation Importance Recorded in the PDA and Assessment Area (outside PDA)

Species	Conservation Status (Fellowes *et al.* 2002)	Status in Hong Kong (Tam *et al.* 2011)	Max. count	Area/ Habitats
Blue-spotted Dusk-hawker *Gynacantha japonica*	Local Concern	Common	1	Assessment Area (Watercourse)
Blue Chaser *Potamarcha congener*	Local Concern	Common	2 (1 at PDA and 1 at Assessment Area)	PDA (Mixed Dry & Wet Agricultural Land) Assessment Area (Watercourse),
Emerald Cascader *Zygonyx iris*	Potential Global Concern	Abundant	4	Assessment Area (Watercourse)

Table 8.44 Butterfly Species of Conservation Importance Recorded in the PDA and Assessment Area (outside PDA)

Species	Conservation Status (Fellowes *et al.* 2002)	Status in Hong Kong (Chan *et al.* 2011)	Max. count	Area/ Habitats Recorded
Common Dart *Potanthus pseudomaesa*	Local Concern	Rare	1	Assessment Area (Secondary Woodland)
Pale Palm Dart *Telicota colon*	Local Concern	Rare	1	Assessment Area (Village Area)
Grass Demon *Udaspes folus*	-	Rare	2	Assessment Area (Dry Agricultural Land, Seasonally Wet Grassland, Village Area)
Purple and Gold Flitter *Zographetus satwa*	Regional Concern	Rare	1	Assessment Area (Secondary Woodland)
Spotted Angle *Caprona dlida*	Local Concern	Very Rare	1	Assessment Area (Grassland/ Shrubland)

Species	Conservation Status (Fellowes *et al.* 2002)	Status in Hong Kong (Chan *et al.* 2011)	Max. count	Area/ Habitats Recorded
Forget-me-not *Catochrysops strabo*	-	Very Rare	2	PDA (Watercourse, Dry Agricultural Land, Grassland/ Shrubland, Orchard)
Metallic Cerulean *Jamides alecto*	-	Very Rare	5 (2 at PDA, 3 at Assessment Area)	PDA (Grassland/ Shrubland, Secondary Woodland, Orchard) Assessment Area (Seasonally Wet Grassland, Secondary Woodland, Orchard, Village Area)
Tiny Grass Blue *Zizula hylax*	-	Very Rare	1	PDA (Orchard)
Courtesan *Euripus nyctelius*	-	Very Rare	1	Assessment Area (Grassland/Shrubland)
Swallowtail *Papilio xuthus*	-	Rare	1	PDA (Orchard, Developed Area)
Small Cabbage White *Pieris rapae*	-	Rare	1	Assessment Area (Watercourse)

Table 8.45 Stream Fauna of Conservation Importance Recorded in the PDA and Assessment Area (outside PDA)

Species	Conservation Status (Fellowes *et al.* 2002)	IUCN Red list Status (IUCN 2017)	Max. count	Area/ Habitats
Predaceous Chub *Parazacco spilurus*	Vulnerable (China Species Red List)	-	5	Assessment Area (Watercourse)
Small Snakehead *Channa asiatica*	Local Concern	-	1	PDA (Watercourse)
Crab *Cryptopotamon anacoluthon*	Potential Global Concern	Vulnerable	1	PDA (Channelised Watercourse)
Crab *Somanniathelphusa zanklon*	Global Concern	Endangered	29 (26 at PDA, 3 at Assessment Area)	PDA & Assessment Area (Channelised Watercourse, Watercourse)

Species	Conservation Status (Fellowes *et al.* 2002)	IUCN Red list Status (IUCN 2017)	Max. count	Area/ Habitats
Tawny Hooktail (Larva) *Paragomphus capricornis*	Regional Concern	-	5	PDA (Watercourse)
Club-tailed Cruiser (Larva) *Macromia urania*	Global Concern	-	10	PDA (Watercourse)
Dancing Shadow-emerald (Larva) *Idionyx victor*	Local Concern	-	3	PDA (Watercourse)
Scarlet Basker (Larva) *Urothemis signata*	Local Concern	-	1	PDA (Watercourse)
Emerald Cascader (Larva) *Zygonynx iris*	Potential Global Concern	-	1	Assessment Area (Watercourse)

Table 8.46 Mammal Species of Conservation Importance Recorded in the PDA and Assessment Area (outside PDA)

Species	Conservation Status (Fellowes *et al* 2002)	Max. count	Area/ Habitats
Leschenault's Rousette *Rousettus leschenaulti*	(Local Concern)	5	PDA (Dry Agricultural Land)
Least Horseshoe Bat *Rhinolophus pusillus*	Potential Regional Concern/(Regional Concern)	1	Assessment Area (Secondary Woodland)
Himalayan Leaf-nosed Bat *Hipposideros armiger*	(Local Concern)	5	PDA (Dry Agricultural Land)
Chinese Noctule* *Nyctalus plancyi*	Potential Regional Concern/(Regional Concern)	4	PDA (Dry Agricultural Land, Mixed Dry & Wet Agricultural land) Assessment Area (Mixed Dry & Wet Agricultural land, Wasteland)
Red Muntjac *Muntiacus muntjak*	Potential Regional Concern	1	PDA (Watercourse) Assessment Area (Secondary Woodland)

* Simmons (2005) states that the *Nytalus* species found in China should be separated as *N. plancyi*.

8.6 Prediction and Evaluation of Ecological Impacts

8.6.1 Assessment Methodology

8.6.1.1 The potential terrestrial and aquatic ecological impacts arising from the construction and operation of the proposed Project, including loss of habitats, removal of vegetation and disturbance to animals have been assessed in accordance with Annexes 8 and 16 of the Technical Memorandum-EIAO. Estimates of habitat lost and identification of areas to be affected by development have been calculated as accurately as possible. Prior to assessing these impacts however, it is necessary to describe the design features of the proposed Project that have been incorporated to avoid or minimise ecological impact.

8.6.2 Avoidance and Minimisation of Design Measures

8.6.2.1 A number of principles were adopted to avoid and minimise ecological impacts as part of the design process, based on consideration of the habitats present in the wider Assessment Area and the results of field surveys as these became available. These comprised the following.

(i) Avoidance of the Secondary Woodland at Shan Ha Tsuen: the proposed alignment of the distributor road (Road D1) will run through the already disturbed grounds in orchards and avoid encroachment upon the mature secondary woodland at Shan Ha Tsuen.

(ii) In the RODP, three lower sections of watercourses, including Stream Catchment 7 (~700m) and Stream Catchment 6 (~650m) in Area 1 and Stream Catchment 4 (~450m) in Area 3 of the PDA have records of the crab *S. zanklon*; these sections are proposed to be retained with 15m 'Green Belt' buffers on both sides of the watercourses; A total buffer width of 30m. This specification is based on buffer distances proposed for other similar study recommendations in the New Territories, notably Ma Tso Lung Stream (AEIAR-175/2013).

(iii) Following the results of flightline surveys in 2014 and verification surveys in 2016, the RODP has been adjusted several times to avoid impacts to major flightlines. The development layout has been designed to maintain gaps/visual corridors between buildings to preserve sight lines for egrets flying within the residential zone. Non-Building Areas (NBA) are provided to the north through land allocated for District Open Space (DO). Additional NBAs are positioned to the south and the west. To the east, the existing developed area is outside of the PDA and flightlines in this direction will be maintained. The major flight lines are shown in **Figure 8.9** overlaid on the RODP.

(iv) Fragmentation of the Agricultural Land at Shan Ha Tsuen by the proposed D1 connection road has been minimised through the Study. The agricultural land at Shan Ha Tsuen is mixed (wet, dry & inactive) agricultural land and is considered to be of Moderate ecological value, with existing linkages with good quality habitats to the west and south (shrubland, woodland and watercourses) in western limits of agricultural land. The location of the road has been adjusted to pass though the eastern

limit of the agricultural land away from those areas with good ecological connectivity. The footprint of the segment of Road D1, which encroached on the agricultural land, has been reduced to minimise the disturbance to the habitat. Apart from reducing the provision footpath from 4m to 2m on both side, the total amenity area have been further reduced from 8.8m to 5.3m in accordance with the DevB Technical Circular (Works) No. 02/2012.

8.6.3 Identification of Impacts

8.6.3.1 Potential sources of impacts as listed in the EIA Study Brief No. ESB-246/2014 are summarised as follows:

- Loss of habitats of conservation interest, in particular wetlands including wet agricultural land, marshes and fishponds, wooded areas, natural/ semi-natural streams and drainage channels;
- Noise, glare, dust, traffic and other human disturbance and other deterioration of environmental quality to ecological sensitive areas and wildlife;
- Indirect ecological impacts due to the potential hydrological disruption and/ or deterioration of the water quality in the drainage channels, fishponds or other wetland habitats in the assessment area due to river training/ diversification works, surface runoff, drainage and effluent discharge, waste dumping/ spillage, etc.;
- Potential impacts on breeding egrets and herons due to loss or disturbance to breeding, roosting and feeding grounds;
- Impact of future development and associated access road/ facilities on the flight path of breeding ardeids between breeding/ roosting and foraging sites taking into account seasonal patterns during construction and operation phases;
- Impacts on birds due to collision to buildings and transparent or semi-transparent noise barriers;
- Fragmentation of habitats on ecologically sensitive areas; and
- Cumulative impacts due to other planned and committed development projects in or near the project site.

8.6.3.2 The potential ecological impacts arising from the construction and operation of the Project based on the RODP have been assessed and evaluated in accordance with Annexes 8 and 16 of EIAO-TM and are presented as follows:

Direct Impacts on Habitats

8.6.3.3 Based on the proposed land use budget of the RODP as presented in Chapter 2, the Project area in PDA covers approximately 223.5ha of the Yuen Long South area. The proposed development will not encroach upon Tai Lam Country Park area. As such, no direct ecological impacts will be resulted from habitat loss within Tai Lam Country Park.

8.6.3.4 Among the proposed land uses, the area zoned as "Agriculture (AGR)" aims to preserve the large existing area of agricultural land to the southwest of Shan Ha

Tsuen and one existing chicken farm. The zoning of "Green Belt (GB)" aims to preserve ecologically important areas including secondary woodland and streams and buffer areas. Based on the proposed land use budget for the RODP and the known footprints of proposed works, potential areas of habitats to be affected by the Project are summarised in **Table 8.47** below and illustrated in **Figure 8.8**.

Table 8.47 Summary of Potential Areas of Habitats to be affected by the Project

Habitats	Areas in PDA (ha)[1]		Areas/ Extent of Habitats in PDA to be Affected (ha)[2]	Areas beyond PDA to be Affected (ha)	Habitat Value	Notes
Watercourse (approx. lengths)	Catchment 3 (Sampling point 27)	70m	0	0	Low	This 70m watercourse will be avoided
	Catchment 4 (Sampling point 29 & 31)	475m	0	0	Moderate-high	This 475m watercourse will be retained/avoided with a 15m buffer each side
	Catchment 4 (Sampling point 30 & 40)	185m + 80m	185m + 80m	0	Low-moderate	Two small tributaries will be lost to the reedbed
	Catchment 5 (Sampling point 23)	150m + 50m	150m + 50m	0	Moderate	Two lower sections of watercourse to be lost to the hillside river corridor/retention lake
	Catchment 6 (Sampling point 37)	640m	0	0	Moderate-high	This 640m watercourse will be retained/avoided with a 15m buffer each side
	Catchment 7 (Sampling point 35)	955m	100m	0	Moderate-high	Much of this 955m watercourse will be retained/avoided with a 15m buffer each side. A short section of 100m will be diverted at its downstream limit.
	Catchment 8 (Sampling point 42)	242m	242m	0	Low	Narrow ditch to be lost
	Catchment 9 (Sampling point 15, 20 & 34)	65m +39m + 162m +94m + 59m	65m +39m + 162m +94m + 59m	0	Low	Fragmented sections of watercourse within PDA to be lost
	Catchment 9 (Sampling point 18 & 19)	328m + 8m	328m + 8m	0	Low-moderate	Fragmented sections of watercourse within PDA to be lost
Channelised Watercourse	5.03		1.47	0.58	Low to Moderate	Channelized watercourses will be subject to

Habitats		Areas in PDA (ha)[1]	Areas/ Extent of Habitats in PDA to be Affected (ha)[2]	Areas beyond PDA to be Affected (ha)	Habitat Value	Notes
						proposed decking/ revitalisation and affected by the project.
Pond		0.41	0.41	0	Low to Moderate	-
Inactive Fish Pond		0	0	0	Low to Moderate	-
Dry Agricultural Land (scattered through PDAs)		2.03	2.03	0.03	Low to Moderate	
Dry Agricultural Land (near Shan Ha Tsuen)		7.54	1.35	0	Moderate	Much of this to be zoned AGR-
Wet Agricultural Land (scattered through PDAs)		0.03	0.02	0	Low to Moderate	-
Wet Agricultural Land (near Shan Ha Tsuen)		0.73	0.2	0	Moderate	Much of this to be zoned AGR -
Mixed Dry & Wet Agricultural Land		2.33	2.33	0.01	Low to Moderate	-
Inactive Agricultural Land		1.54	0.19	0.01	Low to Moderate	-
Seasonally Wet Grassland		1.26	1.24	0	Low to Moderate	-
Reedbed		0.28	0.28	0	Low to Moderate	-
Marsh		0	0	0	Low	-
Grassland		0.30	0.15	0	Low	-
Grassland/Shrubland		3.36	1.44	0	Low to Moderate	-
Secondary Woodland	All areas of woodland	8.62	5.49	0.53	-	-
	Woodland near Shan Ha Tsuen	0.63	0	0	Moderate to High	-
	Hillside woodland	3.71	2.42	-	Moderate	-
	Woodland near Lam Hau Tsuen and Lam Hau Wai Tsuen	0.06	0.06	0.19	Low to Moderate	-
	Woodland near Tong Yan San Tsuen	4.20	3.00	-	Low to Moderate	Comprising of numerous planted ornamentals, orchards and remnants of native woodland stands; heavily disturbed and fragmented by village developments in woodland understory and limited ecological value

Habitats		Areas in PDA (ha)[1]	Areas/ Extent of Habitats in PDA to be Affected (ha)[2]	Areas beyond PDA to be Affected (ha)	Habitat Value	Notes
	Other scattered woodland	0.02	0.02	0.34	Low to Moderate	-
	Woodland under Pylon	0	0	0	Low to Moderate	-
Plantation		0.59	0.55	0.30	Low	-
Lychee Woodland		0.06	0	0	Low	-
Orchard		3.73	2.86	0	Low	-
Nursery		0.19	0	0	Low	-
Village Area		16.52	15.45	0.19	Low	-
Wasteland		5.25	4.87	0.99	Low	-
Developed Area		163.75	146.38	13.22	Low	-
Total		**223.54**	**186.71**	**15.86**	-	-

Notes: [1.] Length in metres for watercourses. 2. Preserved areas including "GB", "GB(1)", "AGR" and "Reedbed" has been excluded.

Watercourses

8.6.3.5 Watercourses within the PDA have been retained as far as possible, though there will still be some loss of some wet, lower sections of watercourses of Low and Low-moderate ecological value. For ecological value of those watercourses within the PDA see **Figure 8.3a**.

8.6.3.6 The crab *S. zanklon* has the potential to occur in streams throughout the Assessment Area and it is recommended that where possible lowland streams are retained and enhanced. The Critically Endangered shrimp *C. apodosis* was previously recorded in a mountain stream near Tai Tong village according to Cai & Ng (1999); while the exact location of the stream from which the specimens were obtained is not known and it may already have been incorporated into the urban landscape and be ecologically unsuitable for this species; there is a chance that this species does still occur. This further strengthens the recommendation for retention and enhancement of lowland streams where practicable. Chinese Pond Herons may also utilise small watercourses when foraging, though usually only single birds were observed given the small size of the watercourses.

8.6.3.7 Within Catchment 3, the short section of the watercourse will be retained within the GIC zone despite this habitat being low ecological value. There will be potential for value of this watercourse to improve with change of adjacent land-use and removal of point pollution. Catchment 8 is a very small channel, less than 1m in width and is essentially a drainage ditch passing through disturbed habitats; this Low value habitat will be lost as part of the development.

8.6.3.8 The following impact assessment tables are for those watercourses within the PDA which will be subject to habitat loss.

Table 8.48a Potential Direct Ecological Impacts to Watercourses in Catchments 3 and 8 in the Absence of Mitigation Measures

	Assessment	
Criteria	**Catchment 3** (Sampling point 27)	**Catchment 8** (Sampling point 42)
Habitat Quality	Low.	Low.
Species	Low diversity and abundance of aquatic fauna in both.	
Size/Abundance	Approx. 100m passes through or close to the PDA.	Approx. length of 240m in PDA (<1m wide) which is not wet throughout the year.
Duration	N/a	Permanent.
Reversibility	N/a	Irreversible.
Magnitude	N/a	Loss of a these sections of low quality watercourses would be of low magnitude.
Overall Impact Severity	**Low** as this section of watercourse will be retained and incorporated into the scheme as GIC zone	**Low.**

8.6.3.9 Within Catchment 4, the main channel of the watercourse will be retained with a 15m 'GB (1)' Buffer on either side of the watercourse. The two tributaries will be lost as part of the development of the Reedbed and the Local Open Space (LO).

Table 8.48b Potential Direct Ecological Impacts to Watercourses in Catchment 4 in the Absence of Mitigation Measures

Criteria	**Catchment 4**	
	Main Channel (Sampling point 29 & 31)	**Tributaries** (Sampling point 30 & 40)
Habitat Quality	Moderate to High	Low to Moderate
Species	Low diversity and abundance Endemic crab *S. zanklon* recorded.	Low diversity and abundance Endemic crab *S. zanklon* recorded.
Size/Abundance	Approx. 475m passes through the PDA.	Narrow channels, <1m wide. Approx. length of 185m and 80m, though extent of permanent water appears to be variable.
Duration	N/a	Permanent loss
Reversibility	N/a	Irreversible
Magnitude	N/a	Permanent loss of two tributaries (totaling c.210m) to the reedbed would be of low to moderate magnitude
Overall Impact Severity	**Low – this section will be retained**	**Low to Moderate** where any remaining *S. zanklon* occur

8.6.3.10 The provision of the 575m long Hillside River Corridor and the Retention Lake at the western boundary of the PDA will result in approximately 200m of two lower sections of Catchment 5 being lost. These are short sections of watercourses; whilst they have natural substrate and banks, they are small narrow watercourses with low fauna diversity and abundance. The endemic crab *S. zanklon* was recorded here. However, there is potential for ecological gain with the creation of the Hillside River Corridor and the Retention Lake.

Table 8.48c Potential Direct Ecological Impacts to Watercourse in Catchment 5 in the Absence of Mitigation Measures

Criteria	**Catchment 5** (Sampling point 23)
Habitat Quality	Moderate
Species	Low numbers of common species. Endemic crab *S. zanklon* recorded.
Size/Abundance	Short section of approx. 200m of two narrow watercourses
Duration	Permanent loss
Reversibility	Irreversible for habitat loss.
Magnitude	Permanent loss of two narrow watercourses to the hillside river corridor/retention lake would be of small magnitude.
Overall Impact Severity	**Low to Moderate** where any remaining *S. zanklon* occur.

8.6.3.11 For Catchments 6 and 7, the main channel of the watercourses will be retained with a 15m 'GB (1)' Buffer on either side of the watercourse (total of 30m Buffer).

8.6.3.12 A short section of Catchment 7 will be diverted into a channelised watercourse (Yuen Long (West) Nullah) to maintain hydrological linkages with a continuous open water feature and an ecological enhanced channel. This short section passes through heavily disturbed brownfield sites.

Table 8.48d Potential Direct Ecological Impacts to Watercourses in Catchments 6 and 7 in the Absence of Mitigation Measures

Criteria	**Catchment 6** (Sampling point 37)	**Catchment 7** (Sampling point 35)
Habitat Quality	Moderate to High	Moderate to High for much of the length. The section to be diverted is in the lower reaches of the watercourse passing through developed areas and is subsequently of low ecological value than more upstream sections due to lack of connectivity.
Species	Low diversity and moderate abundance Endemic crab *S. zanklon*, rare fish Small Snakehead and good assemblage of dragonfly larvae recorded.	Low diversity and abundance Endemic crab *S. zanklon* and good assemblage of dragonfly larvae recorded in upstream sections
Size/Abundance	n/a	Short section to be diverted to Yuen Long (West) Nullah
Duration	n/a	Permanent loss
Reversibility	n/a	Irreversible
Magnitude	n/a	Low magnitude
Overall Impact Severity	**Low – this will be retained.**	**Low** as only a short section of the lower section to be redirected into channelised watercourse.

8.6.3.13 Within Catchment 9, the two watercourses passing through the PDA will be piped underground. Both sections are heavily modified and appear to be isolated from upstream sections. The western watercourse is heavily polluted with poor water

quality. The eastern watercourse does have some sections which may be suitable for the endemic crab *S. zanklon*, though this was not recorded during surveys

Table 8.48e Potential Direct Ecological Impacts to Watercourses in Catchment 9 in the Absence of Mitigation Measures

Criteria	**Catchment 9**	
	Western watercourse (Sampling point 15, 20 & 34)	**Eastern watercourse** (Sampling point 18 & 19)
Habitat Quality	Low	Low to Moderate
Species	Low diversity and abundance	Low diversity and abundance
Size/Abundance	Made up of separate isolated sections (65m +39m + 162m +94m + 59m)	Total of approximately 340m.
Duration	Permanent loss	
Reversibility	Irreversible.	
Magnitude	Low magnitude	
Overall Impact Severity	**Low**- numerous fragments of channel with poor water quality.	**Low**–this is generally lower ecological value.

Channelized Watercourses

8.6.3.14 The main drainage channels within the PDA, namely Yuen Long Nullah (~2.5km), Tin Tsuen Channel (~1.5km) and Yuen Long (West) Nullah (~0.5km) are proposed for beautification and landscaping works to enhance the overall amenity of the existing channels. Approximately 1.47ha within PDA and 0.58ha outside PDA of channelized watercourses will be subject to proposed decking/ revitalisation and affected by the Project. Decking of these main drainage channels will be avoided as far as practicable. However, the northern section of Yuen Long Nullah is proposed to be partially decked to provide adequate road space with potential for nullah revitalisation. The proposed decking work will result in minor horizontal encroachment at the deck level on both sides of the channel; however, the existing channel bed width will be retained or slightly widened.

8.6.3.15 The overall ecological impacts on channelised watercourses are considered to be low due to minor encroachment by proposed road works/ beautification works and are likely to be potentially positive through ecological enhancement of existing main drainage channels. Low numbers of ardeids form the Pak Sha Tsuen Egretry appear to utilise channelised watercourses as part of their foraging habitat during the breeding season. **Table 8.49** below summarises potential direct ecological impacts to channelised watercourses based on the RODP:

Table 8.49 Potential Direct Ecological Impacts to Channelised Watercourses in the Absence of Mitigation Measures

Criteria	**Assessment**
Habitat Quality	Low to Moderate
Species	Low numbers of wetland dependent birds recorded including Little Ringed Plover (5), Greater Painted-snipe (1), Chinese Pond Heron (4), Little Egret (12). Numbers in parentheses are total numbers recorded during the Study. Single records of the dead crabs *S. zanklon* and *C. anacoluthon* in the channels relate presumably to individuals being washed down from

Criteria	Assessment
	more suitable stream courses following heavy rains and increased flow.
Size/Abundance	The northern section of Yuen Long Nullah will be partially decked over for road widening works. It will also be revitalised and integrated in the urban design and landscape framework. Other sections of existing channels are subject to revitalization/ beautification works and will be provided with landscape features. No net loss of channel bed is anticipated.
Duration	Temporary loss/ disturbance to the channels during beautification works/ other road works.
Reversibility	Irreversible for the proposed decking-over works Reversible after channel beautification works.
Magnitude	Small
Overall Impact Severity	**Low and potentially positive** due to amenity enhancement.

Ponds

8.6.3.16 A total of five small active farm ponds with sizes ranging from 0.03ha to 0.18ha (~0.41ha in total) will be permanently lost in Area 1 of the PDA. These active farm ponds adjoining Sha Tseng Tsuen are entirely man-made ponds for irrigation. The small managed pond (shallow pond) (Pond #17) will be permanently converted into residential development in Area 2 of PDA. Overall ecological impacts resulting from loss of these ponds are summarised in **Table 8.50**.

Table 8.50 Potential Direct Ecological Impacts to Ponds in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low to Moderate.
Species	Moderate floristic diversity and faunal diversity. May be used by foraging Ardeids throughout the year.
Size/Abundance	0.41ha (a total of five farm ponds ranging from 0.03ha to 0.18ha in Area 1 and a small managed pond of ~0.04ha in Area 2 of PDA).
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low**.

Agricultural Lands

8.6.3.17 Approximately 9.57ha of dry agricultural land are located within the PDA, of which ~6.19ha will be preserved in the area zoned as "Agriculture" or "Green Belt (1)" in the RODP. Approximately 3.38ha of dry agricultural land will be permanently lost in the RODP. Some minor fringe areas of dry agricultural lands (~0.03ha) outside of PDA will also be lost due to road works.

8.6.3.18 There is ~0.22ha of wet agricultural land, ~2.33ha of mixed dry and wet agricultural land, and ~0.19ha of inactive agricultural land that will be lost due to proposed building development and infrastructural works. The impact of the overall loss of agricultural land (approximately 6.24ha) is considered to be minor, as the areas to be lost are small, fragmented and scattered and were not found to support large populations of species of conservation importance. **Tables 8.51** to **8.54**

summarise the potential direct ecological impacts to various types of agricultural lands within the Assessment Area.

Table 8.51 Potential Direct Ecological Impacts to Dry Agricultural Land in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Moderate (for those near Shan Ha Tsuen) Low to Moderate (for others scattered through PDA)
Species	Moderate floristic diversity and faunal diversity. Generally low abundance of bird species of conservation importance (Maximum count in parentheses); Chinese Pond Heron (3), Eastern Cattle Egret (3), Little Egret (2), Zitting Cisticola (1), White-cheeked Starling (8), Red-throated Pipit (1).
Size/Abundance	~3.38ha in PDA + ~0.03ha outside PDA
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low** (for scattered agricultural lands). **Low** (for loss of the eastern edge of agricultural land near Shan Ha Tsuen where ecological linkages are limited and these areas are subject to higher human disturbances).

Table 8.52 Potential Direct Ecological Impacts to Wet Agricultural Land in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Moderate (for those near Shan Ha Tsuen). Low to Moderate (for others scattered through PDA).
Species	Support common flora and fauna species and ardeids in low numbers throughout the year.
Size/Abundance	~0.22ha.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Table 8.53 Potential Direct Ecological Impacts to Mixed Dry & Wet Agricultural Land in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low to Moderate (small and scattered).
Species	Support common flora and faunal species and ardeids in low numbers
Size/Abundance	~2.33ha in PDA and ~0.01 outside PDA
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Table 8.54 Potential Direct Ecological Impacts to Inactive Agricultural Land in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low to Moderate (small and scattered).
Species	Dead specimens of Leschenault's Rousette, Short-nosed Fruit Bat were found in this habitat type; however these species are highly mobile and widespread throughout the Assessment Area.
Size/Abundance	~0.19ha in PDA and ~0.01 outside PDA.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Seasonally Wet Grassland

8.6.3.19 Approximately 8.07ha of seasonally wet grasslands were identified within the Assessment Area, including ~1.26ha located within PDA. These seasonally wet grasslands are subject to high level of human disturbance from adjacent actively managed agricultural lands (e.g. those in Sha Tseng Tsuen) or isolated by existing village settlements. During the 2013 ardeid breeding season, only one Little Egret was recorded from this area during surveys. At least 1.24ha of these seasonally wet grasslands will be cleared for development in the PDA. Overall ecological impacts resulting from loss of this habitat type is considered to be low as presented in **Table 8.55**.

Table 8.55 Potential Direct Ecological Impacts to Seasonally Wet Grassland in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low to Moderate (semi-natural but highly disturbed and/or isolated).
Species	Low floristic and faunal diversity.
Size/Abundance	~1.24ha.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Reedbeds

8.6.3.20 The reedbed areas (~0.28ha in total) to be lost in the RODP are small and isolated by existing village developments and supports low floristic and faunal diversity. Overall ecological impacts resulting from loss of this reedbed are summarised in **Table 8.56** below.

Table 8.56 Potential Direct Ecological Impacts to Reedbed in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low to Moderate (semi-natural but highly disturbed and/or isolated).
Species	Low floristic and faunal diversity.
Size/Abundance	~0.28ha.

Criteria	Assessment
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Grassland

8.6.3.21 An area of grassland is located within the Assessment Area, and a small portion of if (lowland grassland within PDA) will be encroached upon by proposed building developments. This habitat is considered of low floristic and faunal diversity and is highly disturbed in lowland areas within existing village developments. Overall impacts to this habitat are considered to be negligible as presented in **Table 8.57**.

Table 8.57 Potential Direct Ecological Impacts to Grassland in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low.
Species	Low floristic and faunal diversity.
Size/Abundance	~0.15ha.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Very Small.
Overall Impact Severity	**Negligible.**

Grassland/ Shrubland

8.6.3.22 Approximately 1.44ha of the lowland shrubby grassland along the lower slopes of hillsides within the Assessment Area will be lost. This semi-natural habitat supports common and widespread species of low to moderate floristic diversity. Overall ecological to this habitats are summarised in **Table 8.58**.

Table 8.58 Potential Direct Ecological Impacts to Shrubland/ Grassland in the Absence of Mitigation Measures.

Criteria	Assessment
Habitat Quality	Low to Moderate (semi-natural habitat).
Species	Low floristic and faunal diversity.
Size/Abundance	~1.44ha.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Secondary woodland

8.6.3.23 Approximately 58.67ha of secondary woodland was identified in the Assessment Area, including ~8.62ha within the PDA and ~0.53ha in proposed infrastructural/ road works area outside of the PDA. Of the ~8.62ha woodland in the PDA,

approximately 3.13ha separating the retained natural watercourses from the proposed development will be retained as green buffers and preserved within "GB", "GB(1)" ,"AGR" or the proposed reedbed areas. Approximately 5.49ha of secondary woodlands within PDA (either woodland fringe areas or woodland fragments within PDA) will be permanently lost due to the Project. Approximately 0.53ha of minor woodland fringes outside of PDA will be encroached by proposed road works.

8.6.3.24 Overall ecological impacts resulting from the cumulative loss of secondary woodland fragments/ fringe areas in RODP are summarised in **Table 8.59** below.

Table 8.59 Potential Direct Ecological Impacts to Secondary Woodland in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low to Moderate (~3ha woodland fragments near Tong Yan San Tsuen (comprising of numerous planted ornamentals, orchards and remnants of native woodland stands; heavily disturbed and fragmented by village developments in woodland understory and limited ecological value) and ~0.08ha other woodland fringes) Moderate for ~2.42ha hillside woodlands (only lowland woodland fringes will be affected)
Species	Several seedlings of *Aquilaria sinensis* (protected by Cap. 586) were recorded in woodland adjoining Shan Ha Tsuen but located away from the PDA.
Size/Abundance	~5.49ha in PDA; ~0.53ha outside of PDA).
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Moderate** for loss of ~2.42ha hillside woodland. **Low** for loss of numerous, disturbed woodland fragments (combined total of ~3.08ha). **Low** for loss of other woodland fringes (~0.53ha) outside PDA.

Plantation

8.6.3.25 Approximately 0.59ha of plantation woodland were identified within PDA, of which ~0.04ha will be retained in "GB" in the RODP. Overall total of ~0.55ha of plantation within PDA would be permanently lost in the RODP. A small area of plantation (~0.3ha) near the south-western end of Long Hon Road is reserved for the proposed Reclaimed Water Service Reservoir. Plantation recorded in this Study comprised mainly exotic species in the overstorey and lacked significant native regeneration in the understory in most of the upland areas with poor soil. The overall ecological impacts resulting from loss of plantation are considered low as summarised as **Table 8.60**.

Table 8.60 Potential Direct Ecological Impacts to Plantation in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low (lowland fragmented plantation and some upland plantation with sparse understory).
Species	Dominated by exotic tree species and lack of native species in the understory in most upland areas; support low faunal diversity.

Criteria	Assessment
Size/Abundance	~0.55ha in PDA; ~0.3ha outside of PDA (for the proposed Reclaimed Water Service Reservoir).
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Orchards

8.6.3.26 Approximately 2.86ha of orchards within PDA will be permanently converted into road, Amenity Area and Other Specified Use and Residential uses. As this habitat is entirely man-made and comprises mainly fruit trees subject to a high level of human disturbance from active management, the resulting ecological impacts due to loss of orchards are considered to be low as presented in **Table 8.61**.

Table 8.61 Potential Direct Ecological Impacts to Orchards in the Absence of Mitigation Measures.

Criteria	Assessment
Habitat Quality	Low.
Species	Dominated by exotic/ ornamental/ fruit tree species and lack of native species in the understory; support low faunal diversity. Two trees of *Artocarpus hypargyreus* were recorded at the orchard near the western boundary of Area 1. This species is regarded as "Vulnerable" in China Red Data Book (Pang et al. 2011) and IUCN (IUCN 2017), and "Near Threatened" by AFCD (2003). However, it is commonly found in lowland forest and not locally protected.
Size/Abundance	~2.86ha.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Village Area

8.6.3.27 Approximately 242.23ha of village areas were located in the Assessment Area, including ~15.45ha within the PDA and ~0.19ha to be encroached upon by proposed infrastructural/ road works outside of the PDA. These village areas support moderate diversity of plant species of low floristic value. The habitat is also subject to high levels of human disturbance and supports low faunal diversity in the Assessment Area. The resulting ecological impacts due to loss of village areas are considered to be low, as presented in **Table 8.62**.

Table 8.62 Potential Direct Ecological Impacts to Village Area in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low (man-made habitat and frequently disturbed).
Species	Support moderate diversity of plants species of low floristic value;

Criteria	Assessment
	support low fauna diversity.
Size/Abundance	~15.45ha within PDA; ~0.19ha outside of PDA.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low.**

Wasteland

8.6.3.28 Approximately 76.50ha of wasteland were identified in the Assessment Area, including ~5.25ha within the PDA and ~0.99ha within proposed footprint of infrastructural/ road works outside of the PDA. Most areas of the wasteland within the PDA (~4.87ha) will be permanently converted to other land uses in the RODP. This habitat type is derived from overgrown fallow fields and is dominated by weedy vegetation and is considered of low ecological value. The overall ecological impacts resulting from conversion of wasteland to other developments are considered to be low and could be potentially positive, taking into account the potential for ecological enhancement through re-vegetation in new developments.

Table 8.63 Potential Direct Ecological Impacts to Wasteland in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low (man-made habitat and frequently disturbed).
Species	Overgrown with weedy vegetation. One tree of *Artocarpus hypargyreus* was recorded at the wasteland near the western boundary of Area 1. This species is regarded as "Vulnerable" in China Red Data Book (Pang et al. 2011) and IUCN (IUCN 2017), and "Near Threatened" by AFCD (2003). However, it is commonly found in lowland forest and not locally protected.
Size/Abundance	~4.87ha within PDA; ~0.99ha outside of PDA.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Small.
Overall Impact Severity	**Low and potentially positive.**

Developed Area

8.6.3.29 The majority (~163.75ha) of the PDA belongs to developed area, of this ~146.38ha will be re-developed in the RODP. There will be ~13.22ha of developed area to be encroached upon by proposed infrastructural/ road works outside of the PDA. As this habitat is entirely man-made and disturbed/ developed area and supports low floral and faunal diversity of common urban species, redevelopment in this habitat type is considered to result in negligible impacts as summarised in **Table 8.64** below.

Table 8.64 Potential Direct Ecological Impacts to Developed Area in the Absence of Mitigation Measures

Criteria	Assessment
Habitat Quality	Low (man-made habitat and frequently disturbed).
Species	Supports low flora and faunal diversity. A mature and planted specimen of the rare tree species *Ailanthus fordii* was recorded on Tong Yan San Tsuen Road within existing rural industrial area and within PDA. The wild population of this tree species is considered "rare" in Fellowes et al. (2002) and protected by Cap. 96A. Approximately 10 young saplings of the protected tree species *Aquilaria sinensis* (protected under Cap. 586) were found planted in ground in a small private nursery near the southern end of Lam Tei West Road (Figure 8.5f); assumed that these have been planted/cultivated.
Size/Abundance	~146.38ha within PDA; ~13.22ha outside of PDA.
Duration	Permanent loss.
Reversibility	Habitat loss would be irreversible.
Magnitude	Negligible.
Overall Impact Severity	**Negligible.**

Direct Impacts on Species of Conservation Importance

Flora mortality

8.6.3.30 Except for the mature tree specimen of *Ailanthus fordii* identified in the industrial area of Area 1 of the PDA, three mature tree specimens of *Artocarpus hypargyreus* identified in orchard or wasteland areas of Area 1 in PDA, and the approximately 10 young saplings of *Aquilaria sinensis* planted in the private nursery located in the developed area near the southern end of Lam Tai West Road (within the proposed road connecting Area 1 and Areas 2/3 of PDA), all other recorded plant species of conservation importance are outside of the PDA or any associated works areas and will not be affected by the proposed works of the RODP. The planted individual of *A. fordii* and the *A. sinensis* saplings in the nursery are all cultivated specimens and can be transplanted to suitable receptor site(s) within the PDA if they will be affected by the proposed works. The three specimens of the possibly naturally established *A. hypargyreus* shall be avoided/ retained through careful design of development layout at the detailed design phase as far as practicable. If on-site retention is not practicable, these specimens of species of conservation importance should be transplanted to suitable receptor site(s) subject to the detailed design at a later stage of the Project.

Fauna mortality

8.6.3.31 Direct mortality involves the death of organisms as a result of the development. Animals may be hit and killed or injured by rapidly-moving vehicles or by collision with stationary objects such as buildings or noise barriers. Birds and mammals appear to be most susceptible, though herpetofauna are also prone. Should roads pass through areas of high animal population density, or cut across regular lines of movement, such mortality is likely to be greater. The risk of animal mortality arising from road kill and collision with buildings, windows or transparent noise barriers is likely to be greater in rural areas than in already developed, urban areas, as wildlife populations are higher in the former.

8.6.3.32 None of the road infrastructure within the PDA passes through habitats where animals, in particularly mammals and herpetofauna, occur in large densities. The Proposed Road D1 passes through the agricultural land at Shan Ha Tsuen. Herpetofauna species of conservation importance, namely Reeve's Terrapin, Buff-striped Keelback and Common Rat Snake that were recorded from this habitat are at low density As such, it is not anticipated that fauna mortality through road kill will be a significant impact to the populations of these species at a local or regional level. Collision mortality on birds from buildings or other barriers is potentially of significance from the Project. The height of some buildings in the Project Area will be substantial and, as modern buildings tend to have large areas of glass, avian mortality is possible. Much depends on the nature of the glass, however, with transparent or highly reflective surfaces causing greater mortality, especially if adjacent to dense areas of shrubs or trees.

8.6.3.33 Bats are not significantly impacted by most potential causes of direct mortality potentially arising from the Project; however, they may be vulnerable to direct mortality from destruction of roost sites, especially those occupied by nursing females or by winter bat roosts. Whilst no bat roosts were recorded during surveys, bats will readily utilise buildings, holes in trees and fronds of Fan-palms to form roost sites. Potential direct ecological impact (mortality) on roosting bats is considered to be Low to Moderate severity.

8.6.3.34 Nesting birds may be subject to direct mortality during vegetation clearance and site formation. Different species of birds utilise different vegetation and built structures when nesting. Potential direct ecological impact (mortality) on breeding birds is considered to be Low to Moderate severity.

8.6.3.35 Aquatic fauna of conservation importance that are known to occur within the Assessment Area, including Small Snakehead and endemic crabs, may be subject to direct mortality through loss of watercourse habitats. Whilst better quality watercourses have been retained, in some sections of watercourses that are to be lost where these species could occur, potential direct ecological impact (mortality) on these fauna is considered to be of Low to Moderate severity.

8.6.3.36 Other fauna of conservation interest are less likely to be subject to direct mortality as a result of this Project, on account of either low density of occurrence, habitat preference or high mobility and impacts are not considered to be significant.

Direct Impacts on Sites of Conservation Importance (Tai Lam Country Park and Yeung Ka Tsuen EIS)

8.6.3.37 Tai Lam Country Park and Yeung Ka Tsuen EIS are located immediately outside the southern boundary of Zone 3 of PDA. No habitat loss of Tai Lam Country Park or the Yeung Ka Tsuen EIS is anticipated as it is completely outside of the boundary of PDA or any proposed works areas. As such, no direct ecological impacts are anticipated on either Tai Lam Country Park or Yeung Ka Tsuen EIS.

Indirect Impacts

Fragmentation of Agricultural Land

8.6.3.38 Potential indirect impacts of the Project upon habitats include fragmentation of agricultural land due to the proposed Road D1 joining Area 1 and Area 3 which will run through the eastern edge of this habitat behind Shan Ha Tsuen. This impact is considered of low severity as the proposed road alignment in the RODP

has been reviewed and revised from previous proposals to retain the integrity of the mosaic of habitats within this area and maintains linkages with more natural, less disturbed habitats with higher ecological value to the west and south, namely Grassland/Shrubland, Watercourses and Secondary Woodland. The proposed road in the current RODP will only pass through the eastern edge where ecological linkages are limited and these areas are subject to higher human disturbances. The resulting loss of agricultural land has been minimized and the fragmentation impact on this habitat is considered to be low.

Impacts to flight lines and foraging habitats of breeding ardeids

8.6.3.39 The majority of the breeding ardeids at the Tai Tong (Pak Sha Tsuen) egretry were recorded to fly to habitats to the north, northeast and east of the egretry to unknown foraging grounds beyond the 500m Study Area of this Project.

8.6.3.40 A significant NBA, designated as 'DO' and approximately 100m wide has been incorporated into the RODP to the north of the egretry. This will maintain linkages with the channelised watercourse which runs directly north. Ardeids are known to follow waterways. A flight corridor to the agricultural land and secondary woodland of Shan Ha Tsuen to the northwest is maintained through 'LO' and 'DO' allocated land. Within these areas, no buildings above 3-storeys high are permitted and the land will be used for recreational uses (i.e. Local Park).

8.6.3.41 The layout of the RODP will impact on flightlines to the northeast, approximately 100-250m from the egretry, where tall residential buildings are proposed.

8.6.3.42 Flightlines to the west and south on the egretry, whilst of lesser significance to breeding birds, maintain linkages through a NBA corridor and road to the west and a combined road and NBA corridor to the south.

8.6.3.43 Flightlines directly to the east of the egretry remain unaffected by the development.

8.6.3.44 It is considered that for flightlines to the north, south, east and west of the egretry, there are no significantly impacts.

8.6.3.45 For flightlines to the northeast of the egretry, whilst there will be some minor adjustments to the departure direction of breeding ardeids from the egretry, it is considered that there is sufficient space between the egretry and the buildings for birds to make minor adjustments at the start of the flightline and still continue to foraging grounds apparently outside of the Assessment Area with minimal disturbance. Birds still have the ability to head east and then divert onto a north-eastern trajectory or fly north from the egretry and then divert to the northeast through the 'DO' spaces, 500m to the north. It should be noted that the current land use, i.e. industrial storage units, may already cause periodic disturbances to flightlines due to the operational activities of these units through the use of tower cranes and long-boomed vehicles. As such, the impacts to these flightlines to the northeast are considered to be low given the amount of open areas still available to the low numbers of birds utilising these flightlines.

Disturbance to breeding ardeids

8.6.3.46 The Tai Tong (Pak Sha Tsuen) egretry is over 120m from proposed site formation and/or construction works; thus these are unlikely to affect this egretry and, as such, indirect impacts are considered to be low, if conducted during the breeding season.

8.6.3.47 Furthermore, given the paucity of records of foraging ardeids using suitable habitats within the PDA during the breeding season, disturbance impacts to egrets foraging in these habitats as a result of this Project are not considered to be significant.

Disturbance to other habitats and species of conservation importance

8.6.3.48 Other potential indirect impacts upon other habitats (including Tai Lam Country Park and Yeung Ka Tsuen EIS) and species of conservation importance may occur through increased disturbance (including human activity, noise, vibration, glare or dust), during the construction and operational phase. However, most recorded species of conservation importance recorded within the Project Area were recorded in low numbers not considered significant in a Hong Kong context. Furthermore, many of these species are highly mobile and/or tolerant of human disturbance. Any resulting impacts are unlikely to have a significant effect on population numbers, therefore the impact is considered as being negligible.

Impacts to water quality, sedimentation of watercourses, hydrology and/or aquatic fauna

8.6.3.49 Impacts during the construction phase could generate surface run-off containing lubricants, chemicals and pollutants, the water bodies of most concern being the retained watercourses within the RODP. Construction run-off is potentially destructive to aquatic communities.

8.6.3.50 Dust and exposed earth from construction operations may also enter the waterbodies via run-off, particularly during periods of heavy rain, or be wind-blown. This can lead to high turbidity from soil particles (which can block the gills of aquatic organisms) and eutrophication as a result of nutrient enrichment. Aquatic macrophytes may be reduced as a result of reduced light penetration or increased free-floating algae populations following eutrophication. Severe eutrophication can lead to oxygen depletion and the impoverishment of aquatic communities (including *S. zanklon* and Small Snakehead) as well as animals that prey on them (e.g. waterbirds). Such effects are usually the greatest in the construction phase of a project. Such indirect impacts are considered to be of Low to Moderate severity.

8.6.3.51 Aquatic fauna of conservation importance that are known to occur within the Assessment Area, including Small Snakehead and endemic crabs, may be subject to mortality indirectly as a result of impacts to the hydrology of the retained watercourses. Such indirect impacts are considered to be of Low to Moderate severity.

8.6.3.52 There are potential indirect impacts to the hydrology and water quality of Deep Bay if the watercourses passing through the assessment area and which subsequently feed into Deep Bay become polluted or are impacted hydrologically. However given the distance between the assessment area and Deep Bay and the low scale of work required in the proximity of watercourses and the mitigation proposed, it is considered any impacts to water quality or hydrology of this system is likely to be small or negligible during the construction phase of this Project. This is further discussed in **Section 6.7.2.**

8.6.4 Cumulative Impacts

8.6.4.1 The potential cumulative ecological impacts that may arise from the concurrent projects are summarised in the following Table. Other concurrent projects as described in Chapter 1 and illustrated on **Figure 1.3**.

8.6.4.2 The key concurrent project which may generate cumulative ecological impacts during construction and operation phases in the vicinity is the Hung Shui Kiu New Development Area (HSK NDA) Planning and Engineering Study affects. According to the HSK NDA EIA, major ecological impacts have been avoided as the Project Area and boundary have evolved and adjusted. Approximately 90% of the Project area is of habitats considered to be of Low ecological value. Approximately 45% of the development area is brownfield sites; it is anticipated that modernisation of these areas will improve the general quality of the environment.

8.6.4.3 Riverside channels are proposed to be revitalised and ecological enhancement measures are proposed through landscape planting and provision of foraging habitat for breeding ardeids in flood retention ponds. There is the potential for an ecological gain from these projects cumulatively.

8.6.4.4 Given the habitat loss impacts from YLS development is predicted to be similar i.e loss of brownfield sites of low ecological value, and ecological enhancement promoted throughout the project, thus, no negative cumulative impacts are predicted.

8.6.4.5 The Project would not generate a net increase in pollution loading to the receiving water in the Deep Bay WCZ and North Western WCZ as discussed in Section 6.7.2. Hence, the cumulative impact with other concurrent projects is not anticipated.

Table 8.65 Potential impacts of concurrent projects

Concurrent Project	Project Proponent	Construction due to commence	Operation/Completion of development	Potential Cumulative Impacts	
				Construction Phase	Operational Phase
Hung Shui Kiu New Development Area (HSK NDA) Planning and Engineering Study	CEDD	2019	2024 (First year of population intake)	Low	Low Potential ecological gain

8.6.5 Summary of Ecological Impacts

8.6.5.1 Potential ecological impacts arising from the construction and operation of the Project and whether mitigation is required are summarised in **Table 8.66** below. Impacts assessed as "Low" or "Negligible" are not considered to require mitigation.

Table 8.66 Summary of Potential Ecological Impacts based on the RODP

Potential Impact	Avoidance and Minimisation of Design measures	Impact Severity	Further Ecological Mitigation Proposed
Permanent Direct Impacts on Habitats			
Watercourses			
Catchment 3 (Sampling point 27)	Avoided - within GIC zone	Low	No
Catchment 4 (Main Channel – sampling points 29 &31)	Avoided - GB(1) Buffer	Low	No
Catchment 4 (Tributaries - sampling points 30 & 40)	-	Low - Moderate	Yes – Hillside River Corridor. Enhancement through Reedbed.
Catchment 5 (Sampling point 23)	-	Low - Moderate	Yes – Hillside River Corridor. Enhancement through Retention Lake.
Catchment 6 (Sampling point 37)	Avoided - GB(1) Buffer	Low	No
Catchment 7 (Sampling point 35)	Avoided - GB(1) Buffer	Low	No Enhancement through diversion of lower section.
Catchment 8 (Sampling point 42)	-	Low	No[1]
Catchment 9 (western watercourse - sampling points 15, 20 & 34)	-	Low	No[1]
Catchment 9 (eastern watercourse - sampling points 18 & 19)	-	Low	No[1]
Channelised Watercourses	-	Low	No mitigation Enhancement through revitalization
Pond	-	Low	No
Dry Agricultural Land	The integrity of the mosaic habitats and ecological linkages to the west and south of Shan Ha Tsuen have been avoided by careful design of Road D1	Low	No
Wet Agricultural Land		Low	No
Mixed Dry & Wet Agricultural Land		Low	No
Inactive Agricultural Land	-	Low	No
Seasonally Wet Grassland	-	Low	No
Reedbed	-	Low	No
Grassland	-	Negligible	No
Shrubland/	-	Low	No

Potential Impact	Avoidance and Minimisation of Design measures	Impact Severity	Further Ecological Mitigation Proposed
Grassland			
Secondary Woodland	Woodland near Shan Ha Tsuen of Moderate to High Ecological value has been avoided; Loss of woodlands has been minimized through retention of woodland of significance sizes in GB zone	**Moderate** for loss of ~2.42ha hillside woodland	Yes
	-	**Low** for loss of other scattered woodlands/ woodland fringes	No
Plantation	-	Low	No
Orchard	-	Low	No
Village Area	-	Low	No
Wasteland	-	Low and potentially positive	No
Developed Area	-	Negligible	No
Permanent Direct Impact on Species			
Alianthus fordii (planted/ potted)	-	Low	No – precautionary measures could be adopted to transplant this specimen.
Aquilaria sinensis (planted seedlings/ saplings in ground)	-	Low	No – precautionary measures could be adopted to transplant the affected specimens.
Artocarpus hypargyreus (possibly naturally established)	-	Low	No - precautionary measures could be adopted to retain the specimens on site through careful design of layout or to transplant the affected specimens subject to the detailed design at a later stage
Fauna mortality			
Collision mortality – Mammals	-	Low	No
Collision mortality – Birds	-	Low - Moderate	Yes
Collision mortality – herpetofauna	-	Low	No
Bat mortality in roosts	-	Low - Moderate	Yes
Nesting Birds	-	Low - Moderate	Yes
Aquatic Fauna (habitat loss)	-	Low - Moderate	Yes - Translocation of native species to new watercourses
Other species of Conservation importance	-	Low	No

Potential Impact	Avoidance and Minimisation of Design measures	Impact Severity	Further Ecological Mitigation Proposed
Direct Impacts on Sites of Conservation Importance			
Tai Lam Country Park	Avoided	Negligible	No
Yeung Ka Tsuen EIS	Avoided	Negligible	No
Indirect Impacts			
Fragmentation of agricultural land adjoining Shan Ha Tsuen	-	Low	No
Indirect impacts to flightlines and foraging habitats	Avoided and minimised	Low	No
Disturbance impacts to egretries	-	Low	No
Disturbance to other habitats or species of conservation importance	-	Low - Negligible	No
Impacts to water quality, sedimentation of watercourses and/or aquatic fauna	-	Low - Moderate	Yes
Impacts to the hydrology of the retained watercourses	-	Low - Moderate	Yes - Translocation of native species to new watercourses
Impacts to the hydrology and water quality of Deep Bay	-	Low - Negligible	No
Cumulative Impacts			
Cumulative ecological impacts resulting from other concurrent/ planned projects in the vicinity	-	Low	No

Note: [1] Whilst mitigation is not required for this loss of, or indirect impacts to, these watercourses, it should be noted that surveys and translocation of aquatic fauna (notably the endemic crab *S. zanklon*) will be conducted prior to any habitat loss of these watercourses.

8.6.6 Mitigation Measures

Mitigation of Impacts of Loss of Hillside Secondary Woodland

8.6.6.1 For impacts as listed in **Table 8.66**, the resulting ecological impacts from loss of scattered woodlands/ woodland fringe areas (all are considered of Low-to-Moderate ecological value) are considered to be low given their small size, isolated nature, and lack of connectivity with other ecologically valuable habitats, diversity and highly disturbed nature. Only the impact resulting from the cumulative loss of hillside secondary woodlands (~2.42ha in total) is considered of moderate significance and mitigation is considered as necessary. Given the limited space

within the RODP, on-site woodland compensation is considered not practicable. In view of the urban design of the entire YLS development, discrete areas of district open space and local open space have been assigned among the development parcels and comprehensive woodland compensation is not preferred as it will lose the function of connectivity and compatibility. As such, off-site mitigation measure is the only solution to compensate for the loss of hillside secondary woodland. The proposal for off-site woodland compensation has been developed in accordance with Clauses 5.4.3 to 5.4.5 of Annex 16 of the EIAO TM as presented in **Appendix 8.9**.

8.6.6.2 Loss of hillside secondary woodlands is to be compensated by woodland planting of at least 1:1 in terms of area ratio in the adjacent hillside area to the west of the PDA as shown in **Figure 8.10**. The proposed woodland compensation site is approximately 12ha in the hillside area currently zoned as "Conservation Area". The proposed planting site is not uniformly sloping. It includes some steeper areas with large boulders and some areas along the ridges are more gently sloping (slope gradient <35˚) that are considered suitable for tree planting. The proposed site has been sparsely planted with exotic plantation species of *Pinus elliottii*. The general ground area is sparsely covered with shrubby grassland species such as the fern species *Dicranopteris pedata* and the shrub species *Baeckea frutescens.* The site is considered to be moderately eroded and exposed. As the proposed planting site is located between burial grounds, the planting design should consider establishment of firebreaks consisting of a non-planting zone and a tree belt densely planted (spacing <1m) with fast growing tree species to slow down or stop the progression of hill fire that can potentially impact the planting area. The design should also include a management regime including but not limited to ground preparation works such as clearance of excessive ground herbs, fertilization application, weeding and mulching after planting.

8.6.6.3 Plant species should be selected to include a mix of species with pioneering characteristics (fast-growing/ light-tolerant/ drought-tolerant/ wind-tolerant, etc.) and native species. The planting design should avoid intense competition between native species and exotic species otherwise the former would be out-competed by the latter. In general, fast-growing exotic species such *Melaleuca cajuputi* subsp. *cumingiana* and *Acacia* sp. can be planted along the most severely eroded ridges and wind-exposed areas. Native species should be planted in groups instead of inter-planting between exotic species to minimize competition between exotic and native species. Based on the *Guiding Principles on Use of Native Plant Species in Public Works Projects (Appendix C)* published by Greening, Landscape and Tree Management Section of Development Bureau in October 2010, a list of species with pioneering characteristics for eroded hillside areas and native species of hillside planting are recommended for the proposed woodland compensation planting as displayed in **Table 8.67** below. Woodland mix species shall be planted at an area ratio of at least 1:1 within the proposed planting site (~12ha). Shrub species should be planted at spacing of 600mm – 1,000mm and tree species at spacing of 1,000mm to 1,500mm. Depending on the latest site conditions, the planting design should also explore the applicability of planting in phases, including an initial phase to establish greenery cover using fast-growing pioneer species, and an enhancement phase during which selective thinning of planted exotic species will be carried out as appropriate to allow enrichment planting of native woodland mix species.

Table 8.67 Recommended List of Plant Species for Woodland Compensation

Recommended List of Exotic Species with Pioneering Characteristics for Eroded Hillside Area	
Species (Botanical Name)	**Size/ Standard**
Acacia auriculiformis	Whip
Acacia confusa	Whip
Acacia mangium	Whip
Casuarina equisetifolia	Whip
Lophostemon confertus	Whip
Melaleuca cajuputi subsp. *cumingiana*	Whip
Schima wallichii	Whip
Recommended List of Native Plant Species for Woodland Compensation*	
Aporusa dioica	Whip/ Seedling
Bridelia tomentosa	Whip/ Seedling
Catanopsis fissa	Whip/ Seedling
Cyclobalanopsis edithiae	Whip/ Seedling
Polyspora axillaris	Whip/ Seedling
Ilex rotunda var. *microcarpa*	Whip/ Seedling
Litsea cubeba	Whip/ Seedling
Litsea glutinosa	Whip/ Seedling
Machilus breviflora	Whip/ Seedling
Machilus chekiangensis	Whip/ Seedling
Machilus thunbergii	Whip/ Seedling
Mallotus paniculatus	Whip/ Seedling
Melastoma malabathricum	Whip/ Seedling
Melastoma sanguineum	Whip/ Seedling
Ormosia emarginata	Whip/ Seedling
Phyllanthus emblica	Whip/ Seedling
Reevesia thyrsoidea	Whip/ Seedling
Rhaphiolepis indica	Whip/ Seedling
Rhodomyrtus tomentosa	Whip/ Seedling
Sapium discolor	Whip/ Seedling
Schefflera heptaphylla	Whip/ Seedling
Schima superba	Whip/ Seedling
Sterculia lanceolata	Whip/ Seedling
Syzygium hancei	Whip/ Seedling
Viburnum odoratissimum	Whip/ Seedling

*Notes: Core species that are selected to formulate the plant mix for planting under various site conditions and commonly used in projects (DevB 2010).

8.6.6.4 Before commencement of construction works of the Project, a detailed Woodland Compensation Plan shall be prepared by a qualified botanist/ plant ecologist with relevant experience in reforestation and shall be submitted and agreed with the relevant authorities. The Plan shall be developed based on the latest baseline condition of the proposed woodland planting site prior to commencement of the Project. The Plan shall include proposals on site preparation works, planting design and layout, planting period (early wet season), planting methodology, site supervision of planting, post-planting monitoring, replanting/ replacement of

unsuccessful specimens and maintenance programme. Planting work shall be conducted in the early wet season (i.e. March/ April), and should be supervised by a qualified plant ecologist/ botanist/ Certified Arborist with relevant experience in restoration.

Mitigation for Watercourse Loss

8.6.6.5 To mitigate for the cumulative loss of watercourses in Catchment 4 and Catchment 5 (total length ~465m) which are generally isolated and the loss of these watercourses is considered to result in Low-moderate ecological impacts, it is proposed that a new watercourse be created along the hillside of the western boundary of PDA. This will be approximately 575m long. By intercepting several hillside streams, this new watercourse will provide hydrological and ecological connectivity to downstream sections.

8.6.6.6 The watercourses will be zoned as 'Other Specified Uses (Hillside River Corridor with Scenic Cycle Track') ('OU'). The new watercourse will be designed to maximize ecological opportunities for aquatic flora and fauna and provide new linkages to several fragmented/isolated watercourses.

8.6.6.7 Its channel bed and banks should be designed to replicate those streams that it is fed by, to provide suitable habitats for native species which are likely to occur here. Natural substrates will be used as far as practicable to facilitate natural succession and colonisation of native flora and fauna. In-channel modifications to create microhabitats are recommended; this can be achieved by adding different substrates that can provide a diverse environment with different biotic and abiotic components for different species which further enhance channel complexity (DSD 2015). Conceptual cross sections of the channel can be seen in Figure 8.11a. Within the channel, inclusion of riffles or short cascades can be implemented to allow for changes in elevation as the watercourse flows downstream.

8.6.6.8 A weir will be installed at the point at which the watercourse meets the Retention Lake in order to try to reduce the likelihood of the upstream section being colonised by exotic fish species, in particular Tilapia.

8.6.6.9 At the detailed design stage, a Hillside River Corridor Management Plan (HRCMP) will be prepared by a qualified ecologist with full details of the design, construction methodology, and hydrology; along with an implementation programme for creation, establishment, monitoring and maintenance programme. Specific reference should be made to DSD Guidance Notes (DSD 2015) and other international practices to maximise wildlife opportunities for this habitat creation.

8.6.6.10 Within the HRCMP, consideration of the hydrological linkages of other watercourses within the Assessment Area is required in order to ensure that this new watercourse does not impact water supplies of other watercourses of significance. This issue should be further assessed in the detailed design stage

8.6.6.11 The Plan should be submitted to and approved by EPD prior to commencement of works in areas affecting watercourses. The approved habitat creation will be supervised by a qualified ecologist with relevant experience in wetland creation.

Mitigation for Disturbance to Aquatic Fauna of Conservation Importance

8.6.6.12 Mitigation for impacts on streams and stream-associated fauna (including the fish Small Snakehead and the endemic crab *S. zanklon*) is primarily avoidance and implementations of a suitable buffer which can be further enhanced and maintain

the integrity of the riparian corridor. The buffer can have a multiple function role, minimising impacts to the stream, providing additional and enhanced habitats to support higher populations of fauna thus maintaining populations and also an aesthetic role for landscaping purposes. Three watercourses are proposed to be retained (and enhanced and /reinstated where possible) to increase the ecological value of the PDA and to help maintain species diversity through reduction of habitat fragmentation (even possibly enhancing ecological linkages from the current condition).

8.6.6.13 Zoning of these watercourses and associated buffers as 'Green Belt' will aid to preserve the integrity of the good quality watercourses and their riparian corridors which are largely natural/semi-natural with good water quality and have high ecological value providing linkages and corridors. These watercourses have good species diversity and assemblages, and they act as nurseries for several aquatic species. Currently good quality lowland watercourses are one of the most threatened habitats in lowland Hong Kong. It is anticipated the proposed zoning will also help to reduce any unauthorized encroachment and illegal dumping and/or pollution of these high quality systems.

8.6.6.14 Within the "GB(1)" zone, there is scope to enhance the current riparian corridor. Whilst 'GB(1)" is only a planning incentive with more stringent regulation than normal "GB" zone to restrict the future development falling within this area, the Project Proponent should look to revitalize the watercourse within "GB(1)" when opportunities arise in order achieve habitat enhancement.

8.6.6.15 Prior to any commencement of works which would lead to watercourse loss, pre-construction surveys are required to check for presence of any aquatic fauna species of conservation importance in the watercourses to be directly impacted before commencement of works Should any aquatic fauna species of conservation importance occur in any of those fragmented and polluted watercourses in the lowland areas, then an Aquatic Fauna Translocation Programme proposal shall be designed, developed and agreed with the relevant authorities. Potential receptor sites include the above-mentioned three lowland watercourses with buffer zones, or the recreated watercourse that is proposed to run along the western boundary.

Mitigation for Impacts to Water Quality and Hydrology

8.6.6.16 Generally, indirect water impact to any aquatic fauna during the construction phase should easily be avoided by implementing water control measures (ETWB TCW No. 5/2005) to avoid direct or indirect impacts any watercourses and good site practices (further details are discussed in **Section 6**).

8.6.6.17 The hydrology of those watercourses retained with GB(1) Buffers will remain unaffected by the Project as they will maintain links with upland sources and flow will not be intercepted.

8.6.6.18 The source of the water for those low quality watercourses in Catchment 9 is unknown due to the local land use and extent of development. Should there be any hydrological issues to these watercourses, then mitigation should be by way of translocation of fauna as per the previous section, with a particular focus on the endemic crab species and other native aquatic species, if present. Exotic fauna should not be translocated to the Hillside River Corridor or those retained watercourses.

Mitigation for Reducing Risk of Bird Strike with Noise Barriers

8.6.6.19 Poorly designed noise barriers do pose a risk to flying birds, especially those which utilise transparent material. Collisions usually occur as a result of birds perceiving a clear path through an object that is transparent or appears to be transparent at some distance, or if the barrier is highly reflective thus appearing to be composed of the adjacent natural vegetation. Potential risk of collision mortality may be minimised by the use of opaque, non-reflective panels, through which it is obvious there is no passage, and no reflection of the surrounding environment is created. As a precautionary approach, consideration should be given at the detailed design stage to avoiding the use of highly reflective materials in the design, and implementing the use of opaque materials, fritting, breaking up external reflections with stickers or plastic wrap and/or any other bird-friendly design for noise barriers.

Mitigation Measures to Minimize Potential Direct Ecological Impacts (Mortality) to Nesting Birds

8.6.6.20 There will be a certain amount of vegetation clearance and tree felling required. Many of the trees affected are landscape species; these are often introduced exotic species, with little ecological value, and have been planted for aesthetic appeal. The ecological impacts arising from the loss of these trees are not considered to be significant.

8.6.6.21 However, precautionary checks of the vegetation and open areas for the presence of nesting birds should be carried out before vegetation clearance by a suitably experienced ecologist. If active nests are found, site clearance should be postponed until the young have fledged. All workers will be informed that all wild birds, including their nests and eggs, are protected under the Wild Animals Protection Ordinance (Cap. 170).

Mitigation Measures to Minimize Potential Direct Ecological Impacts (Mortality) to Bats

8.6.6.22 Bats are vulnerable to mortality at roosts, especially if the roosts are being utilized by nursing females. Some insectivorous bat species utilize abandoned buildings or trees as roost sites.

8.6.6.23 Mortality impacts on bats are predicted to be low in the absence of mitigation: however, direct ecological impact on roosting bats is considered to be low to moderate. Given that bats are a relatively little known group and identification is difficult, it is recommended that on a precautionary basis, prior to the demolition of buildings or felling or transplanting of trees, sites should be checked for the presence of bat roosts by a suitably qualified ecologist. If roosts are found to be occupied by nursing females, demolition should be postponed until the roost is vacated by nursing females.

Mitigation Measures to Minimize Disturbance Impacts to the Egretry

8.6.6.24 The egretry location shall be checked for any evidence of occupation during the ardeid breeding season by a qualified ecologist of the ET prior to the commencement of any works activity within 250m of the egretry. A monitoring programme shall be developed with the relevant authorities to confirm the requirements for monitoring the conditions of the egretry and potential impacts to egretry flight-lines during construction phase and the first 12 months of Project operation. Furthermore, as a precautionary measure, good site practices and confining artificial lighting within the site will be implemented to minimize

potential impact to the egretry for all construction works within 250m of the egretry.

Good Site Practices

8.6.6.25 In order to demonstrate ecological awareness and to minimise the risk of indirect impacts from water pollution, a series of good site practices should be adopted by site staff throughout the construction phase at each works site. Such measures include the containment of silt runoff within the Project boundary, the containment of contaminated soils for removal from the site, appropriate storage of chemicals and chemical waste away from sites of ecological value and the provision of sanitary facilities for on-site workers. Adoption of such measures should permit waste to be suitably contained within the site for subsequent removal and appropriate disposal. These are as follows:

- Erect signs to alert site staff about any locations which are ecologically sensitive and measures to prevent accidental impacts e.g. along those watercourses to be retained as GB (1), The Hillside River Corridor and Retention Lake, Shan Ha Woodland and agricultural land at Shan Ha Tsuen;
- Erection of temporary geotextile silt or sediment fences/oil traps around any earth-moving works to trap any sediments and prevent them from entering watercourses;
- Prohibition of soil storage against trees or close to waterbodies;
- Delineation of works site to prevent encroachment onto adjacent habitats and fence off areas which have some ecological value;
- No smoking, hot works or sources of fire close to upland grassland;
- No on-site burning of waste; and
- Waste and refuse in appropriate receptacles.

Ecological Enhancement Opportunities

Reedbed

8.6.6.26 A reedbed with an area of about 4ha is proposed to be built at the south of Area 3 (see **Section 2).** The drainage/stormwater will be treated before being discharged into Yuen Long Nullah. Such a treatment system can provide dual roles i.e. as a clean-up reedbed and providing significant ecological enhancement to the area providing valuable foraging and shelter opportunities for a suite of aquatic fauna such as *S. zanklon*; and in particular wetland dependent birds.

8.6.6.27 Further development and design of the reedbed will be required to optimise enhanced capabilities and also provision of ecological function. It is recommended that a deep water reedbed is formed (i.e. min. 0.5m deep water) to maximise ecological function and also reduce on-going maintenance issues. Other sub-surface flow clean-up reedbeds in Hong Kong have limited ecological function due to the regular establishment of terrestrial flora which reduces clean-up capabilities and overall ecological function and benefits.

8.6.6.28 A freshwater marsh (minimum area of 0.5ha) prior to discharge to create habitat opportunities for common aquatic species and *S. zanklon* will be provided. This

habitat would also create additional foraging opportunities for ardeids, in particular, Chinese Pond Heron. This will be further developed in the detailed design stage.

Retention Lake

8.6.6.29 A Retention Lake, of approximately 0.6ha, is proposed at the northern end of the hillside watercourse. Whilst the primary function of this to prevent flooding further downstream, there will be vegetated, sloping banks and the lake is anticipated to always retain water.

8.6.6.30 The inclusion of berms or shelves within the lake would allow for shallow water to be present for suitable planting of aquatic macrophytes at the edge to create additional habitat. These berms can be designed to be submerged during floods and therefore should not impact on the overall flood capacity. Creation of berms allows more structural diversity within the lake and offer more ecological niches for aquatic fauna. Berms will be planted with vegetation which can tolerate temporary flooding and the berm will retain wet areas during periods of low water levels. Therefore, with appropriate planting and careful design at the detailed design stage, there are potential opportunities for waterbirds e.g. Little Egret, Cattle Egret, Chinese Pond Heron and a range of wetland fauna to utilise this habitat. Conceptual drawings of this lake can be seen in Figure 8.11b.

8.6.6.31 Any stands of suitable bamboos occurring within the vicinity should be retained and the habitat in the immediate environs enhanced (by planting and transplanting of larger trees e.g. *Ficus microcarpus, Celtis sinensis* and bamboos where practicable) to increase nesting opportunities for ardeids in this area. Consideration should be given to the creation of an island within the Retention Lake, planted with a mixture of native trees and bamboo suitable for nesting ardeids. Whilst there is yet to be a successful example of a purpose built egretry in Hong Kong, in this instance the close proximity of an existing egretry, the option of creating a single large pond and the ability to control human access, all combine to provide a unique opportunity to create such an egretry.

8.6.6.32 Installation of floating islands of either native vegetation or gravel substrates (which can allow for natural plant colonisation) can also add ecological benefits to open water bodies, and are recommended for consideration in the Retention Lake. These floating islands can also provide nesting opportunities for smaller waterbird species e.g. White-breasted Waterhen. In addition overhanging and emergent vegetation and submerged root systems can provide shelter for fish fry and invertebrate larvae and additional foraging habitat for ardeid species, in particular Chinese Pond Heron.

Revitalisation of Nullahs

8.6.6.33 Revitalisation is proposed in the open channels of three channelised watercourses: Yuen Long (West) Nullah (~0.5km), Tin Tsun Channel (~1.5km) and Yuen Long Nullah (~2.5km). Approximately 1.47ha (about 4km) within PDA and 0.58ha (about 0.7km) outside PDA of channelized watercourses will be subject to proposed decking and revitalisation.

8.6.6.34 In-channel modifications to create microhabitats are recommended; this can be achieved by adding different substrates to provide a diverse environment with different biotic and abiotic components for different species which further enhance channel complexity (DSD 2015). In-channel cover can provide refuge for fish and other aquatic life protecting them from predators and in times of spate. They may

be constructed from rock, wood, aquatic vegetation or built into hard edge banks. These should be structurally diverse, creating numerous shapes and sizes of holes and crevices which aquatic life can exploit.

Watercourse Diversion

8.6.6.35 A short length of the lower section of the watercourse in Catchment 7 is proposed to be diverted to create a continuous open water feature, connecting this watercourse with the Yuen Long (West) Nullah. This open channel will be designed to have natural substrate as far as practicable and specific reference should be made to making reference to DSD Guidance notes (DSD 2015) and other international practices to maximise wildlife opportunities for this habitat creation

8.6.6.36 Tributaries within Catchment 4, within the footprint of the proposed clean-up reedbed, would either be retained or realigned to maintain their connectivity and ecological function.

Landscape planting

8.6.6.37 Further ecological enhancement can be provided throughout the scheme by planting of suitable species to offer more resources to widespread species which commonly occur close to, or within, human activities.

8.6.6.38 Within the "LO" and "DO" zoning), retention, transplantation and or provision of native plant species should be included to increase ecological value of this development. Species selected may offer food resources through flowers, seeds and or fruits for invertebrates and birds, roosting opportunities (for bats and birds) or suitable nesting substrates.

8.7 Residual Impacts

8.7.1.1 This section addresses the requirements of Appendix F 4 (xi) & (xii) of the Study Brief. Directs impacts to habitats of ecological significance, including large areas of contiguous woodland, watercourses of higher ecological value and the Tai Tong (Pak Sha Tsuen) egretry have been avoided. Direct impact on habitats to be permanently lost by the project are identified and quantified in **Section 8.6.3** and **Table 8.47**. Ecological impacts as evaluated in **Section 8.6.3** and **Table 8.66** range from "Negligible" to "Moderate" severity. With the implementation of mitigation measures, the residual direct impact of habitat loss due to the Project is considered to be minor and acceptable.

8.7.1.2 It is anticipated that through proper implementation of the mitigation and enhancement measures mentioned in **Section 8.6.6**, there will be an overall ecological gain in the PDA and the vicinity. By avoidance of key habitats, notably good quality watercourses, incorporation of watercourse buffer zones in the RODP and the provision of connectivity through restoration/recreation of a new watercourse along the western boundary, greater ecological opportunities will be provided in this currently disturbed and polluted area.

8.7.1.3 The opportunity for ecological enhancement within this scheme comes through the retention and reinstatement of watercourses, incorporating ecological design within existing channelised watercourses, provision of a 0.6ha Retention Lake and a 4 ha Reedbed to create foraging opportunities for wetland species and the creation of a new 575m watercourse to provide greater ecological connectivity. Potentially, with forward-thinking approaches and implementation at the detailed design stage, there

is scope for significant ecological gain particularly for watercourses which have been historically subject to fragmentation and neglect.

8.8 Environmental Monitoring and Audit Programme

8.8.1.1 Implementation of the recommended mitigation measures described in Section 8.6 should be conducted as presented in the standalone Environmental Monitoring and Audit (EM&A) Manual.

Woodland Compensation

8.8.1.2 Details of the monitoring programme for the compensation woodland will be included in the Woodland Compensation Plan. The monitoring programme of the compensatory woodland shall allow on-going review of the effectiveness of the implemented plantings and to ensure long-term establishment of the planted species. Monthly monitoring shall continue throughout the period of planting in Year 1. The monitoring frequency can be reduced to bi-monthly (once in every two months) or quarterly at later stages of the monitoring period depending on the conditions of the planted specimens.

Watercourses

8.8.1.3 During the construction phase, these watercourses should be monitored monthly by a qualified ecologist to confirm they are not significantly impacted by nearby construction activities. Site audits should be undertaken to ensure construction activities are not encroaching or impacting upon the watercourses. These site audits should be conducted when there are construction activities within 100m of any retained watercourses by a suitably qualified ecologist. Aquatic faunal monitoring on monthly basis shall also be conducted in the first 12 months of the operation phase at the three retained watercourses in Area 1 and Area 3 and the new watercourse along the hillside of the western boundary of Area 3.

8.8.1.4 For those watercourses to be lost, translocation of any fauna of conservation importance that may occur should be conducted before commencement of any construction works. Monitoring of any translocated organisms should be conducted for an appropriate period of time by suitably qualified ecologists by conducting aquatic fauna surveys.

Egretry

8.8.1.5 Prior to any construction activities, surveys of the Tai Tong (Pak Sha Tsuen) egretry should be conducted to confirm its location and status.

8.8.1.6 During the construction phase, for any works within 250m of the egretry, weekly monitoring will take place during the Ardeid breeding season i.e. from March to August by a qualified ecologist to confirm it activity during the breeding season and to ensure the egretry is not significantly disturbed by nearby construction activities. A monitoring programme will be developed with the relevant authorities to confirm the requirements for monitoring the conditions of the egretry and potential impacts to the egretry flightlines during construction and the first 12-months of the Project operation.

8.9 Conclusion

8.9.1.1 The RODP has preserved all woodland patches of significant sizes in the PDA and avoided encroachment upon the mature woodland behind Shan Ha Tsuen through adoption of separate road junctions located away from the preserved woodlands. The proposed Road D1 will unavoidably run through the eastern edge of agricultural land behind Shan Ha Tsuen. The integrity of Shan Ha Tsuen Agricultural Land and linkages with more natural, less disturbed habitats with higher ecological value to the west and south will still be retained in the RODP. The resulting loss of agricultural land has been minimized and the fragmentation impact on this habitat is considered to be low.

8.9.1.2 The impacts resulting from loss of hillside secondary woodland (~2.42ha in total) is considered to be of moderate severity and will require appropriate ecological mitigation. On-site woodland compensation appears to be impractical due to limited space within the PDA. An off-site/ near-site for compensatory woodland planting (at a site of ~12ha) has been proposed to provide a compensation area ratio of at least 1:1 to adequately mitigate the impacts from woodland loss.

8.9.1.3 The RODP will also retain three lower sections with buffers of the natural watercourses which have records of the endemic crab *S. zanklon*. Translocation of this species can be conducted to these three streams or the proposed 575m Hillside River Corridor should they occur in those fragmented watercourses in the lowlands.

8.9.1.4 Loss of sections of watercourse which have an ecological value of Low-Moderate, or potentially provide suitable habitat for *S. zanklon* and/or other aquatic fauna of conservation importance, are considered to be of Low-Moderate severity. Mitigation for the loss of this habitat by the creation of the Hillside River Corridor and enhancement by the Retention Lake and Reedbed are proposed. A translocation programme for *S. zanklon* and/or other aquatic fauna of conservation importance can be implemented, using these new created habitats or retained suitable watercourses as receptor sites.

8.9.1.5 Additional feeding opportunities for breeding ardeids may be provided through the creation of the 0.6ha Retention Lake, the Reedbed and the Hillside River Corridor and through the protection and enhancement of the three retained watercourses with GB(1) Buffers and the revitalisation of nullahs. Moreover, additional nesting opportunities for breeding ardeids will be provided through retention of bamboos and planting of large trees at the Retention Lake.

8.9.1.6 For flightlines to the northeast of the egretry, whilst there will be some minor adjustments to the departure direction of breeding ardeids from the egretry, it is considered that there is sufficient space for birds to make minor adjustments at the start of the flightline. It is considered that for flightlines to the north, south, east and west of the egretry, utilised by the majority of the breeding birds, there are no significant impacts.

A reedbed with an area of about 4ha is proposed to be built at the south of Area 3. The drainage/stormwater will be treated before being directly discharged into Yuen Long Nullah can provide ecological enhancement to the area providing foraging and shelter opportunities for a suite of aquatic fauna, and in particular, wetland dependent birds.

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